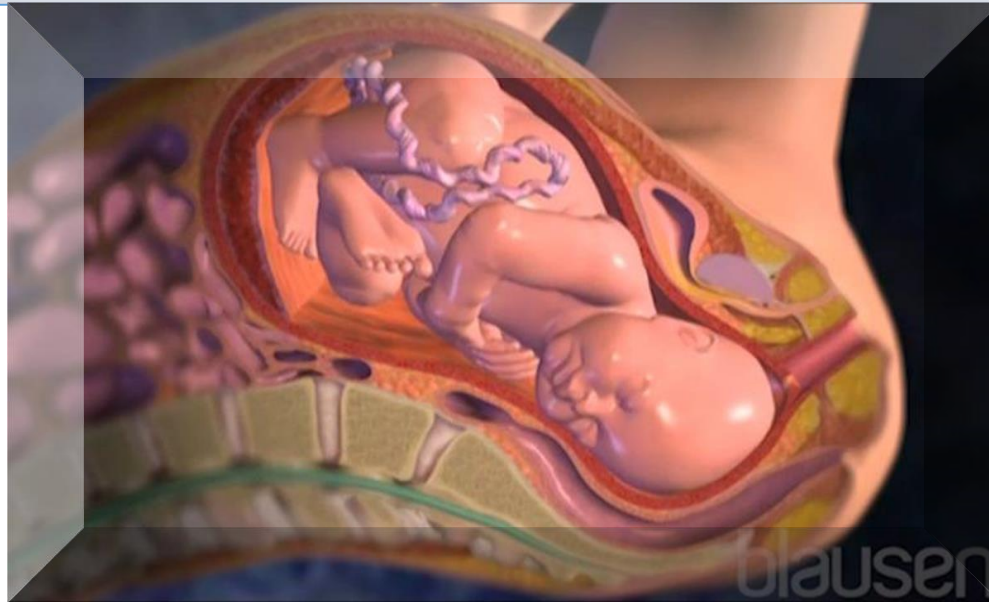




Tropical college of medicine

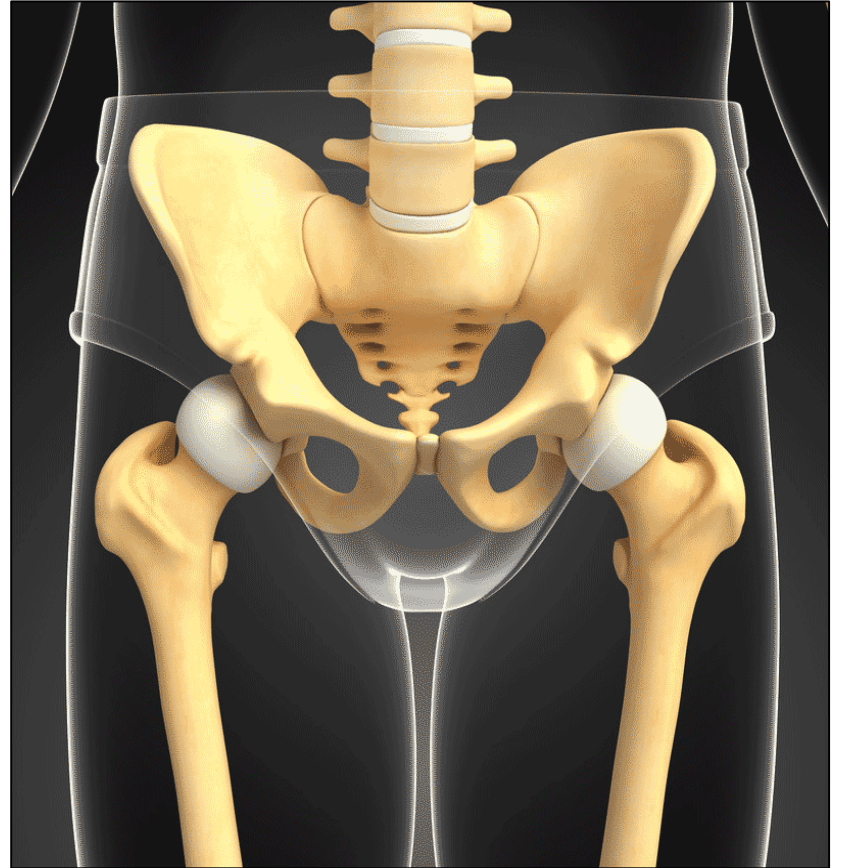


Obstetrics and gynecology Nursing I
for 3rd year Bsc nursing student

Dr Abreham Tizazu(MD)
2016 EC
Debremarkos,Ethiopia

Unit- I

- Review of anatomy and physiology of female reproductive system



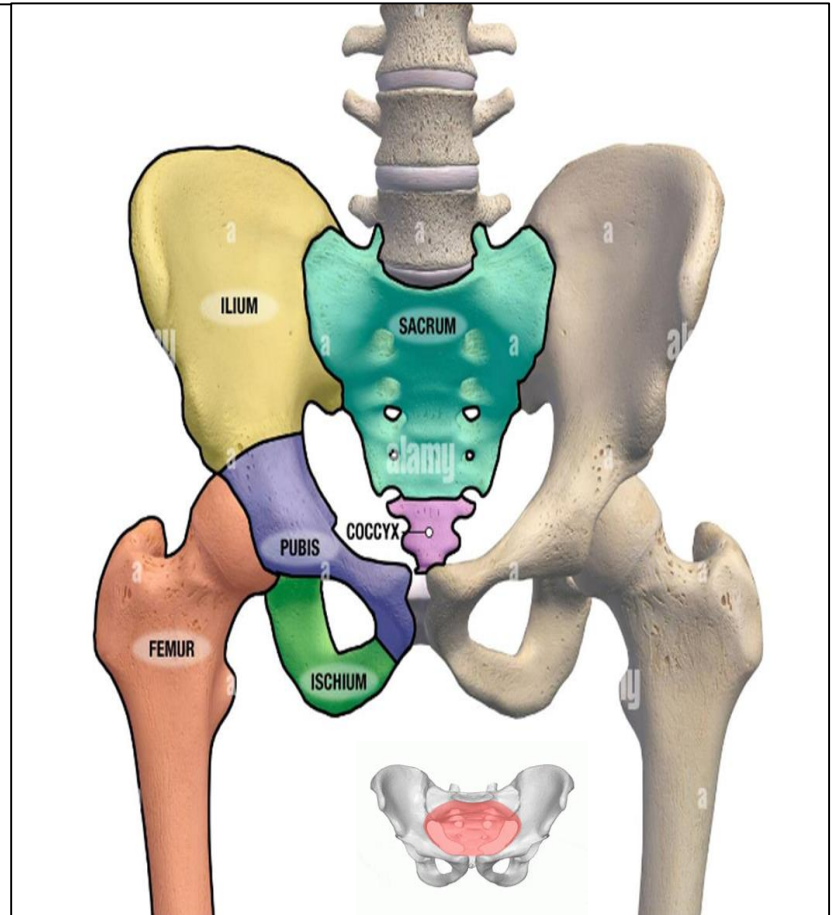
Female pelvis

- Objectives

- ⇒ Define female pelvis.
- ⇒ List the functions of female pelvis
- ⇒ Describe the bones, joints of the pelvis
- ⇒ Describe types of pelvis
- ⇒ Explain the pelvic floor.
- ⇒ Demonstrate pelvis assessment

Female pelvis

- The part of the trunk inferoposterior to the abdomen.
- Area of transition between the trunk and the lower limbs.
- Part of the appendicular skeleton of the lower limb.
- The space or compartment surrounded by the bony pelvis.



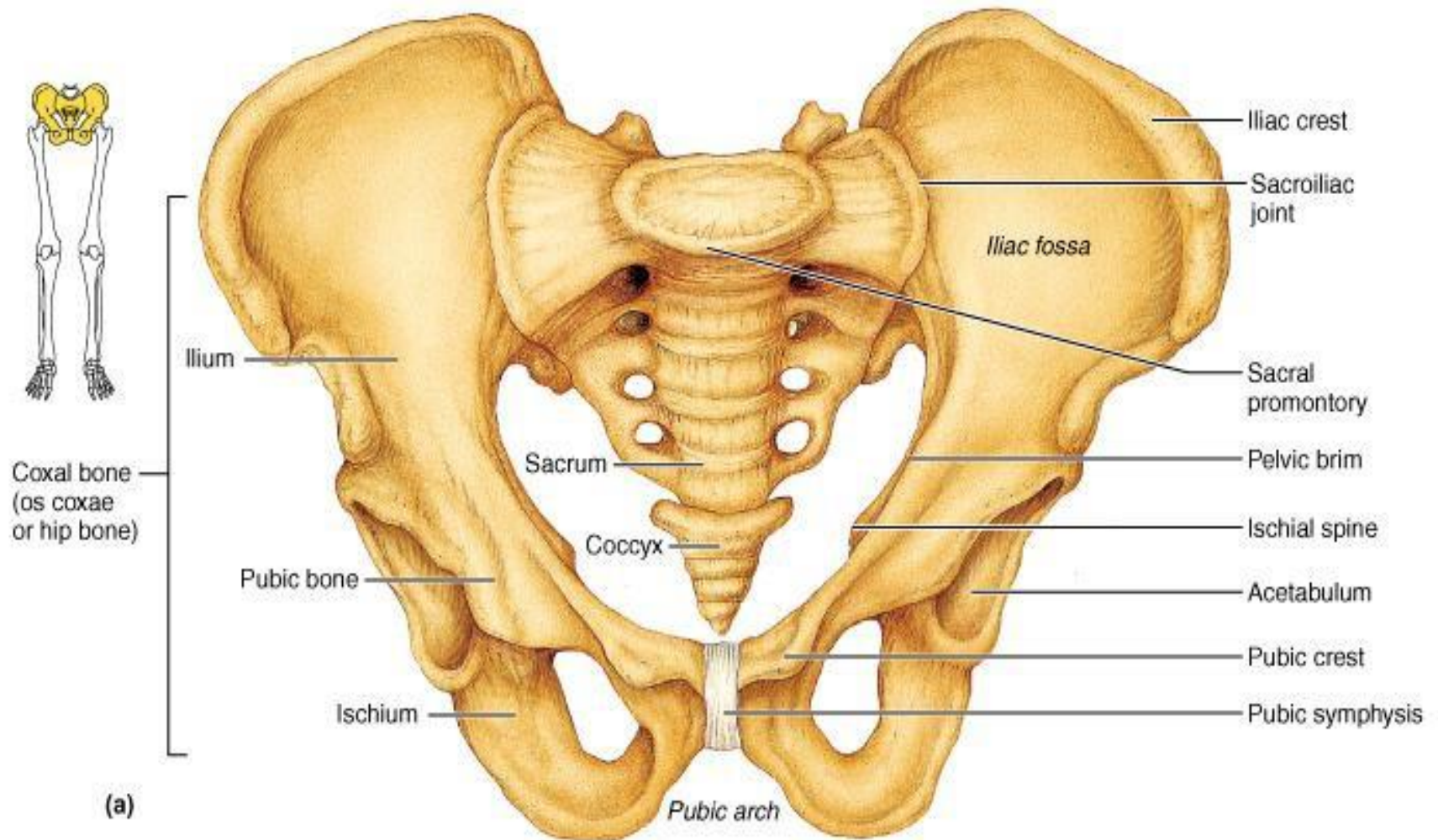
Function of pelvis

- Adapted for child bearing
- Allow movement of the body
- Permits the person to sit and stand
- Transmits the weight of the trunk to the leg
- Protection to pelvic and abdominal organs
- Provide attachment for the powerful muscles
- Provide attachment for the erectile bodies of external genitalia.

The bones of the pelvis

- Is the strong bone which forms the essential part of the skeleton frame work.
- It is composed of four bones:
 1. Two innominate or hip bones
 2. One sacrum
 3. One coccyx

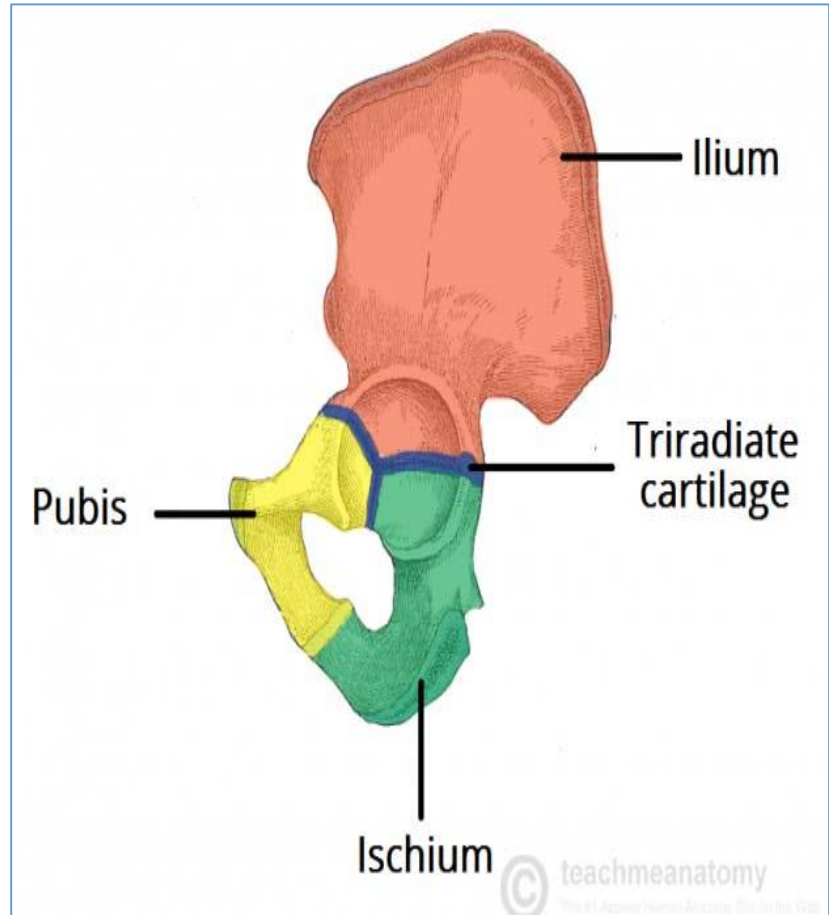
The bones of the pelvis....



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Innominate or hip bones

- Each innominate bone is formed by fusion of three bones
 1. The ilium
 2. The ischium
 3. The pubic bone
- The three bones fused at acetabulum



1. The ileum

- Is the upper expanded part
- Is the large flared out part of the innominate bone
- Is made up of a relatively flat of bone above and part of acetabulum below.
- The superior, fan-shaped part of the hip bone

2. The ischium

- The lowest constituent of innominate bone .
- Formed by the following parts
 - ⇒ Forms the lowest two-fifths of the acetabulum
 - ⇒ Ischial tuberosity the part which the weight of the body rest on when sitting.
 - ⇒ Forms the lower boundary of the obturator foramen.

2. The ischium...

- Internal aspect the ischium forms the side wall of true pelvis.
- Its conspicuous projection known as the ischial spine which is used to estimate the station of fetal head.
- Lesser sciatic notch is situated between the ischial spine above and ischial tuberosity below.

3. The pubis

- The smallest of the three bones forming innominate bone
- Form the lower one fifth of the acetabulum.
- The right and left pubic bones unite at symphysis pubis.
- The upper surface of the body forms a crest, the pubic crest, which ends laterally in the pubic tubercle.

2. The sacrum

- Is five fused sacral vertebral in the posterior part of pelvis
- The sacral alae – are the widened out wings of bone on each side of the 1st sacral vertebra.
- Promontory of the sacrum – is the center point of the upper border of the 1st sacral vertebrae which with the base of the 5th lumbar vertebra, protrudes over the hollow of the sacrum.

2. The sacrum...

- Sacrum articulates
 - ⇒ Laterally with the ileum at sacroiliac joint
 - ⇒ Lower margin with coccyx at sacro-coccygeal joint.

3. The coccyx

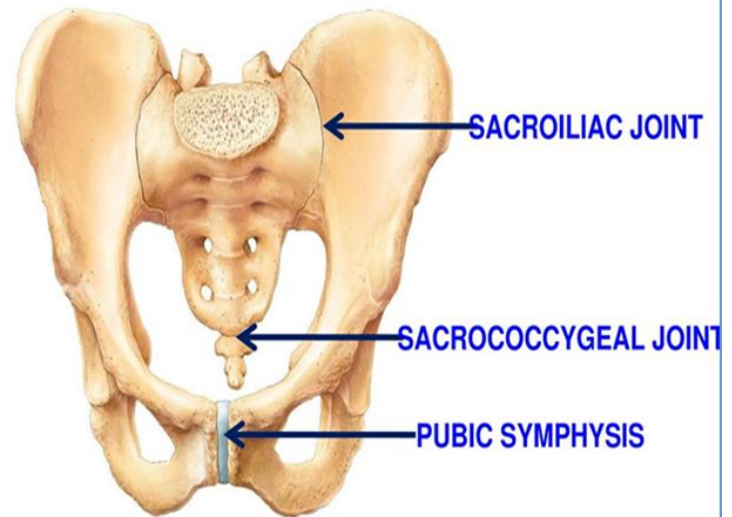
- Consists of four fused coccygeal vertebrae.
- It is triangular in shape with its base uppermost.
- The 1st coccygeal vertebra articulates with lower end of the sacrum at the sacro-coccygeal joint.
- The remaining three vertebrae mere rudimentary nodules of bone

Pelvic joints

A. The sacro – Iliac joints

- The strongest joint in the body.
- Join the sacrum to the ileum
- Connect the spine to the pelvis.
- The joint cavity is very small

THE BONY PELVIS: JOINTS



Pelvic joints...

B. The symphysis pubis joint

- Is formed at the junction of the pubic bones which are united by a pad of cartilages.
- It is reinforced by supporting ligaments which pass from one pubic bone to the other in front, behind, above and below the disc of cartilage.

Pelvic joints...

C. The sacro- coccygeal joint

- Is situated between the lower border of the sacrum and the upper border of the coccyx.
- Slight back ward and forward movement of the coccyx on the lower end of the sacrum occur normally.
- The back ward movement is greatly increased during labor at the time of the actual birth of the head.

Features of normal female pelvis

Sacral promontory is not prominent or unreachable.

Rounded brim

The cavity is shallow(deep) with straight sides

Smooth (blunt) ischial spines

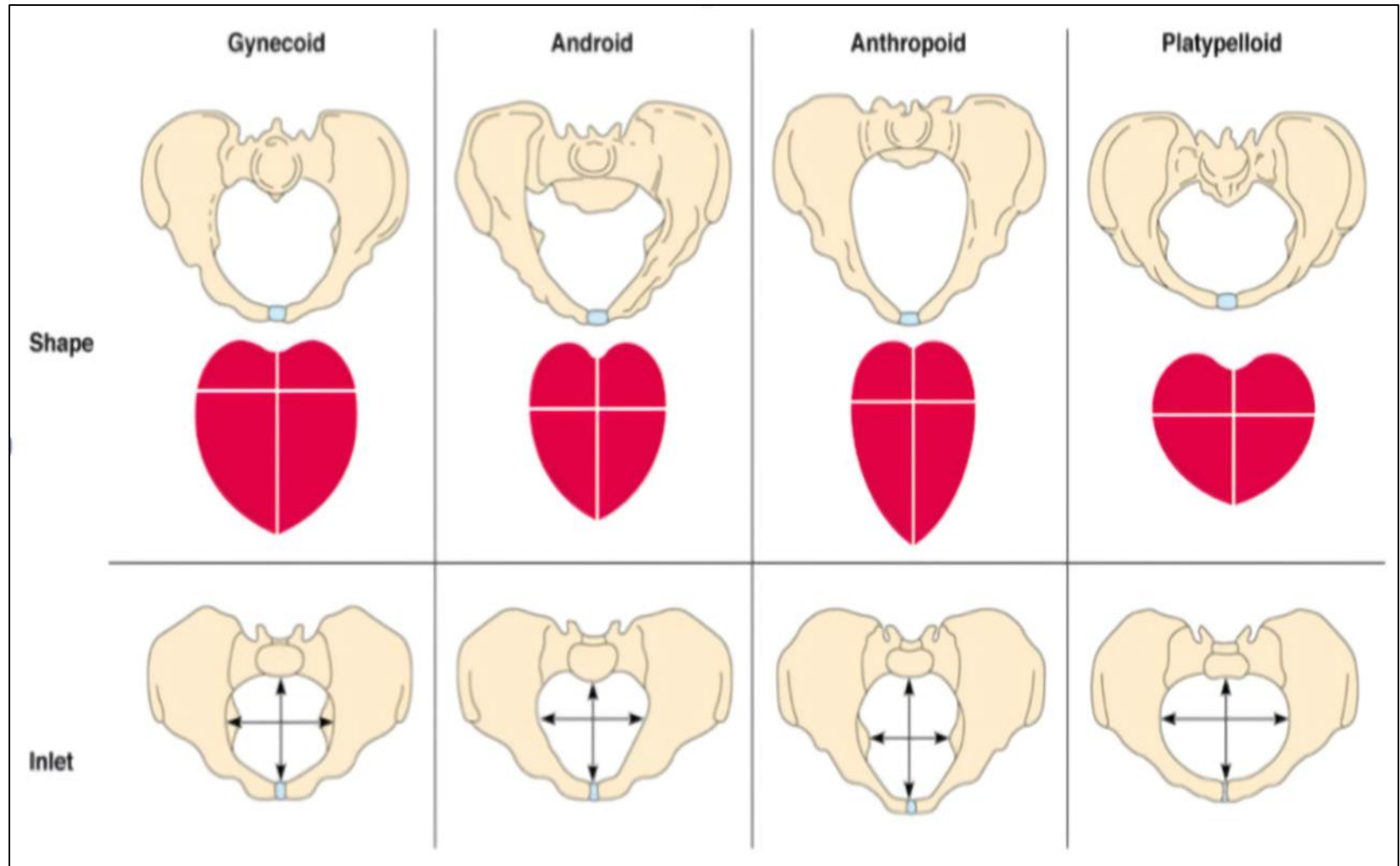
The sacrum is smoothly concave (curved in)

Wide sciatic notch.

The pubic arch is round

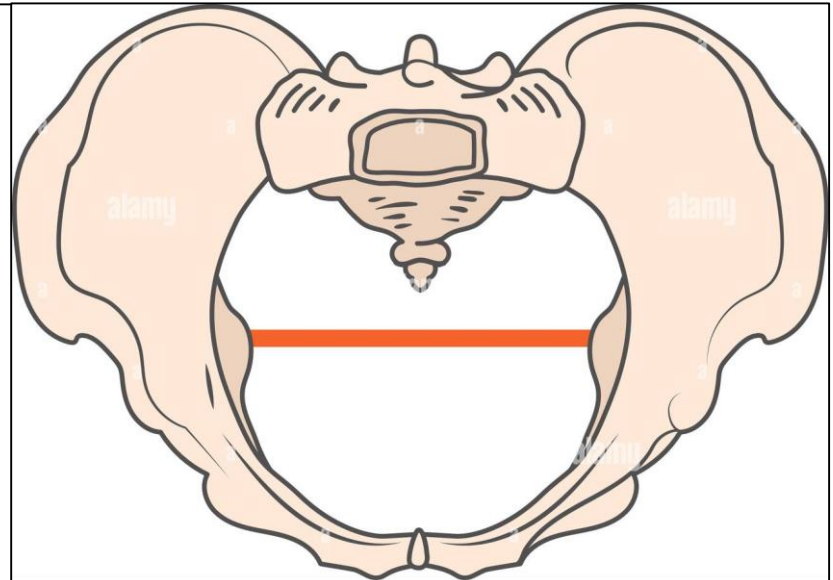
Sub pubic angle 90°

Types of pelvis



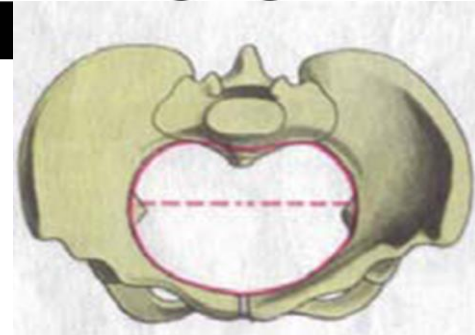
1. Gynecoid pelvis

- 50% of female pelvis
- Is the normal female type.
- Inlet is slightly transverse oval.
- Sacrum is wide with average concavity and inclination.
- Side walls are straight
- Blunt ischial spines.
- Sacro-sciatic notch is wide.
- Subpubic angle is 90-100°.



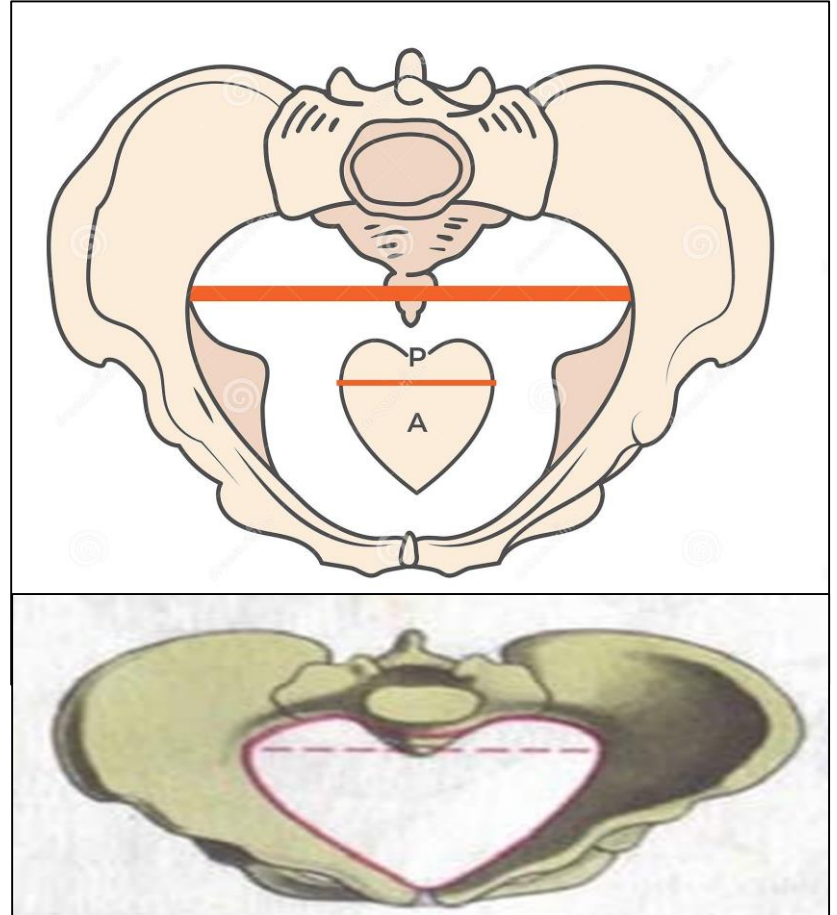
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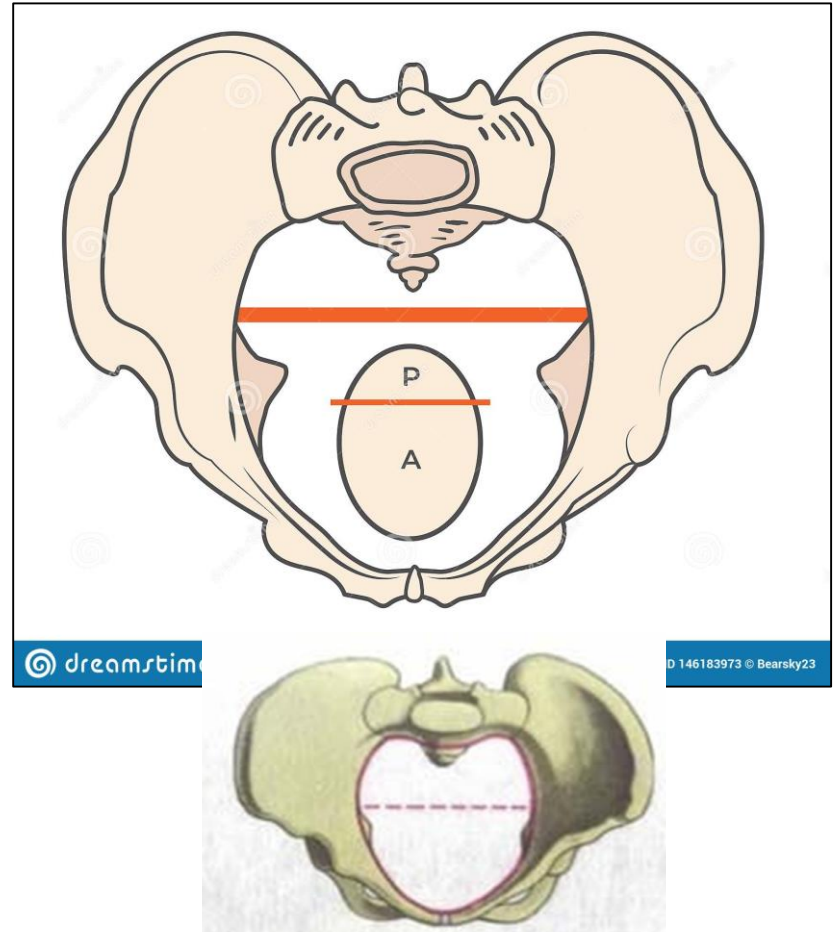
2. Android pelvis

- 20% of female pelvis
- It is a male type.
- Inlet is heart-shaped
- Side walls are converging
- Projecting ischial spines.
- Sacro-sciatic notch is narrow.
- Subpubic angle is narrow $<90^{\circ}$.



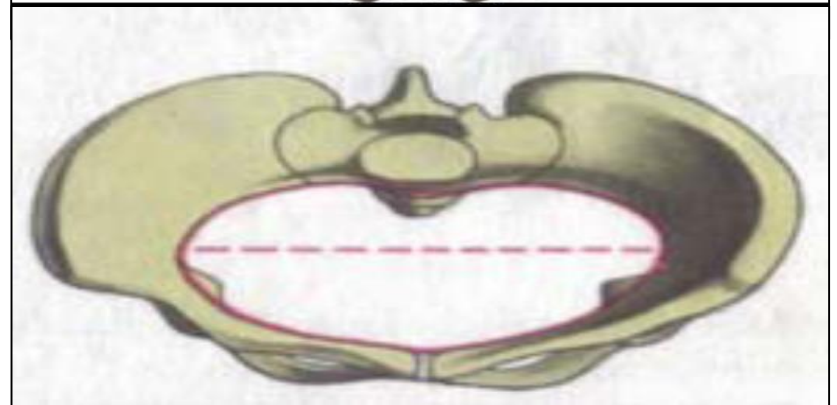
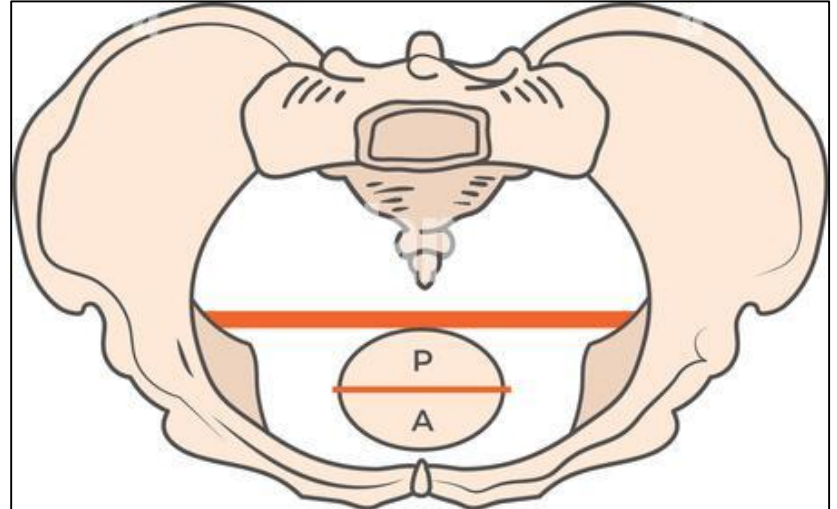
3. Anthropoid pelvis

- 25% of female
- It is ape-like type
- Lone antero-posterior diameters
- Short transverse diameters
- Sacrum is long and narrow.
- Sacro-sciatic notch is wide.
- Subpubic angle is narrow.



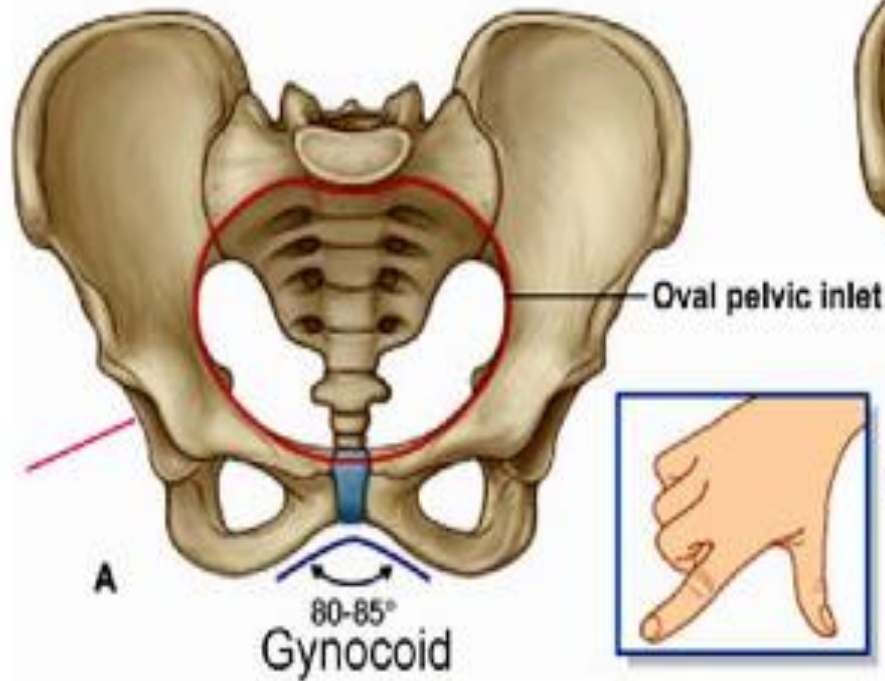
4. Platypelloid pelvis

- 5% of female pelvis
- It is a flat female type.
- It has kidney shaped brim.
- Short antero-posterior diameters
- long transverse diameters
- Sacro-sciatic notch is narrow.
- Subpubic angle is very wide.

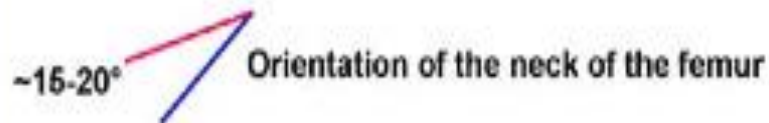
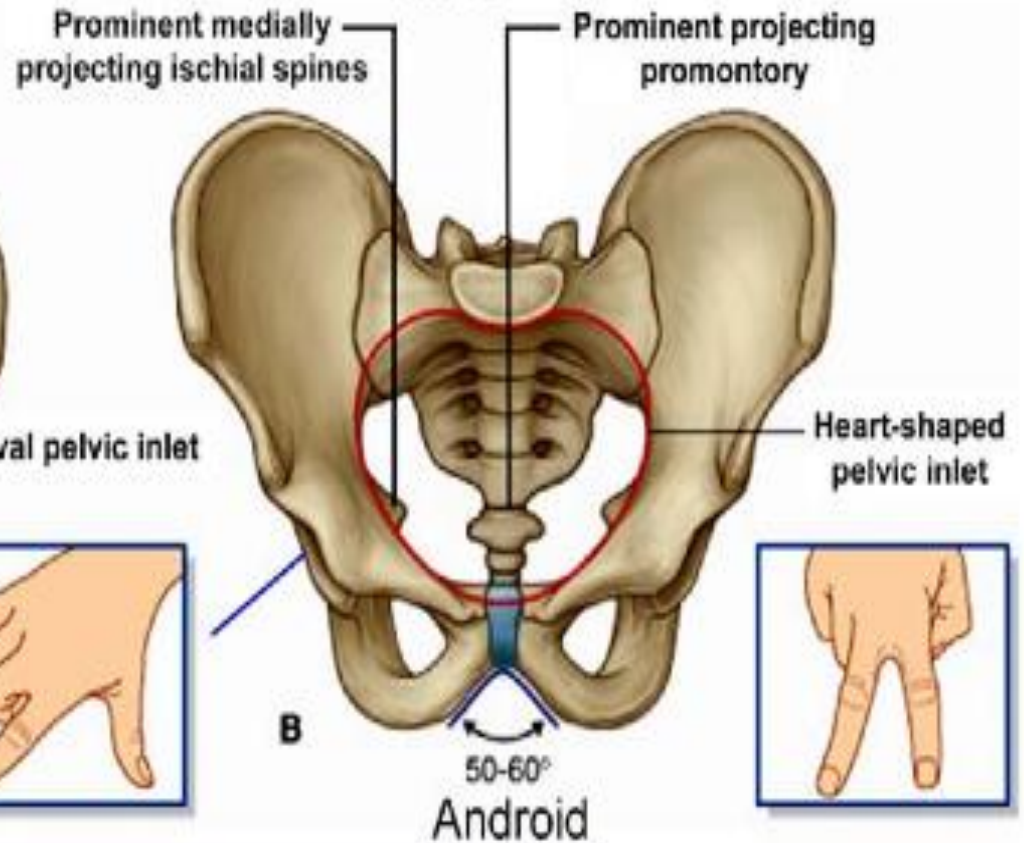


Female & Male Differences

Female



Male



Gray's 5.27

Pelvic floor

- Formed by soft tissues that fill the outlet of pelvis.
- The most important of these is the pelvic diaphragm
 - ⇒ Forms a broad muscular sling
 - ⇒ Provides substantial support to the pelvic viscera.

Pelvic floor....

- Function

- ⇒ Support the abdominal and pelvic organs
- ⇒ For sexual activity
- ⇒ Passive movement during child birth
- ⇒ Voluntary control of micturation and defecation

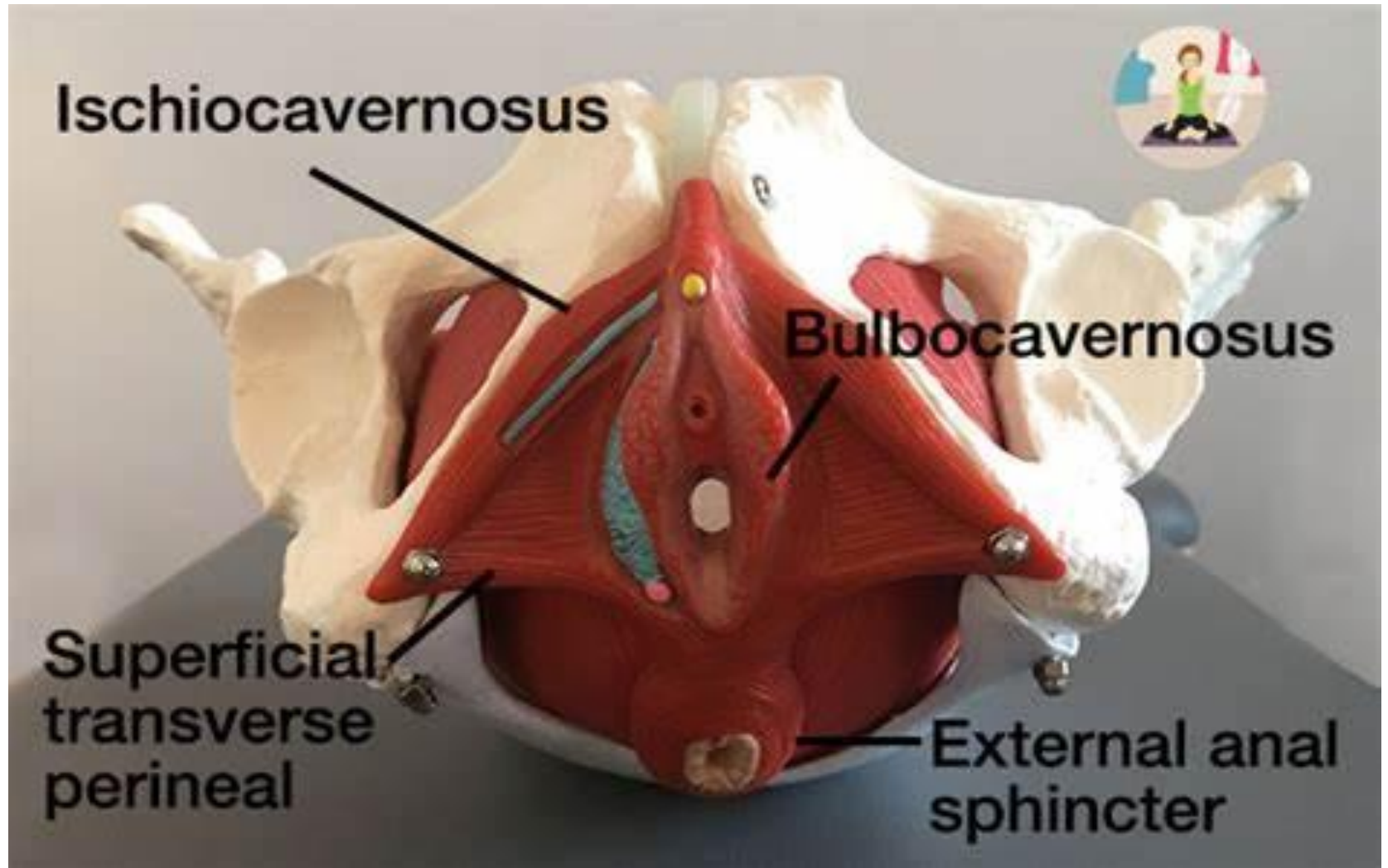
Type of pelvic floor

1. Superficial pelvic floor muscles
2. Deep pelvic floor muscles

Superficial floor muscles

Muscle	Description
Membranous sphincter of urethra	• Composed of muscle fibers passing above and below the urethra • Attached to pubic bones. • Is not true sphincter but acts to close urethra
Bulbocavernosous	• Passes from the perineum forwards around the vagina to the corpora cavernosa of clitoris.
Ischiocavernosus	• From ischial tuberosity along the pubic arch to the coccyx.
External anal sphincter	• Encircles the anus • Attached behind by a few fibers of the coccyx.
Transverse perineal	• Passes from the ischial tuberosities to the center of perineum.

Superficial floor muscles....



B. Deep pelvic floor muscles

- 3 pairs of muscles together known as levator ani muscles

Pubococcygeous
muscle

- From the pubis to the coccyx

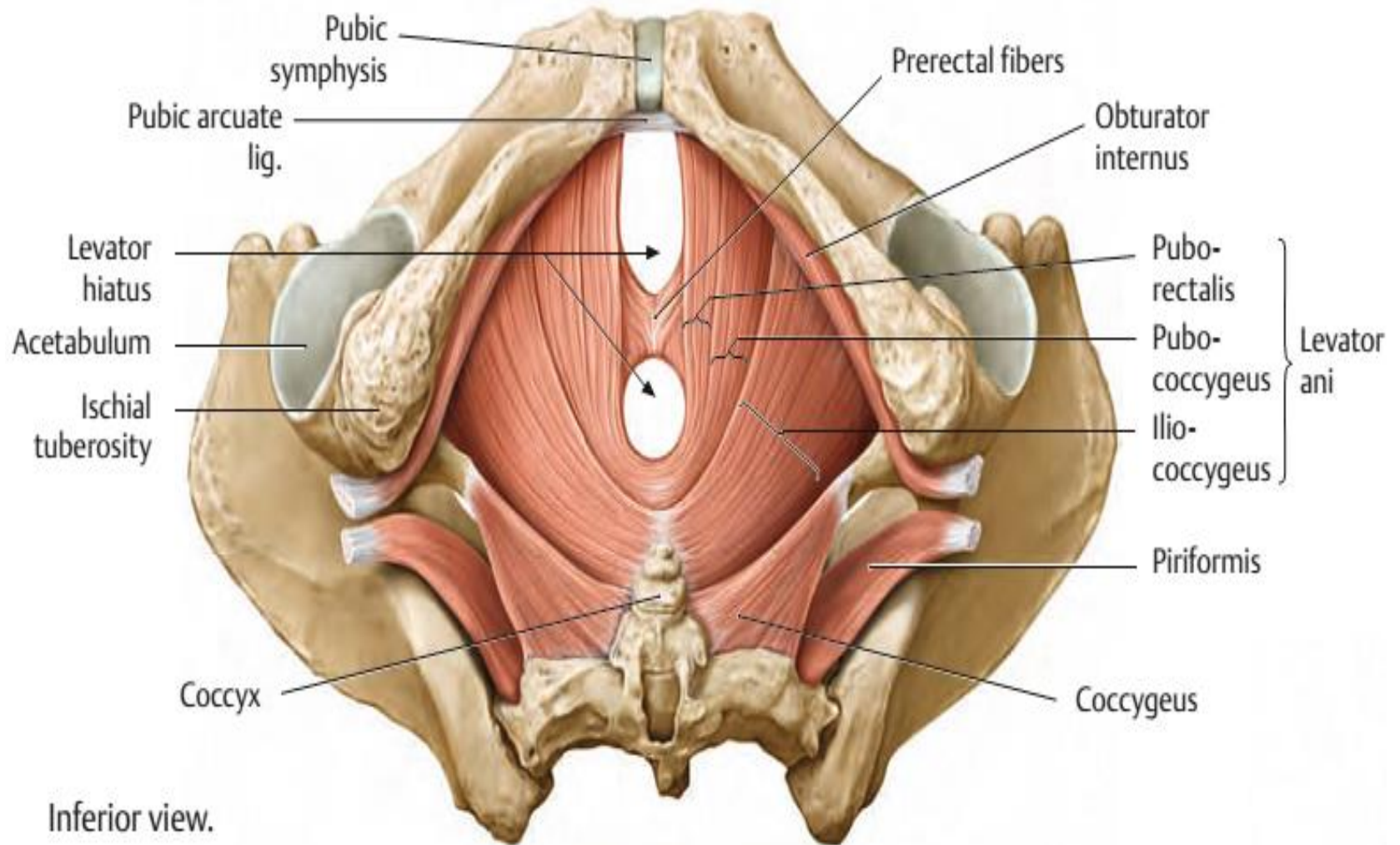
Iliococcygeous

- From fascia covering the obturator internus muscle (the white line of the pelvic fascia) to the coccyx.

Ischiococcygeous

- From ischial spine to the coccyx

Deep pelvic floor muscles...



Deep pelvic floor muscles...

- Function of levator ani
 - ⇒ Support of pelvic viscera in general
 - ⇒ Resist increased abdominopelvic cavity pressure
 - ⇒ Puborectal sling: important for urinary & fecal continence

Pelvic assessment

- It is done after 36 wks of pregnancy.
- Before 36wks the muscles are not relaxed.
- Pelvic examination(before 20weeks)
 - ⇒ To document evidences of pregnancy
 - ⇒ When there is vaginal bleeding
 - ⇒ To do clinical pelvimetry

Pelvic assessment...

- Equipment:

- ⇒ Sterile glove
- ⇒ Cotton swab in antiseptic solution
- ⇒ Waste container
- ⇒ Meter

Pelvic assessment...

- Position is lithotomy
- Preparation of the mother:
- Explain the procedure to the client to relax
- Empty her bladder and bowel before the procedure, because full bladder and rectum restricts the vaginal canal.

Pelvic assessment...

- Position client dorsally with leg flexed at hips and knees
- Inspect vulva for edema, warts, varicose vein, lesions, discharge
- Vaginal examination:
 - ⇒ Insert your index & middle finger which are gloved.
 - ⇒ If necessary use lubricant.
 - ⇒ The middle finger should reach promontory of sacrum

Pelvic assessment...

What to Expect During a Pelvic Exam

1

Lie on back with feet in stirrups



2

Doctor examines outside of vaginal area



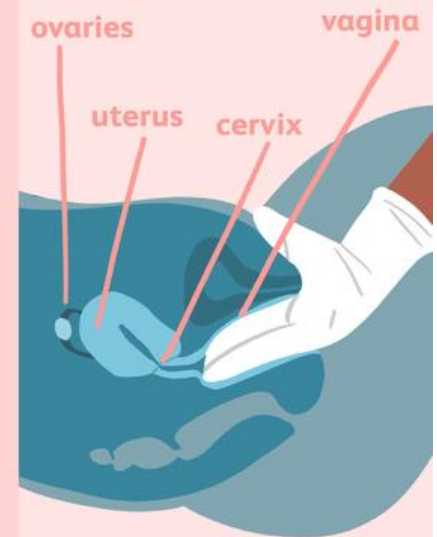
3

Speculum separates vagina walls; doctor examines cervix



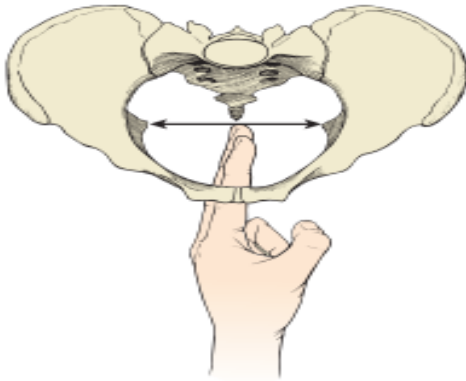
4

Doctor inserts 2 fingers to check uterus and ovaries

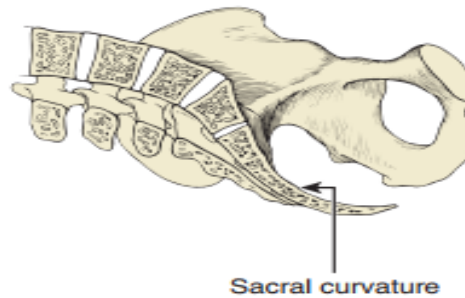


Pelvic assessment...

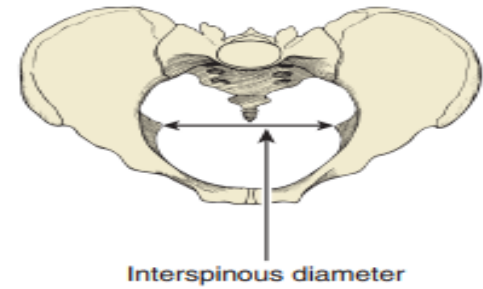
① Estimation of prominence of ischial spines



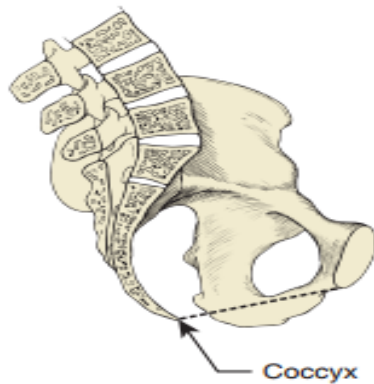
② Assess curvature of the sacrum



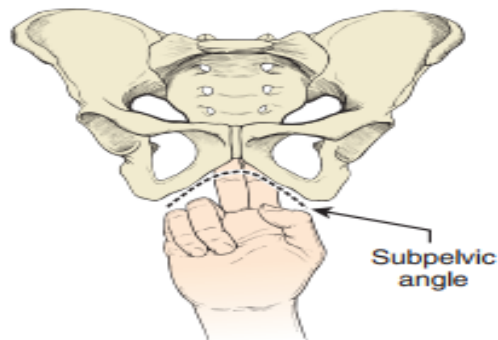
③ Assessment of interspinous diameter



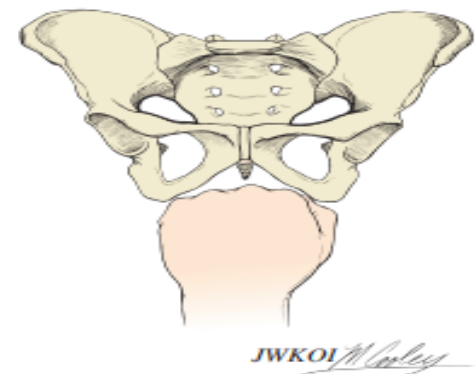
① Estimation of prominence of coccyx



② Estimation of subpelvic angle



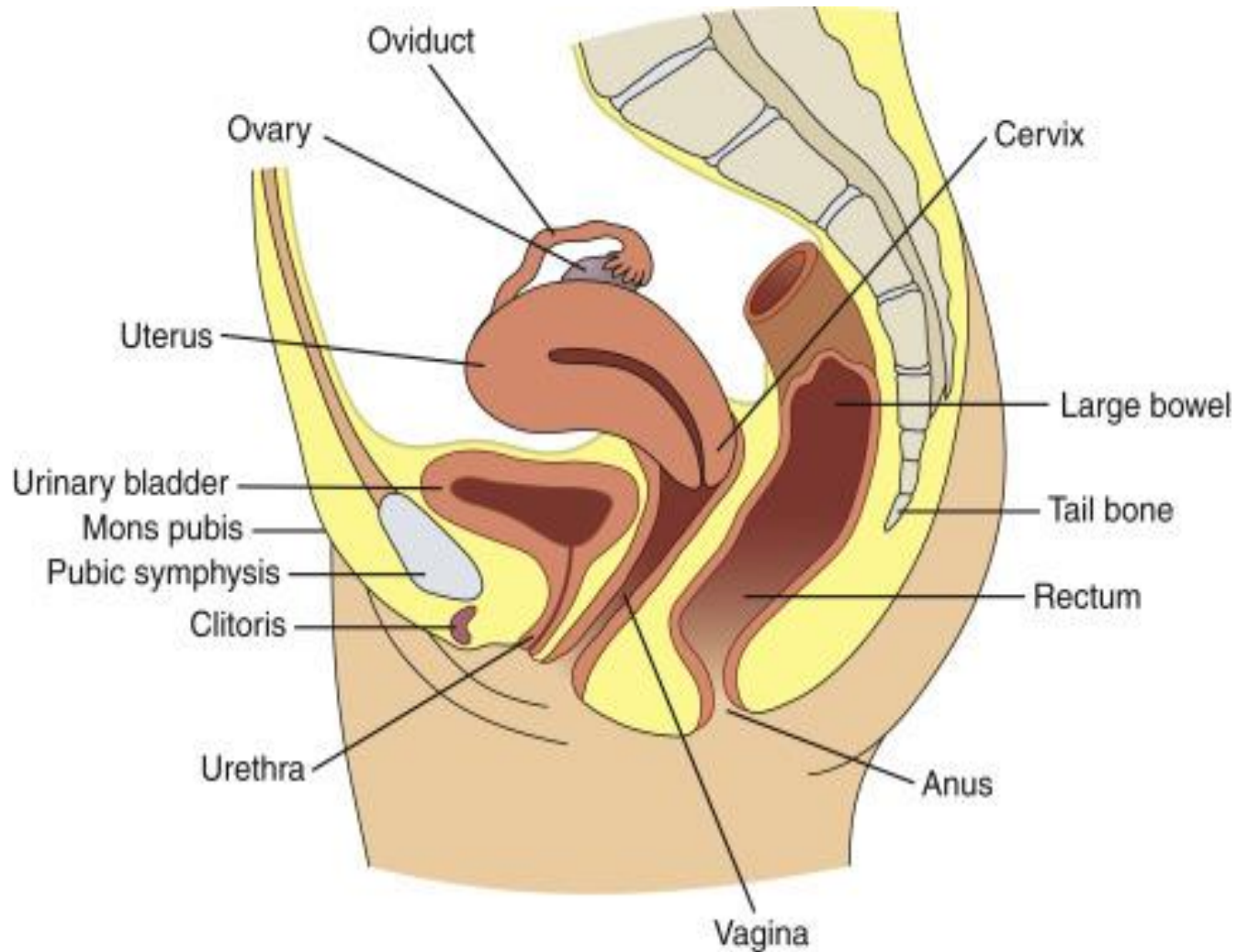
③ Estimation of intertuberosity diameter



Pelvic assessment...

- Clinical features of adequate pelvis
 - ⇒ Blunt ischial spine
 - ⇒ Unreachable sacral promontory
 - ⇒ Sub pubic angle should admit two finger
 - ⇒ Inter ischial tuberosity space should admit one fist

Female reproductive organs



Functions of the female reproductive system

Produce ova

Secrete sex hormones

Transport the ova of zygote

Receive spermatozoa

Sites for fertilization & implantation

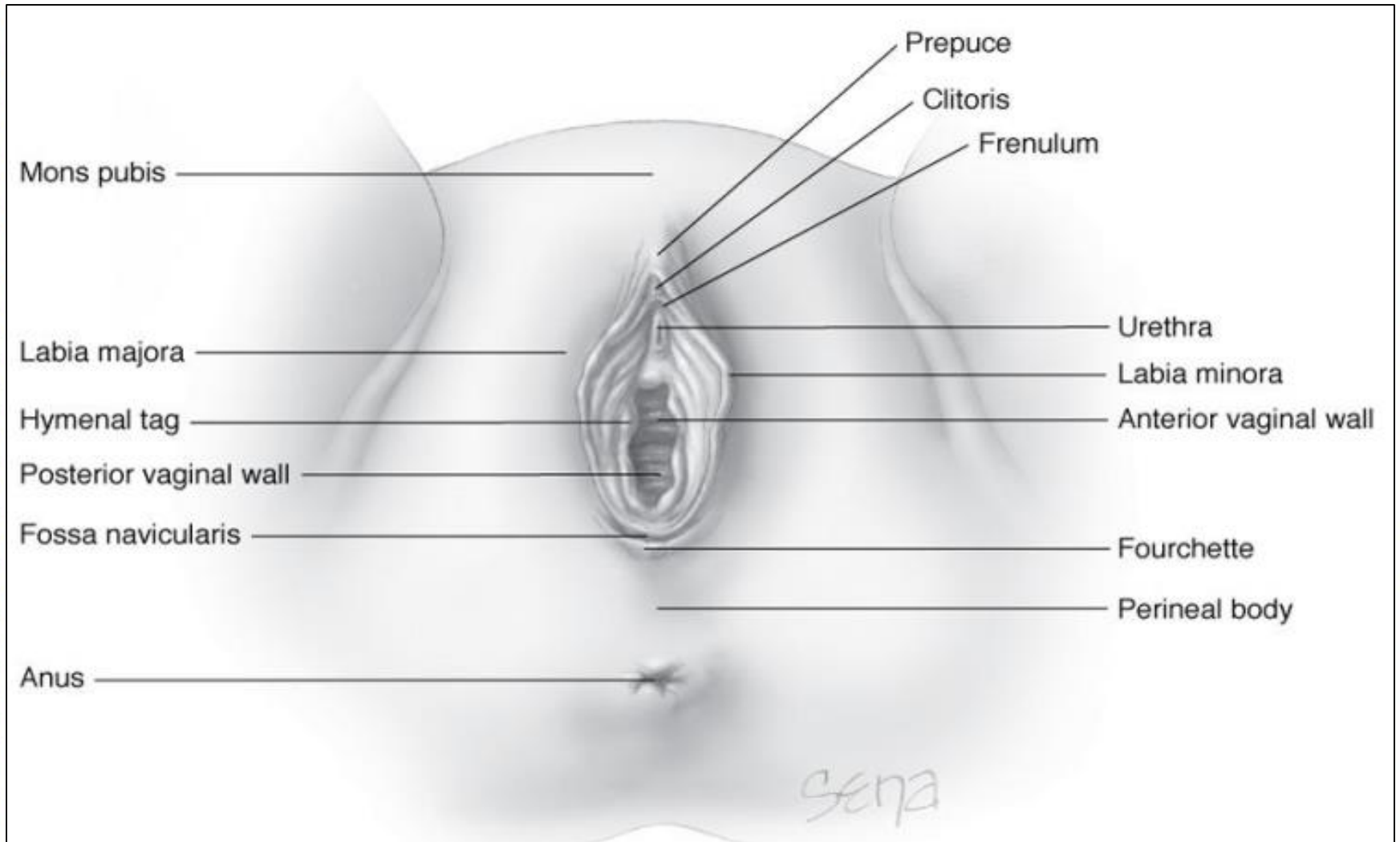
Facilitate delivery of the baby

Provide nourishment for the baby

Components of the female genital organs

1	Primary sex organs	• Ovaries which produce ⇒ Ova (gametes) or eggs ⇒ Sex hormones
2	Secondary sex organs	• Uterus (womb) • Uterine tube (fallopian tube) • Vagina • Mammary glands

Female external genitalia



The mons pubis or mons veneris

- Is a pad of sub-cutaneous adipose connective tissue lying over the symphysis pubis.
- Covered with pubic hair from the time of puberty.
- Protect the junction of the pubic bone from trauma.

The labia Majora (greater lips)

- Are two folds of fat & areolar tissues.
- Outer surfaces are covered by hair & inner are hairless.
- Are analogues of male scrotum.
- Covered by squamous epithelium
- Contain sebaceous glands, sweat glands & hair follicles.
- Adipose tissue is richly supplied by venous plexus.

Labia minora (lesser lips)

- Two thin folds of skin lying between the labia majora.
- Anteriorly they divide to close the clitoris
- Posteriorly they fuse, forming the fourchette
- Do not contain hair follicles
- Contain connective tissues, numerous sebaceous glands, erectile muscle fibers and numerous vessels and nerve.

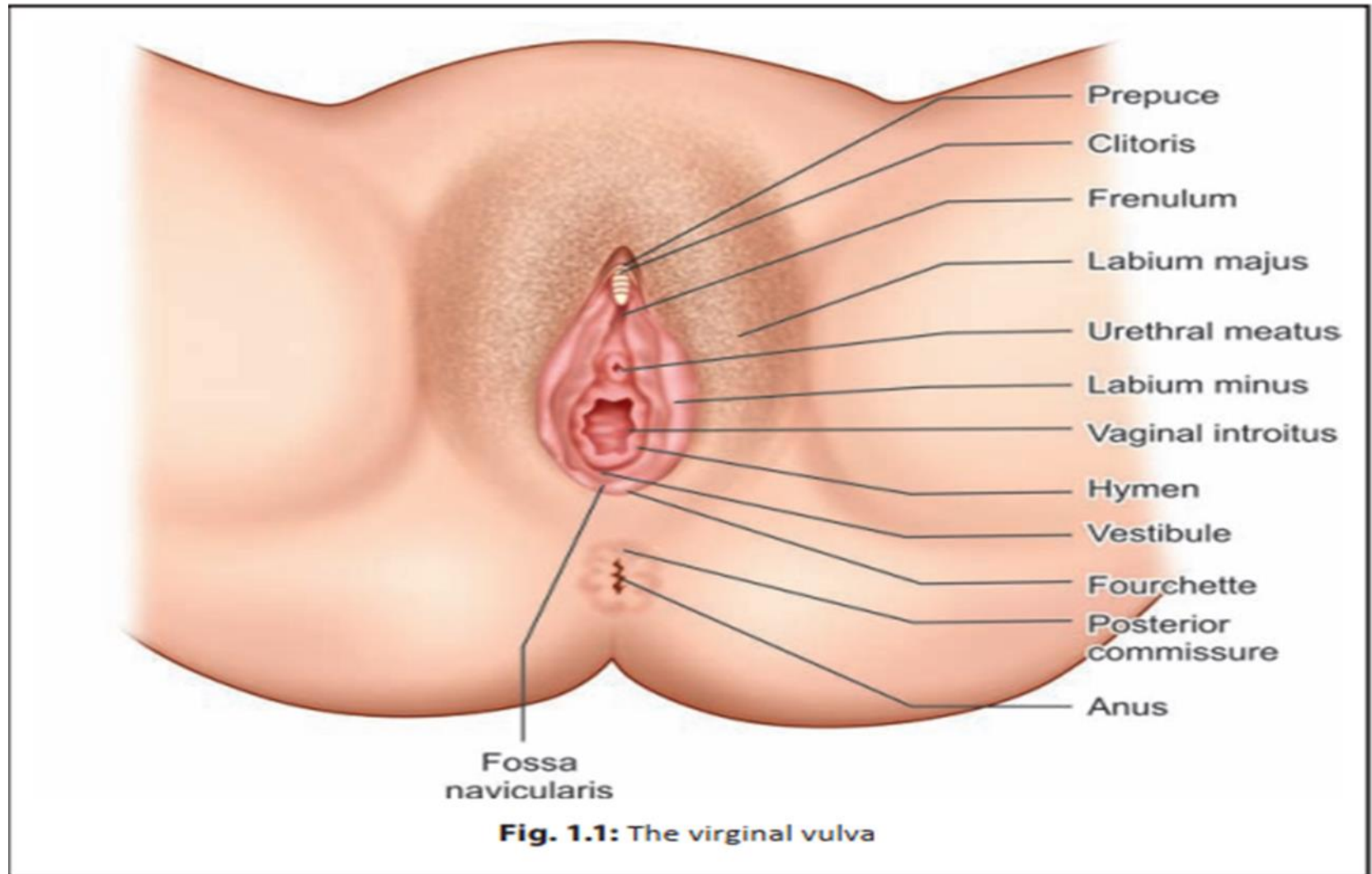
The clitoris

- Is small cylindrical, erectile body measuring about 2.5cm
- Situate in the most anterior part of the vulva.
- Is a rudimentary organ corresponding to the male penis.
- Attached to the under surface of the pubis by the suspensory ligament.
- It is entirely sensitive and highly vascular that plays a great role in orgasm of sexual intercourse.

The vestibule

- Is a triangular space bounded
 - ⇒ Anteriorly by the clitoris
 - ⇒ Posteriorly by the fourchette
 - ⇒ Laterally by the labia minora.
- There are 4 openings into the vestibule.
 - ⇒ The urethral opening
 - ⇒ The vaginal orifice (introitus)
 - ⇒ Bartholin's glands and Hymen

The vestibule...



The internal genitalia

Vagina

Uterus

Fallopian tube

Ovaries

Vagina

- A narrow passage leading from the vulva to the uterus
- Function:
 - ⇒ Excretory duct
 - ⇒ Organ of copulation
 - ⇒ Entrance for spermatozoa
 - ⇒ Exit for menstrual flow and products of conception
 - ⇒ Helps to support the uterus
 - ⇒ Helps to prevent infection

Vagina....

- No glands
- Moistened by the mucus from cervix and transudation that seeps from the blood vessels of vaginal wall.
- PH:is acidic (4.4) due to the presence of lactic acid formed by the action of lactobacilli found on squamous epithelium.
- Blood supply: from branches of internal iliac artery.

Uterus

- Also known as womb or hysteria
- Hollow thick walled muscular organ
- Pear shaped organ located between bladder & rectum
- Embedded, grows & nourished fertilized ovum
- Shape, size & position based on:age,pregnancy and state of rectum & urinary bladder.

Uterus....

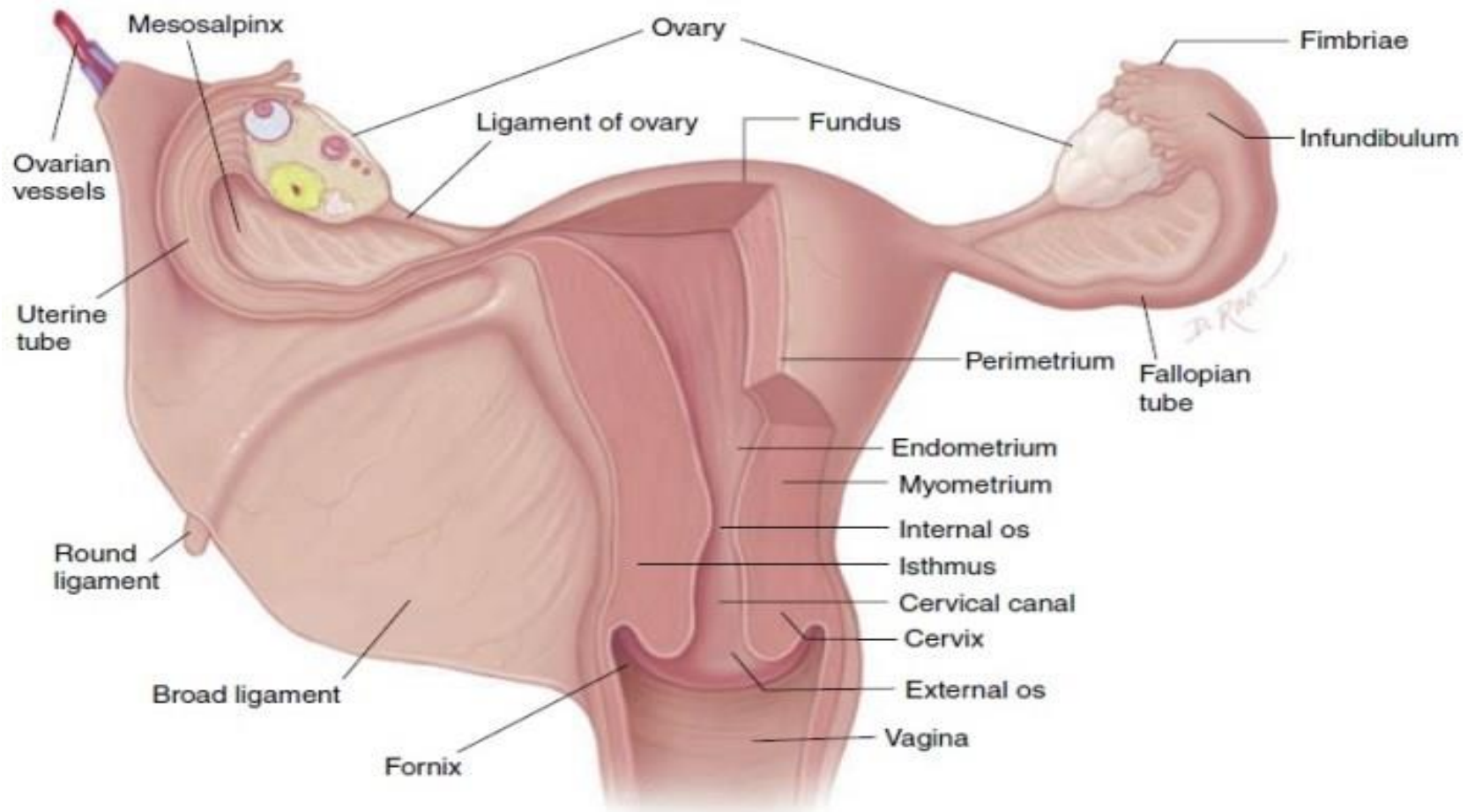


FIGURE 7.6. The Wall of the Uterus

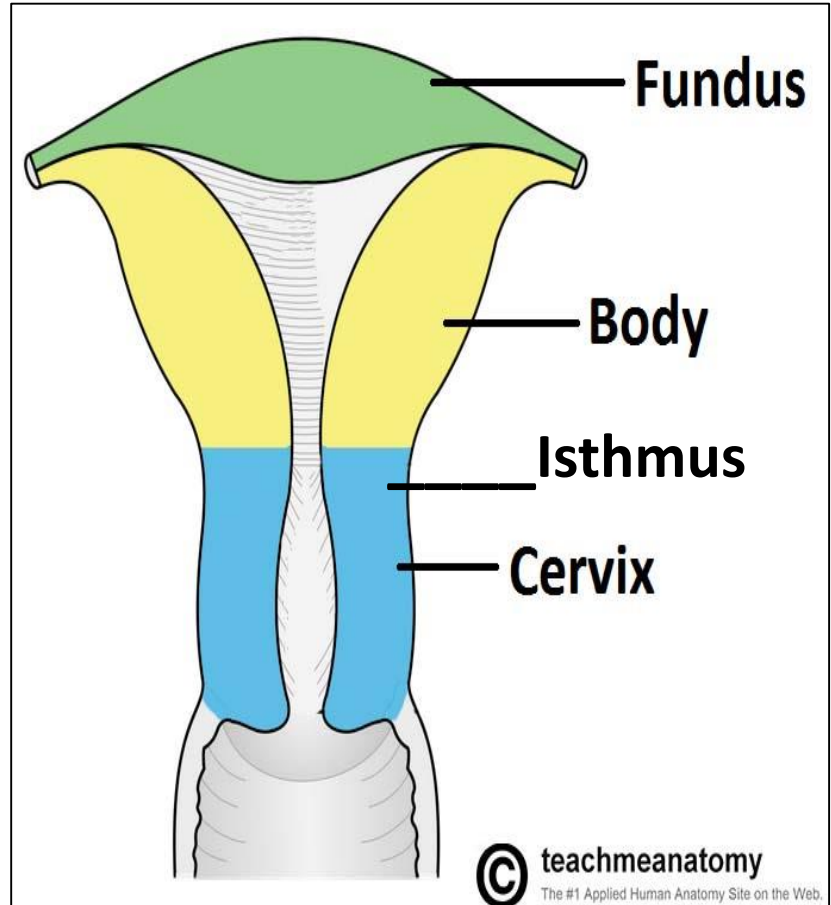
Uterus...

- It is normally anteverted and anteflexed
- It measures 8cm in length 5cm width and 2-3 cm thick.
- Its weight is approximately 57-60 gm.
- Transverse diameter 5 to 7cm at fundus 2 to 2.5cm at cervix.

Uterus...

- Parts of uterus

1. Fundus
2. Body
3. Isthmus
4. Cervix



Uterus...

- **Layers of uterus:**

- **Perimetrium**

- a part of visceral peritoneum

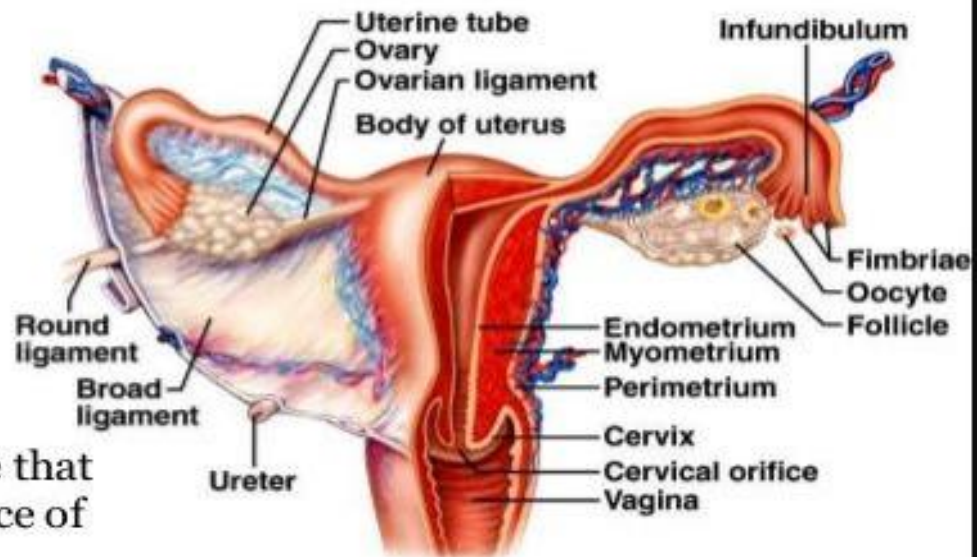
- **Myometrium**

- Bulk of uterus
 - Three layers of muscle that contract under influence of oxytocin during labor.

- **Endometrium** – Highly vascular mucosa

- **Stratum functionalis** – Shed during menstruation

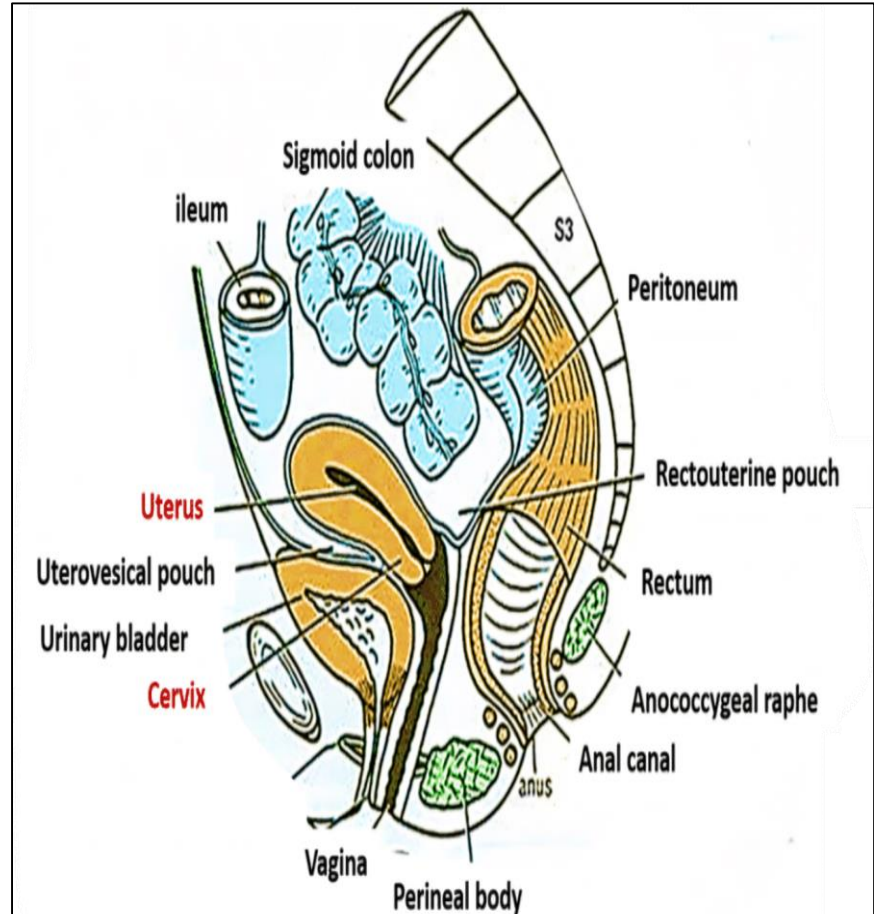
- **Stratum basalis** – Deeper, permanent layer, gives rise to new stratum functionalis after each cycle.



Uterus...

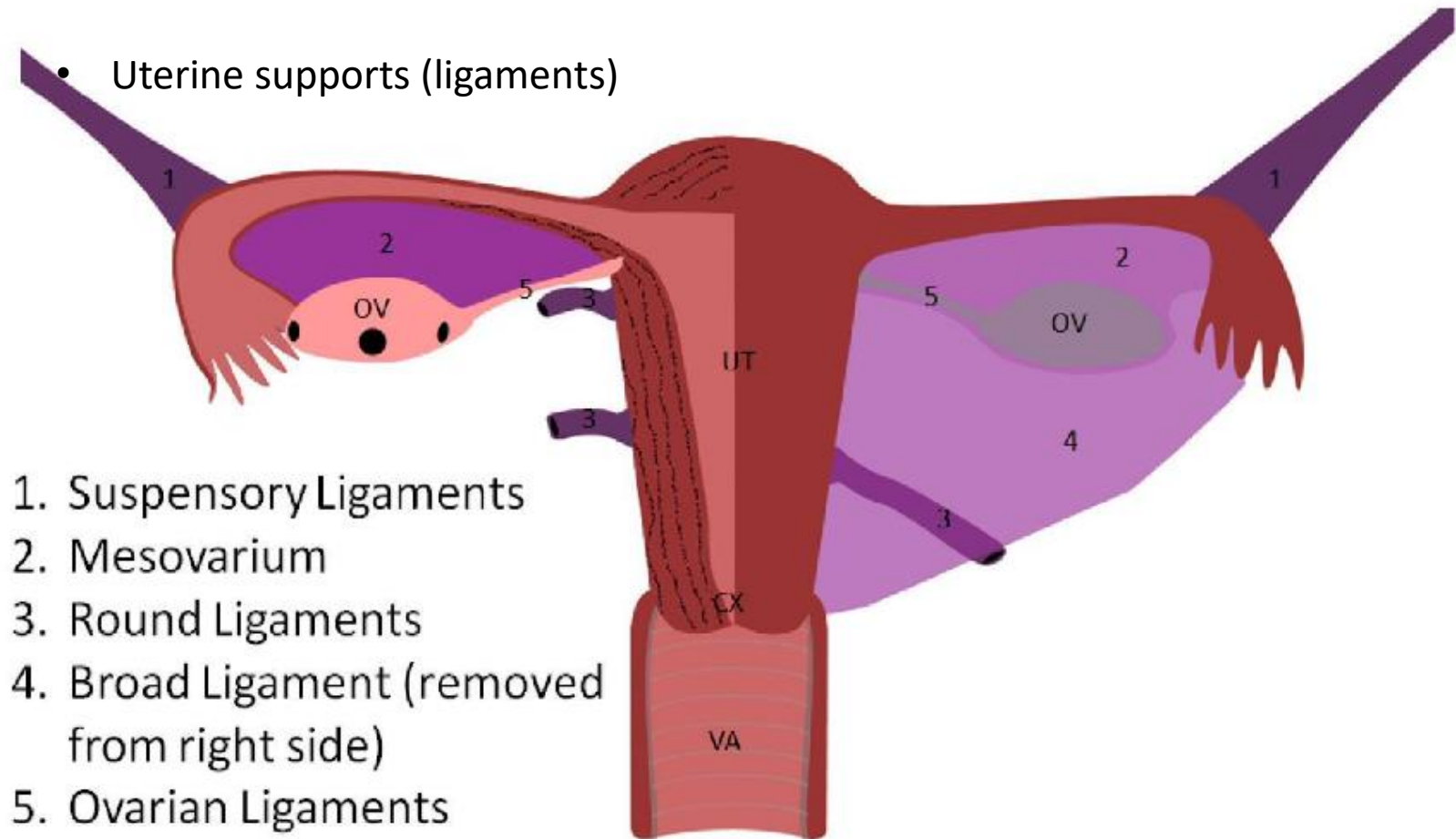
- Relations

- ⇒ Anterior –uterovesical pouch and bladder.
- ⇒ Posterior- recto uterine pouch and the rectum.
- ⇒ Lateral –broad ligaments, the fallopian tubes and ovaries.
- ⇒ Superior- the intestines
- ⇒ Inferior – the vagina.



Uterus...

- Uterine supports (ligaments)



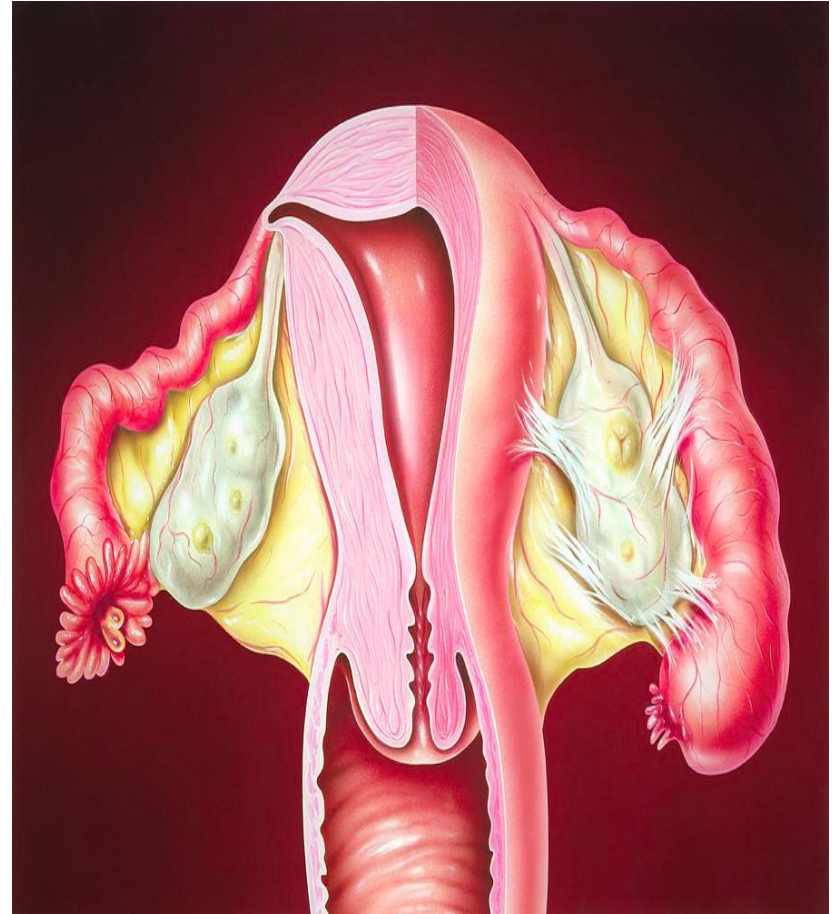
1. Suspensory Ligaments
2. Mesovarium
3. Round Ligaments
4. Broad Ligament (removed from right side)
5. Ovarian Ligaments

Uterus...

- Function of the uterus
 - ⇒ To prepare a bed for the fertilized ovum.
 - ⇒ To nourish the fertilized ovum for the gestation period
 - ⇒ To expel the product of conception at full term.
 - ⇒ To involutes following child birth

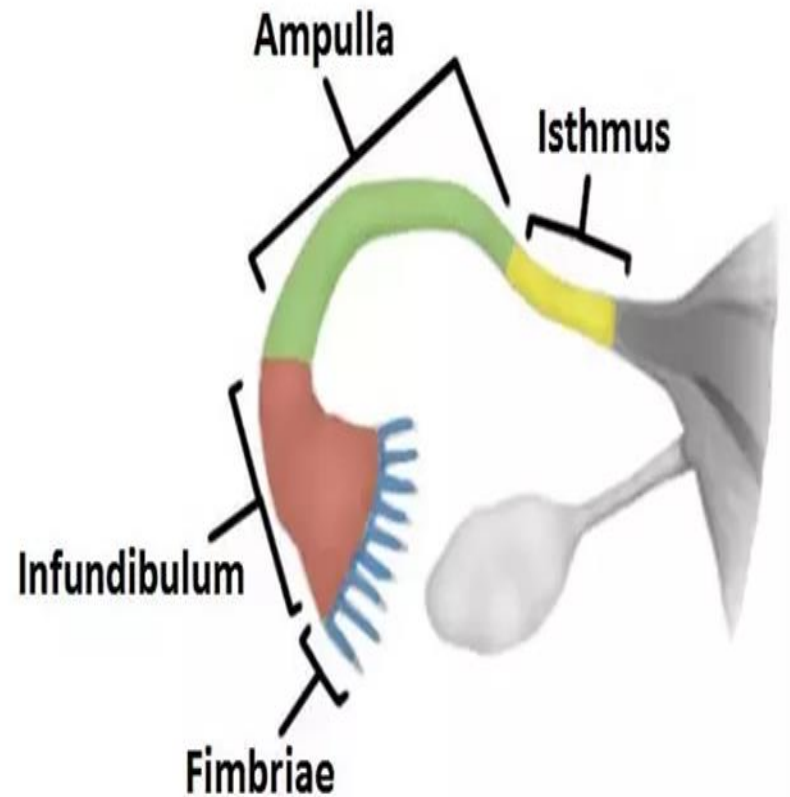
Fallopian tubes

- Are paired structures which are attached with the lateral angle of uterine cavity.
- 10 cm in length.
- Support by the infundibulo pelvic ligaments.



Fallopian tubes...

- Fallopian tubes
 - ⇒ Interstitial portion
 - ⇒ Isthmus
 - ⇒ Ampulla
 - ⇒ Infundibulum



Fallopian tubes...

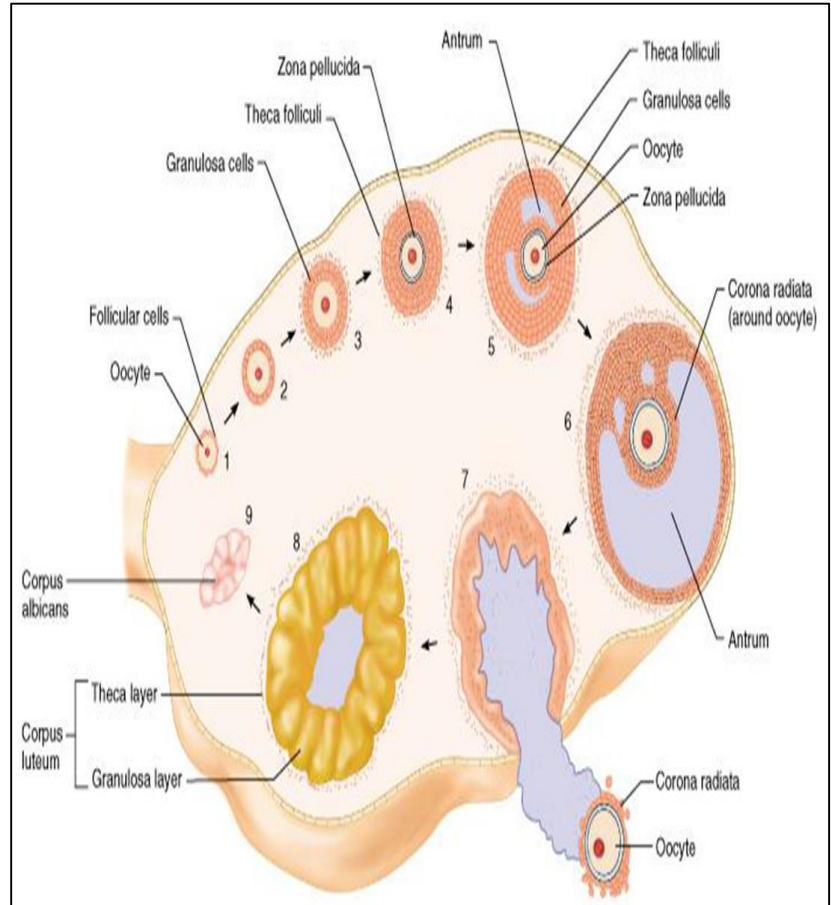
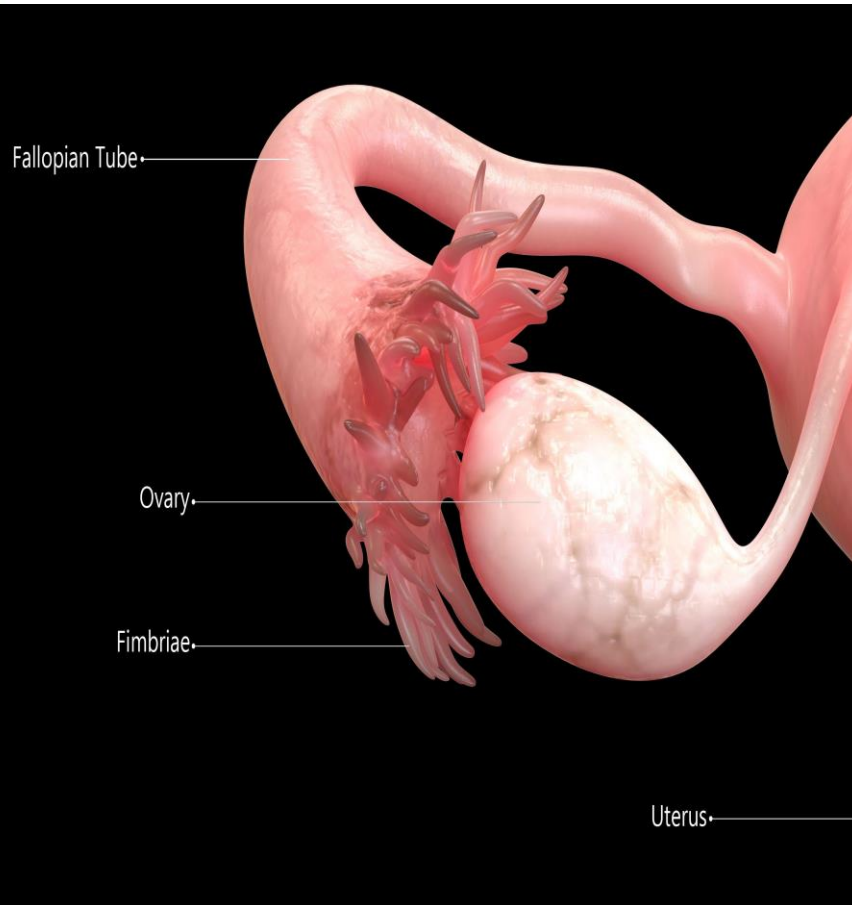
- Functions

- ⇒ Pushes the ovum towards the uterus.
- ⇒ Receives the spermatozoa as they travel up wards
- ⇒ Provides a site for fertilization
- ⇒ Supplies the fertilized ovum with nutrition during its continued journey to the uterus.

The ovaries

- Paired primary sex organs
- Are two nodular bodies, Intraperitoneal organ
- Homologous to the testes
- Oval-shaped pinkish-white glands
- Each ovary is 3cm long, 1.5cm wide & 1cm thick
- Each ovary weighs about 10-15 grams
- Has variable size but smaller at menopause

The ovaries...



The ovaries...

- Function

- ⇒ Produce egg

- ⇒ Produce estrogen & progesterone for:-

- A. Regulation of menstrual cycle

- B. Secondary sex organ to appear

- C. Maturation of primary sex organs

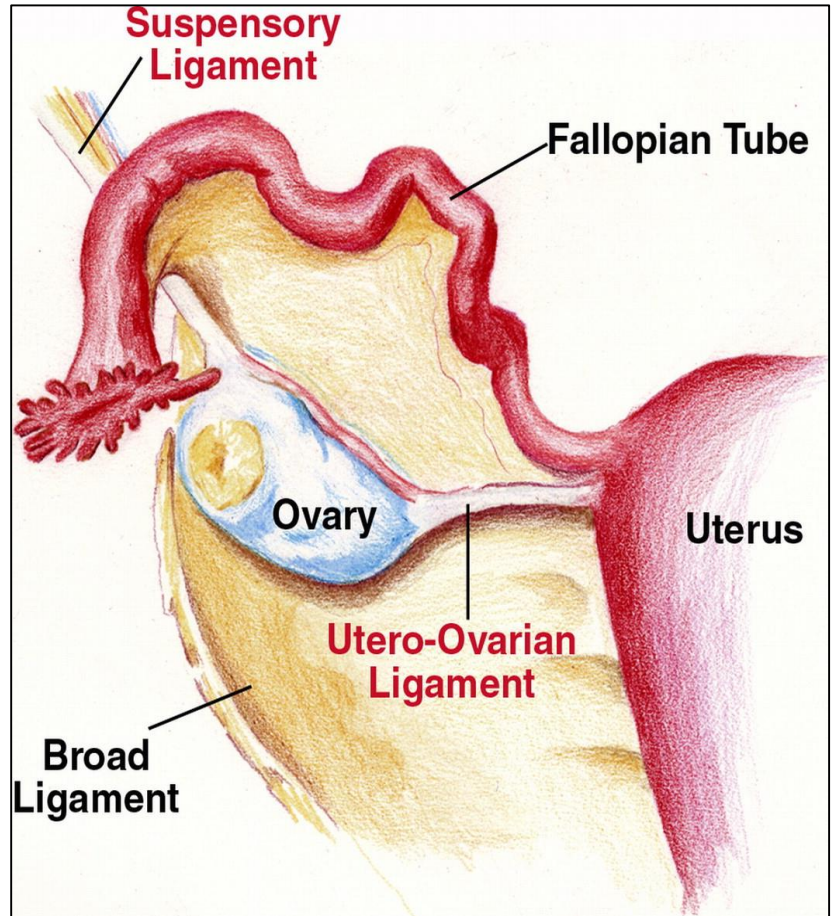
The ovaries...

- Attachments of the ovary

⇒ Suspensory Ligament

⇒ Ovarian Ligament

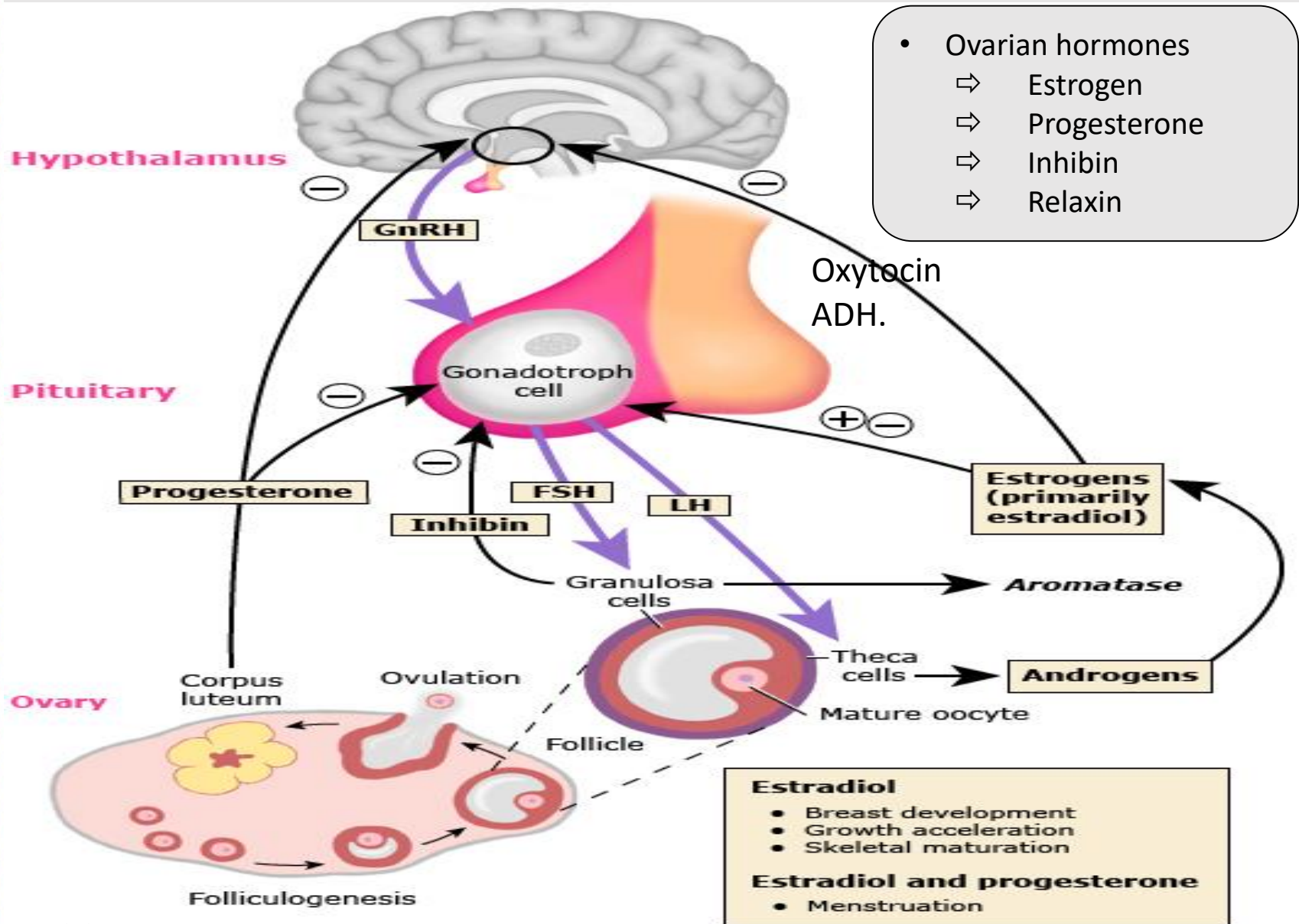
⇒ Mesovarium



Physiology of the female reproductive organs

- Session objective
 - ⇒ Define H-P-O axis's hormones
 - ⇒ List and describe action of hormone
 - ⇒ Define puberty
 - ⇒ Describe normal menstrual cycles
 - ⇒ Phase of menstrual cycles

Hypothalamo- pitutary-ovarian axis



Estrogens

- Produced primarily by ovaries, during pregnancy by placenta.
- Follicle-stimulating hormone (FSH) stimulates production from granulosa cells of the ovarian follicles and corpora lutea.
- Liver, adrenal glands, and breasts also produce small amount.

Estrogens...

- Function of estrogens
 - ⇒ Pubertal changes
 - ⇒ Vaginal epithelium gets thickened, stratified & cornified.
 - ⇒ Increase rhythmic contractions of fallopian tubes & uterus
 - ⇒ Induce a watery alkaline secretion from cervix.
 - ⇒ Sensitize uterus to oxytocin
 - ⇒ Maintain bone mass by retarding bone resorption
 - ⇒ Promotes positive calcium balance
 - ⇒ ↓ LDL cholesterol while ↑ HDL & triglyceride levels.

Progesterone

- Secreted from the corpus luteum.
- Function
 - ⇒ Maintains the uterine endometrium for implant
 - ⇒ Differentiation factor for mammalian glands.
 - ⇒ Growth of ducts and lobules of the breast
 - ⇒ Relax smooth muscles of uterus, blood vessels, GI & urinary tract.
 - ⇒ Secretory changes and decasualization of endometrium.
 - ⇒ Vasodilatation
 - ⇒ Thermogenic (increase BBT).

Puberty

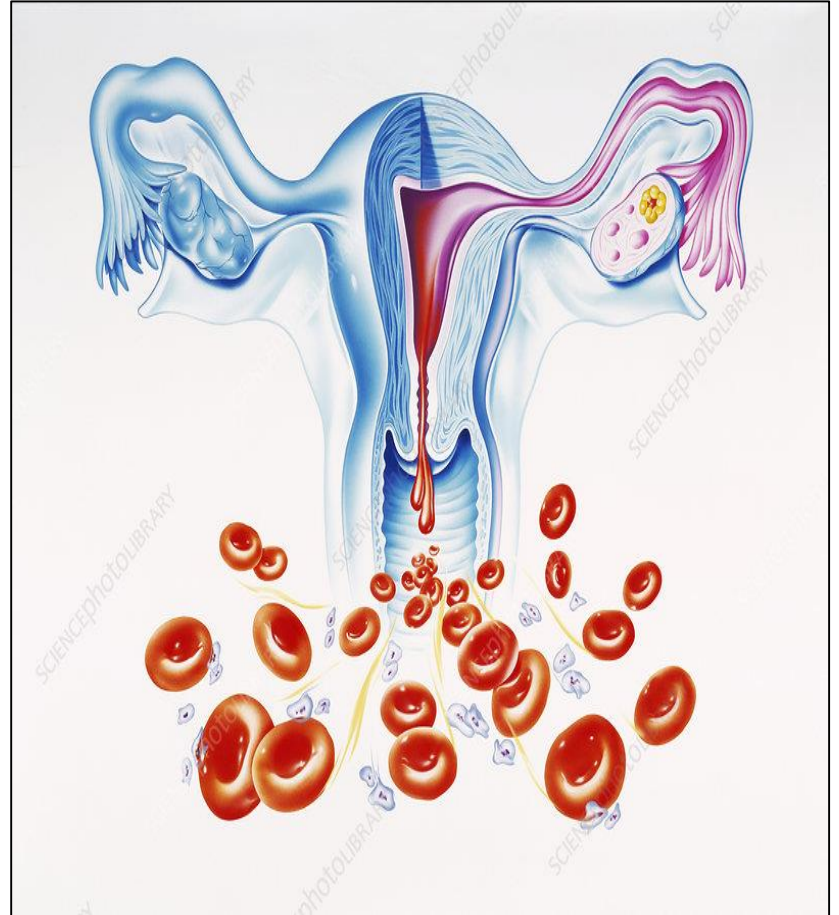
- Culminates in the onset of menstruation, called menarche
- Reproductive organ became mature
- Is stage of life at which secondary sexual characteristics appear
- It is marked by neuro-endocrine changes.
- Pulsative releases of GnRh seen at this time.
- Ovarian steroid hormones start to be secreted.
- The H-P-O axis becomes coordinated.

Puberty.....



The menstrual cycle

- Is a periodic uterine bleeding in response to cyclic hormonal changes.
- It is hallmark(features) of reproductive year.

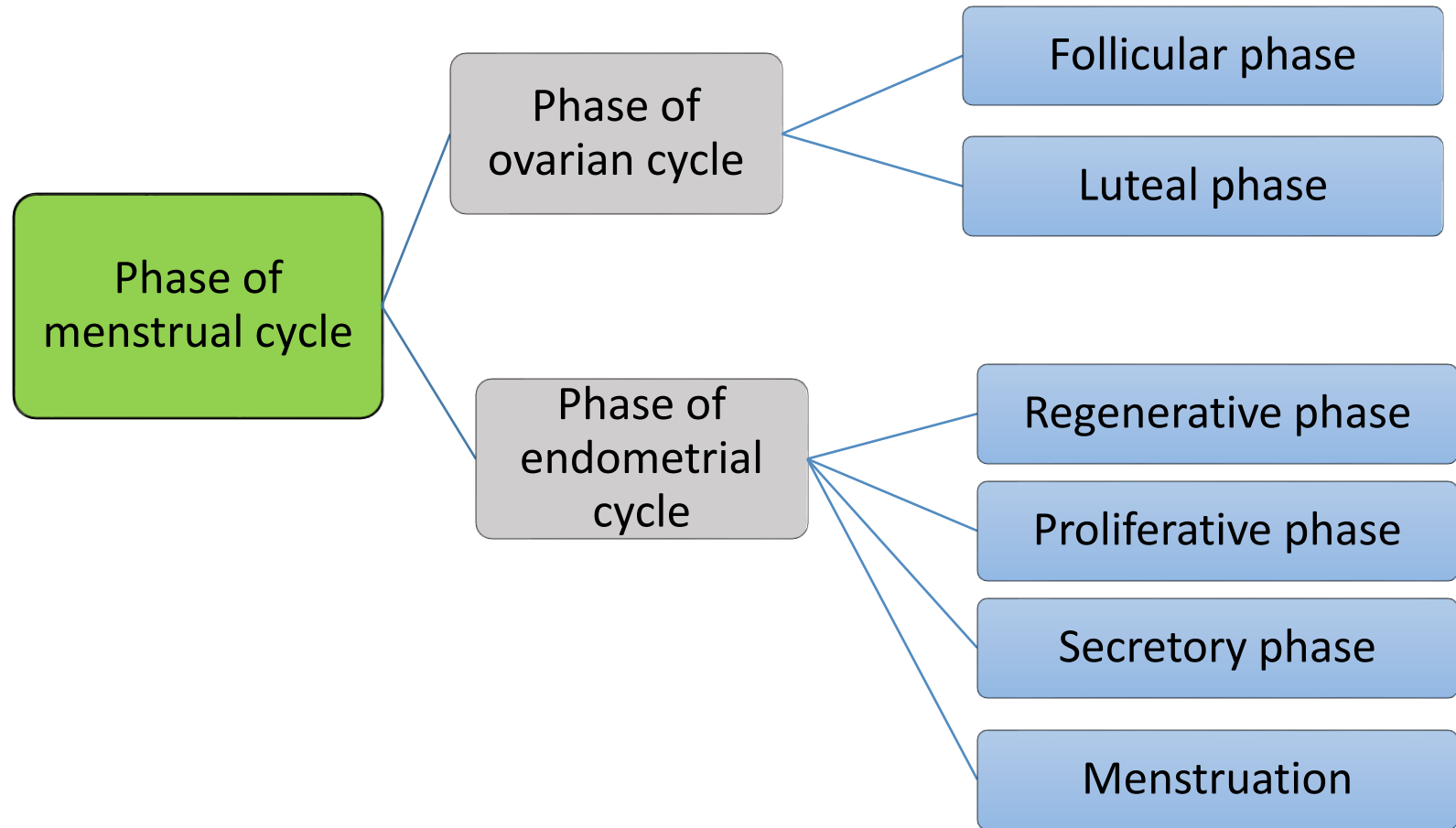


The menstrual cycle...

- Characteristics of normal menses
 - ⇒ Interval 21-35 days
 - ⇒ Amount 30-80 ml
 - ⇒ Duration 1-8 days, average 5 days
 - ⇒ Most blood is arterial (85%) oxygenated
 - ⇒ Color dark red and non clotted in nature

Physiology of menstruation

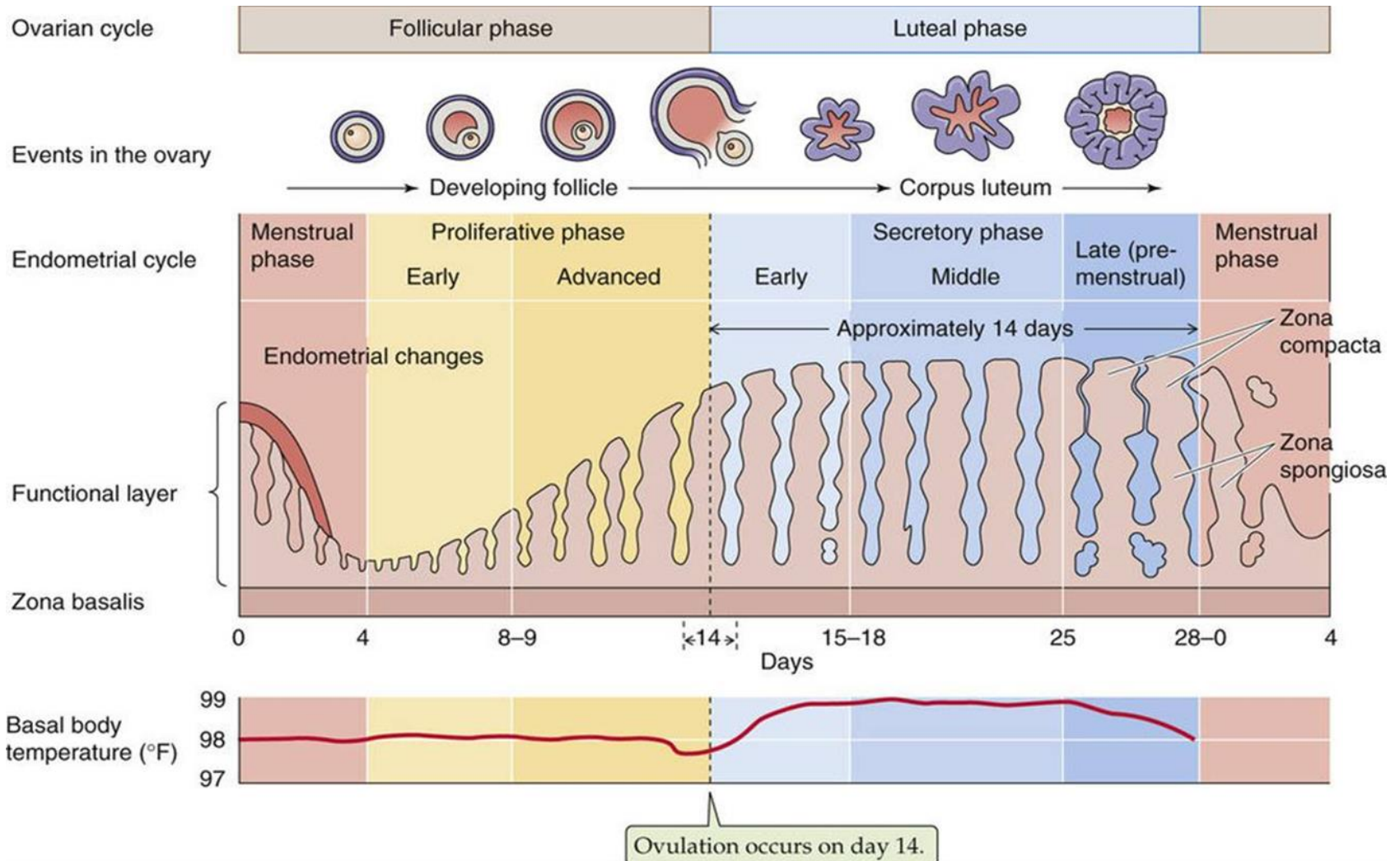
- Ovarian follicular development
 - ⇒ Oocytes number are 6-7 million by 20 weeks of gestation.
 - ⇒ 1-2 million oocytes remain in the ovary at birth.
 - ⇒ At puberty only 300,000 oocytes remains.
 - ⇒ Only 400-500 will ultimately ovulate and they will end up at the time of menopause.



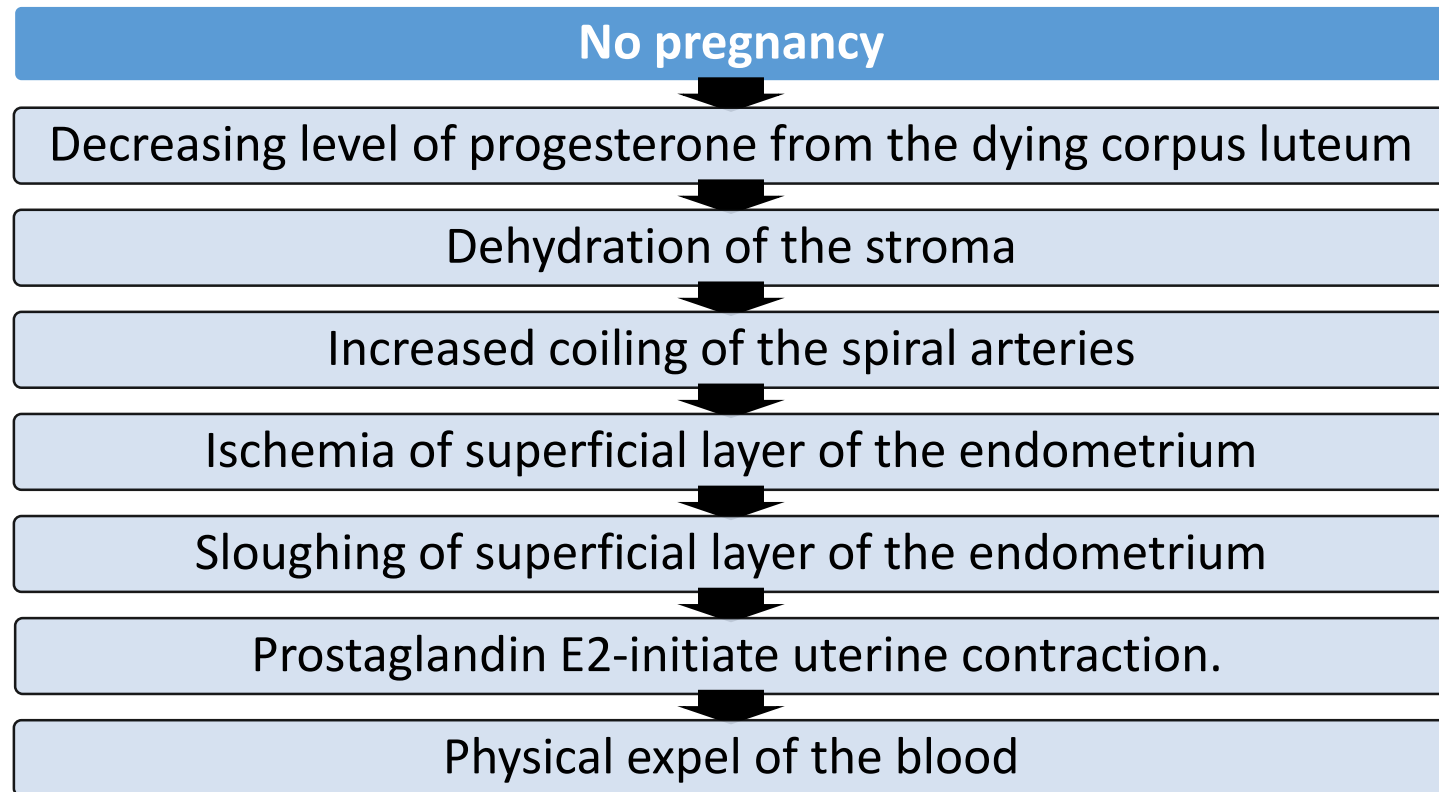
A. Phase of ovarian cycle

Follicular phase	• Begins with the onset of menses • Ends a day before the luteinizing hormone surge. • Development of single dominant follicle. • Due to hormonal feedback • Its average length is 10-14 days.
Luteal phase:	• Begins on the day of the LH surge • Ends at the onset of the next menses. • The time from ovulation to the onset of the menses. • Average length 14 days.

Phase of menstrual cycle



Mechanism of menstruation



Embryology

- Objective

- ⇒ Define important terms

- ⇒ Describe fertilization and fetal development

- ⇒ Describe the structures of conception

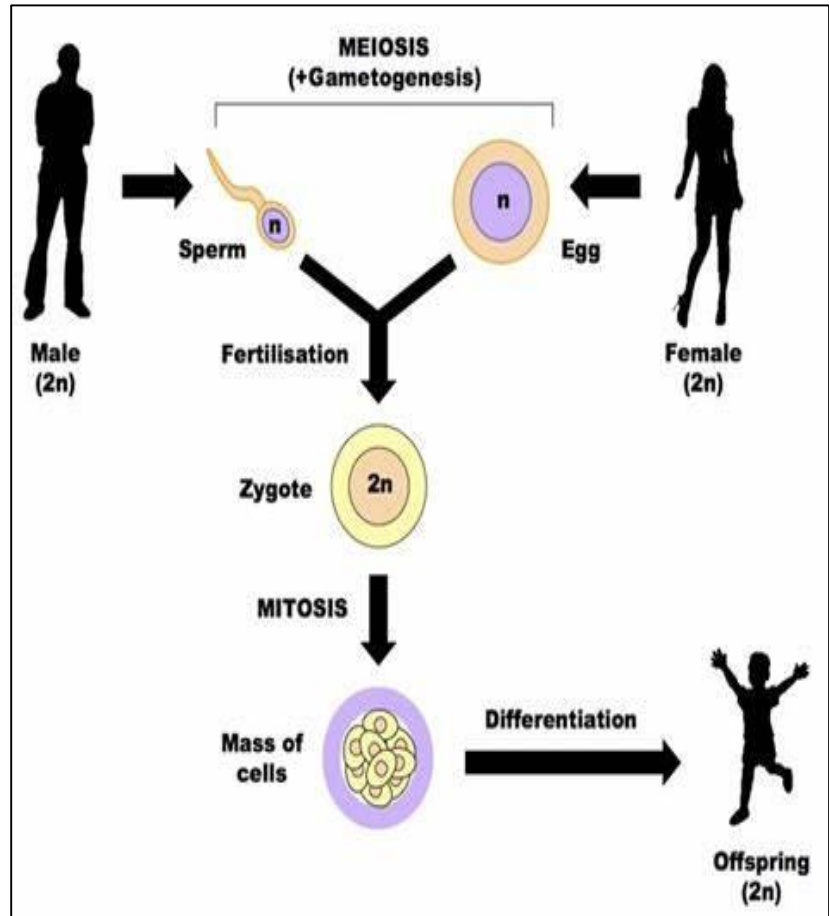
- ⇒ Describe fetal skull

Embryologic terminology

Term	Definition
Oocyte	• (ovum or egg) is female germ or sex cells
Sperm	• (seed) or spermatozoon is male germ cell
Embryology	• Study of the origin and development of individual organism.
Zygote	• Union of an oocyte and a sperm during fertilization
Morula	• Solid mass of blastomeres (12 -32) in 3-4 days after fertilization
Blastocyst	• Morula enters the uterus and surrounded by a fluid-filled cavity
Implantation	• A process of blastocyst attaches to the uterus, and embeds in it
Embryo	• Developing human during first 56 days
Fetus	• Developing human from 9weeks to expulsion or delivery
Conceptus	• The embryo and its associated membranes (placenta, amnion, chorionic sac, and umbilical vesicle or yolk sac)

Fertilization

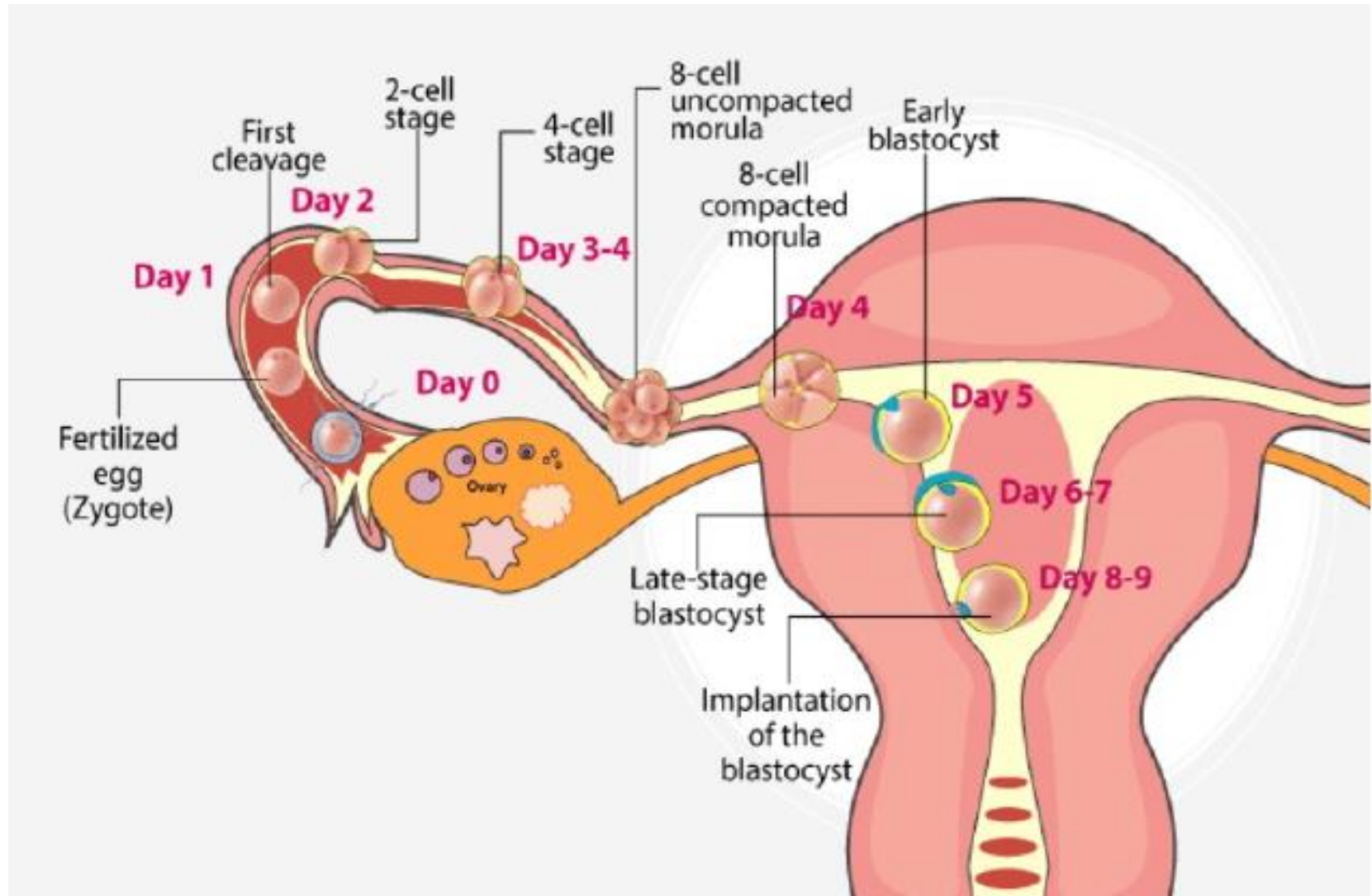
- Is the union of the ovum and spermatozoa.
- After penetration of the ovum by sperm the chromosomal material of the ovum and the spermatozoa fuse.
⇒ The resulting structure is called a zygote.



Fertilization....

- The occurrence of fertilization depends on:
 - ⇒ Maturation of both sperm and ovum.
 - ⇒ Ability of the sperm to reach the ovum
 - ⇒ Ability of the sperm to penetrate the zonapellucida and cell membrane and achieve fertilization.
- Fertilization usually takes place at the ampulla.

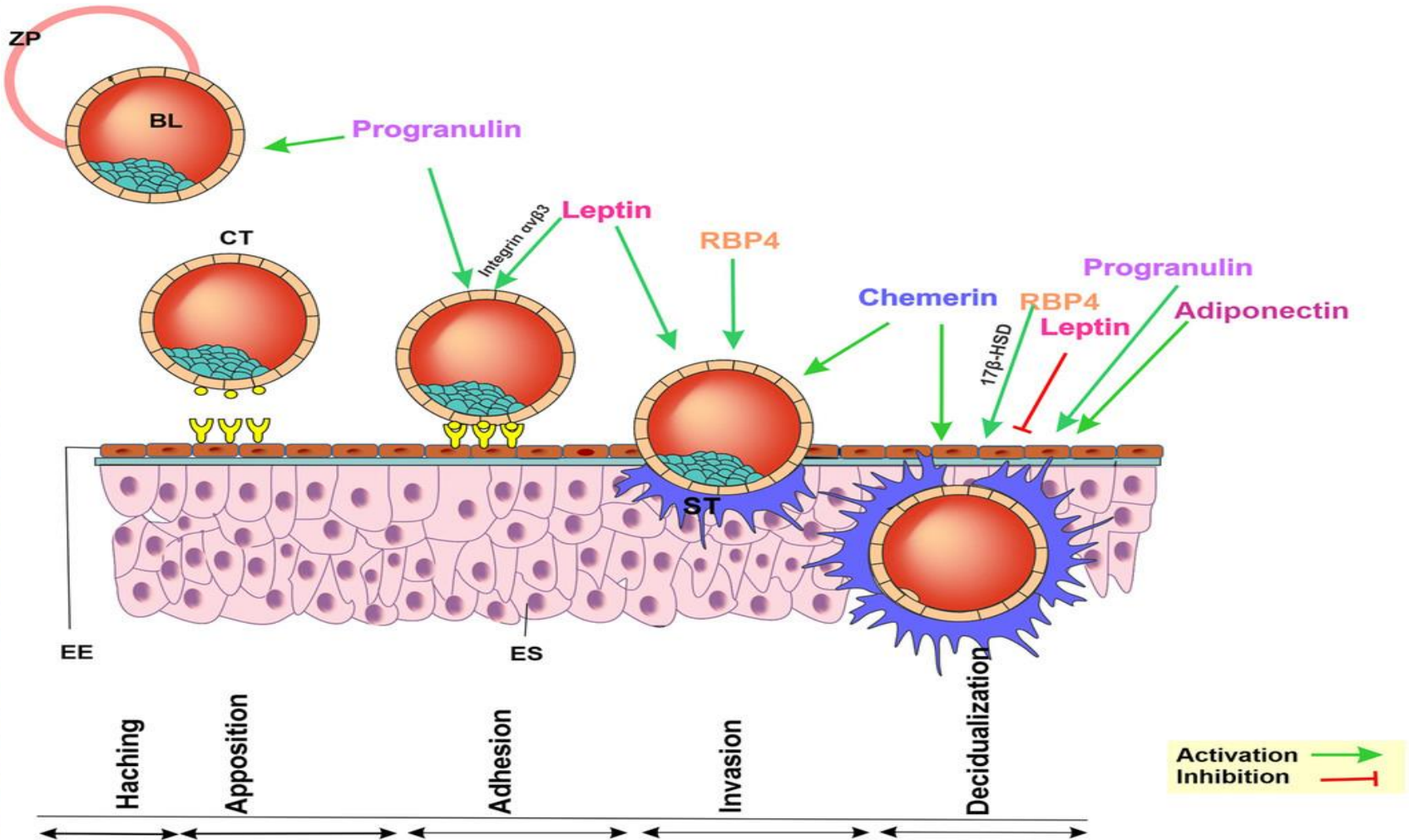
Fertilization....



Implantation

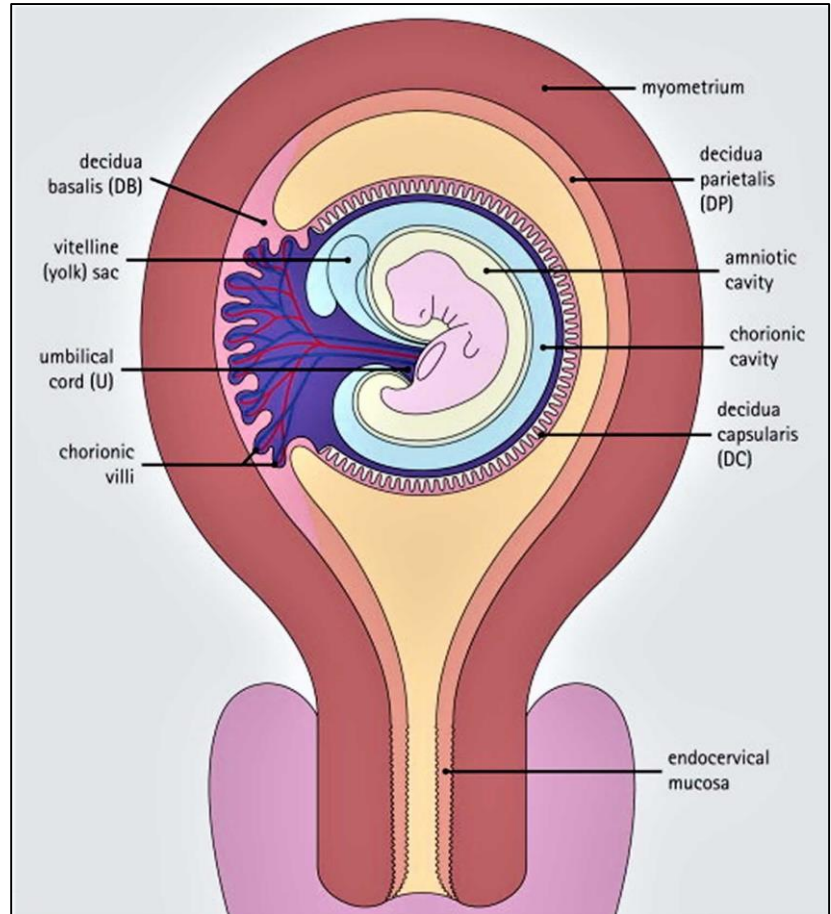
- Is a contact between the blastocyst and the uterine endothelium.
- It occurs approximately 8 to 10 Days after fertilization.
- Once fertilization is complete the zygote migrates towards the body of the uterus aided by muscular contraction of the fallopian tubes.

Implantation



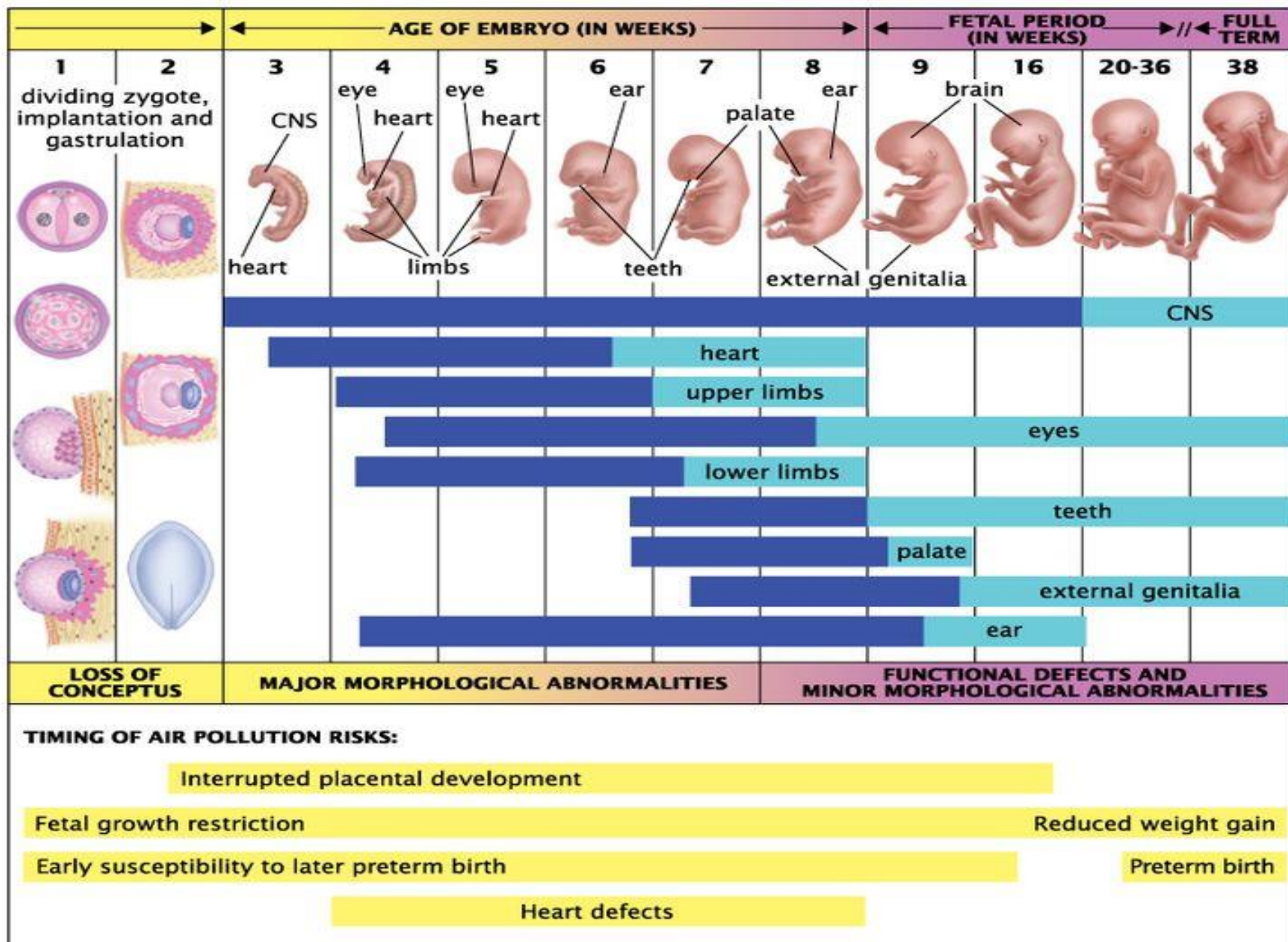
Post implantation changes in the uterus

- Endometrium is called **decidua**
- Increased secretion of estrogen causes the endometrium to grow 4x its original thickness
- Progesterone from the CL stimulates the endometrial glands and increases the size of the blood vessels.



Trimester

Trimester	1 st trimester				2 nd trimester				3 rd trimester			
Month	0	1	2	3	3	4	5	6	6	7	8	9
Week	0-4	5-8	9-12	13	14-17	18-21	22-25	26-27	28-30	31-34	35-38	39-42



Note: Blue bars indicate time periods when major morphological abnormalities can occur, while light blue bars correspond to periods at risk for minor abnormalities and functional defects.

Early development of the fetus



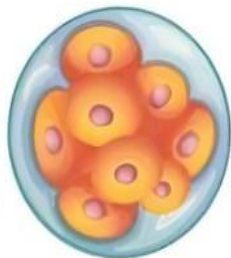
Fertilized egg



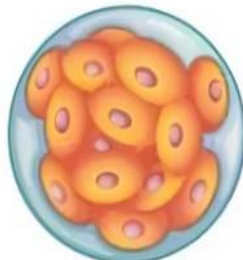
2-cell stage



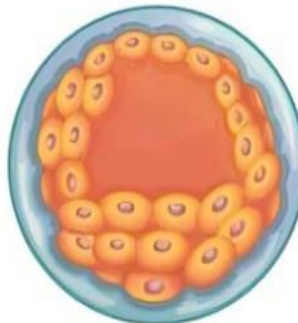
4-cell stage



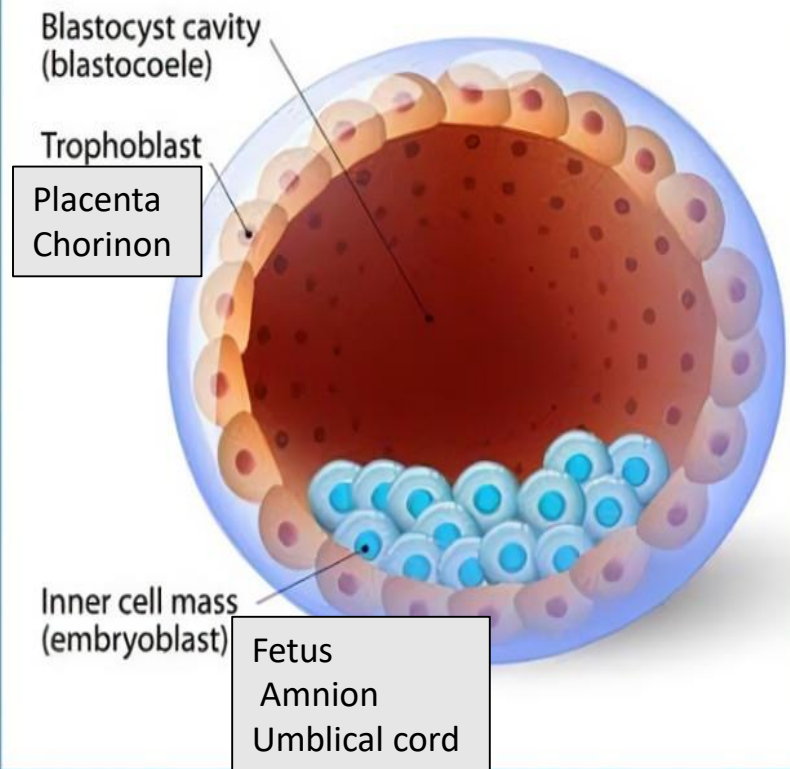
8-cell stage



16-cell stage



Blastocyst



Trophoblastic cells

- Trophoblastic cells differentiate into 3 layers

Syncytiotrophoblast

- Also called syncytium
- Erodes the walls of blood vessels of decidua

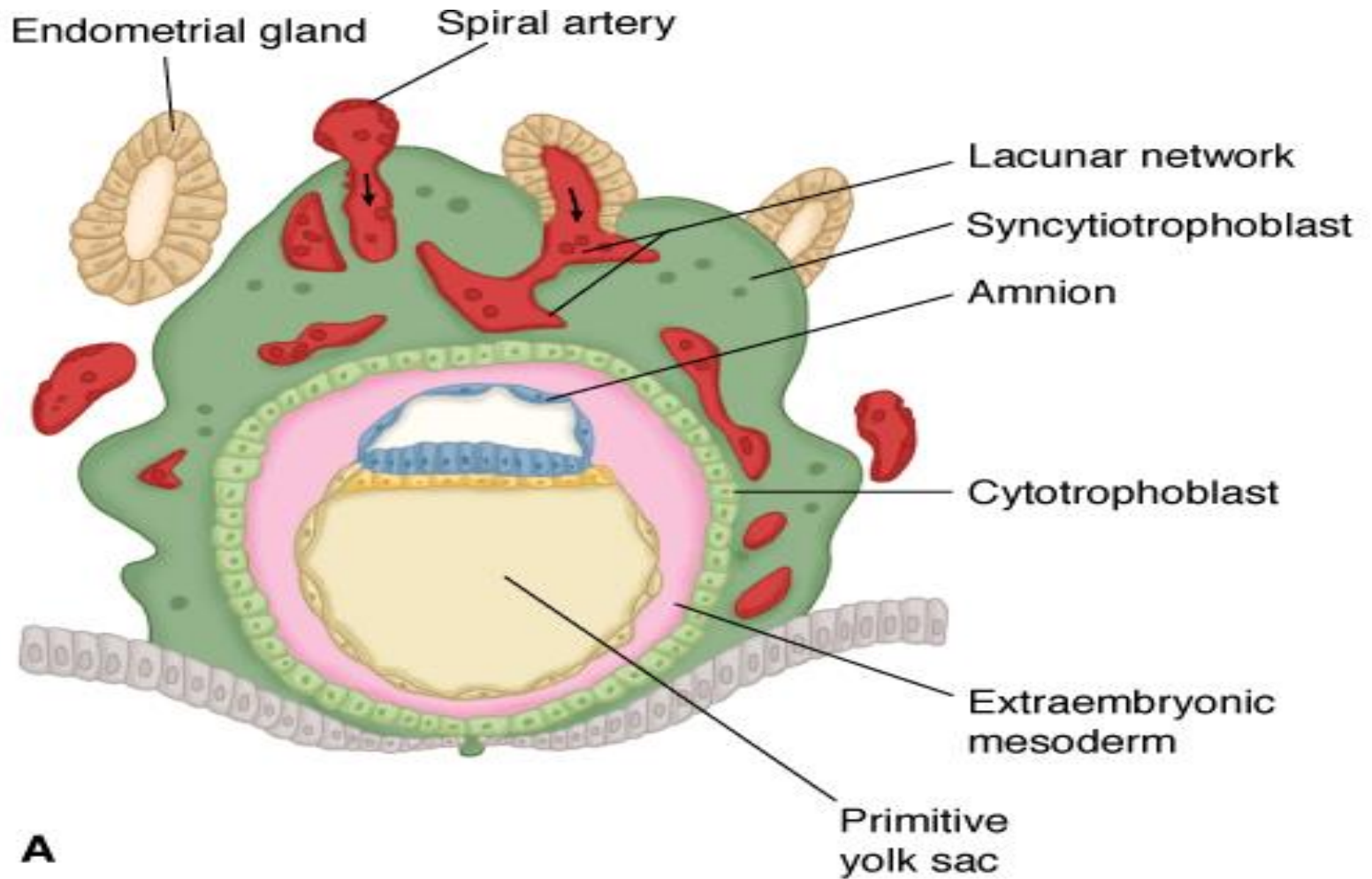
Cytotrophoblast

- Produces HCG, HPL, progesterone and estrogen.
- Progesterone maintains decidua (no menstruation)
- High level of estrogen suppresses FSH.

Primitive mesenchyme or mesoderm

- Develops to chorionic vesicle with its membrane (chorion).
- Forms the umbilical cord later on.

Trophoblastic cells

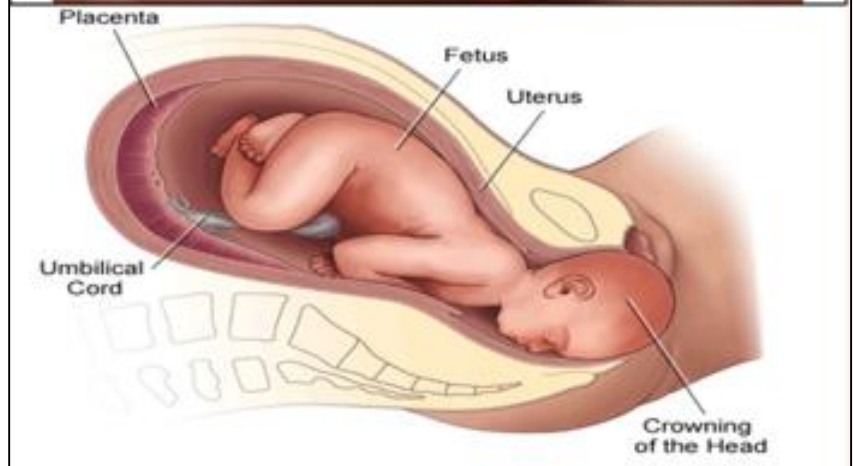


Inner cell mass

Ectoderm	Mesoderm	Endoderm
• Outer part of inner mass • Forms ⇒ Nervous system ⇒ Skin ⇒ Hair ⇒ Nails ⇒ Sense organs ⇒ Mucous membranes of the mouth and anus	• Intermediate part • Forms supporting structures of the body ⇒ connective tissue ⇒ Cartilage ⇒ Muscle ⇒ Bone and tendons • Derived organ ⇒ Urinary system ⇒ Reproductive system ⇒ Heart ⇒ Circulatory system ⇒ Blood cells.	• Inner part of inner mass • Forms ⇒ Mucous membranes ⇒ Gland (lining of the GIT, respiratory tract, tonsils, parathyroid, thyroid, thymus gland).

Placenta

- Is an intermediate organ of fetal organs to maternal tissues for physiologic exchange.
- It has two surfaces
 - ⇒ Maternal
 - ⇒ Fetal surface



Placenta...

- Functions of placental

1. Respiration
2. Nutrition
3. Storage
4. Excretion
5. Protection
6. Endocrine function
(HCG,estrogen,progesteron,HPL..)



The fetal sac

- Chorion

- ⇒ Is thick opaque

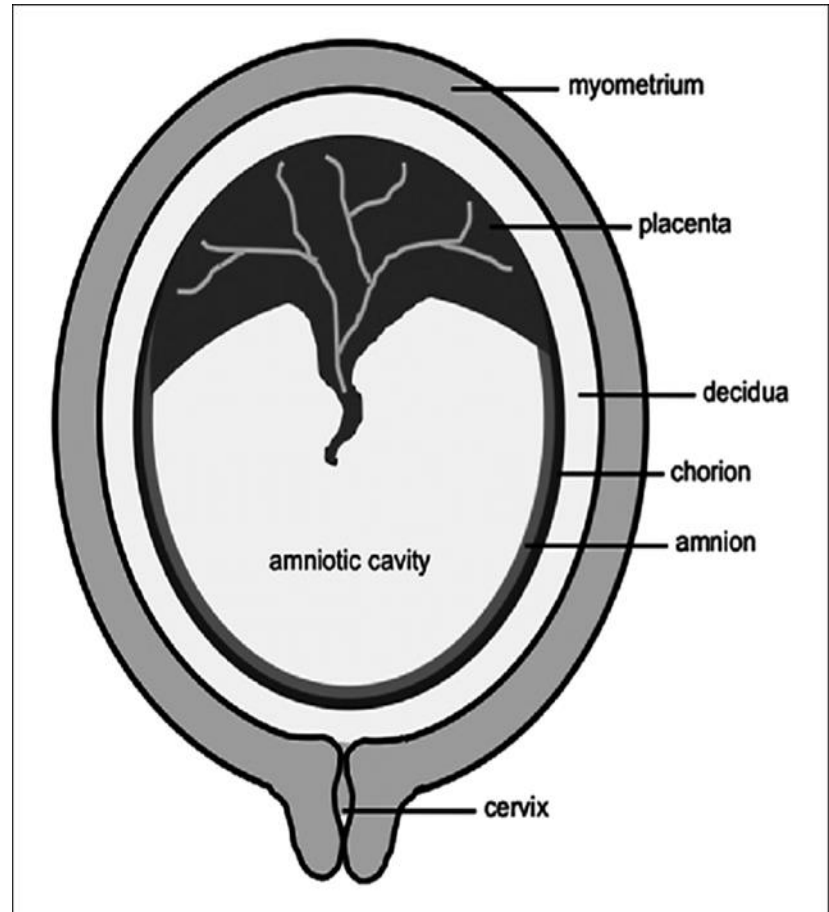
- ⇒ Friable membrane

- ⇒ Forms the base of placenta

- Amnion

- ⇒ Is smooth, tough, translucent

- ⇒ Is thought to function in production of amniotic fluid

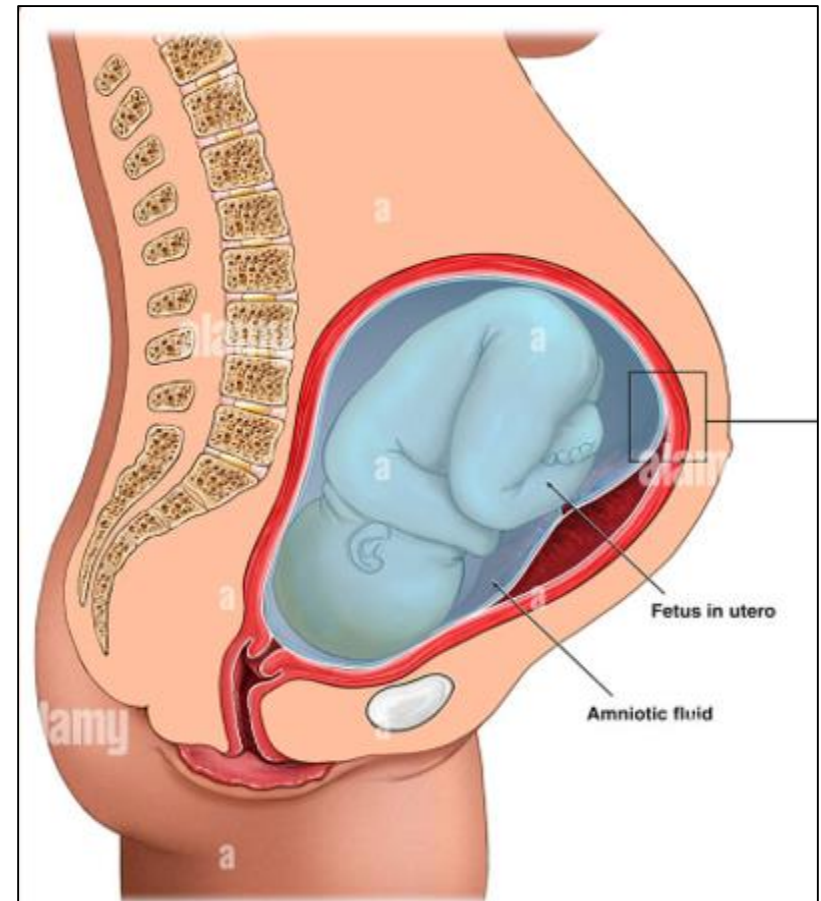


Amniotic fluid

- Is a clear fluid in which through the fetus floats
- The normal amount is from 500cc-1500cc.
- It is alkaline in reaction and has 99% of water.
- Origin
 - ⇒ Is from amnion.
 - ⇒ Maternal vessels in the decidua.
 - ⇒ Fetal vessels in the placenta.
 - ⇒ Fetal urine.

Amniotic fluid...

- Functions of amniotic fluid
 - ⇒ Allows free movement.
 - ⇒ Limb development
 - ⇒ Equalizes intra uterine pressure
 - ⇒ Acts as a shock absorber.
 - ⇒ Stablises intrauterine temperature.



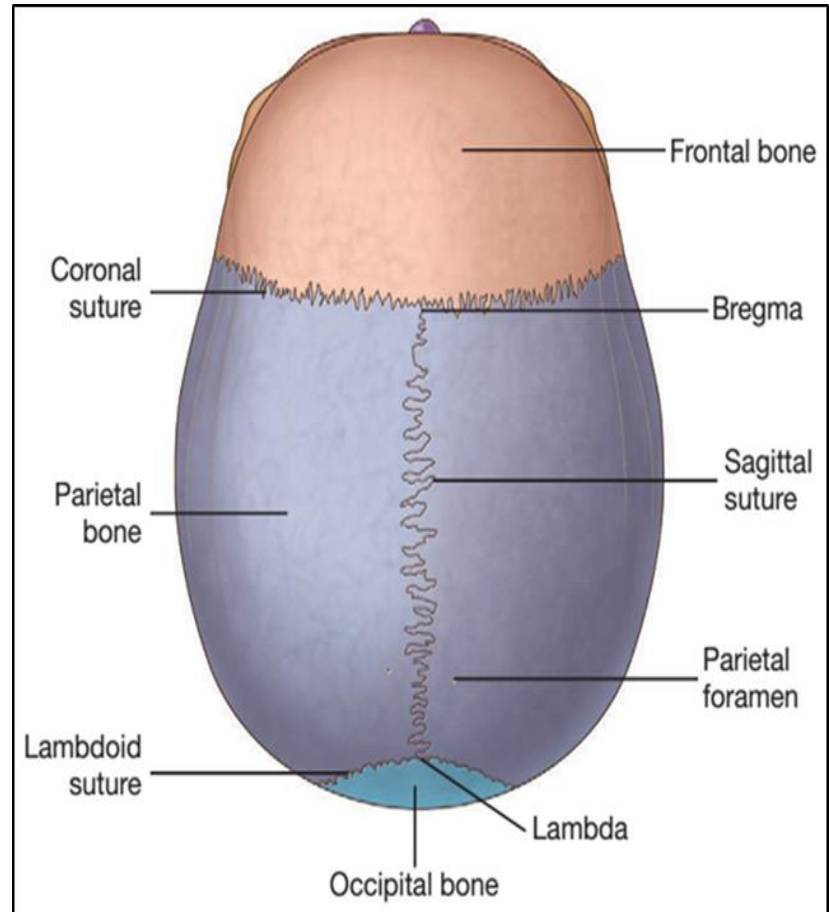
The umbilical cord (funis)

- Connects the fetus to placenta
- Contain 2 arteries and 1 vein
- Average length=50cm
- Functions
 - ⇒ Transport O₂ and nutrients
 - ⇒ Excrete waste
 - ⇒ Allows fetal mobility



The fetal skull

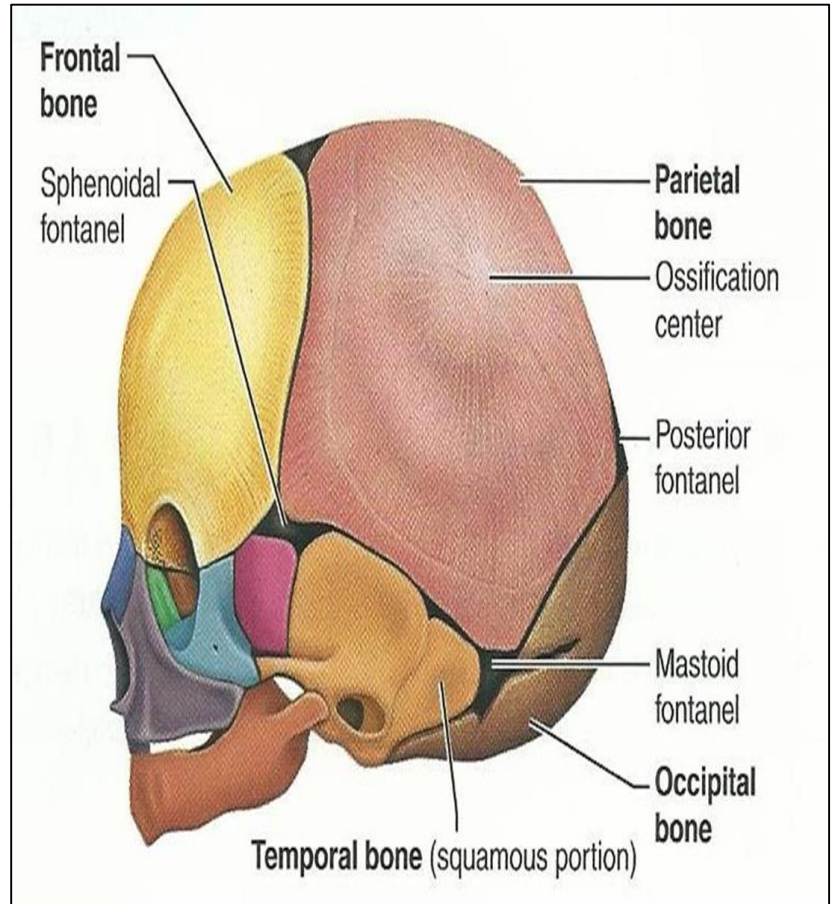
- Is a bony box like cavity
- Contains and protects brain.
- Most important part because
 - ⇒ It contain the delicate brain
 - ⇒ Least compressible part
 - ⇒ Most difficult part to deliver
 - ⇒ It is the largest part of fetus.



Bones of the vault....

- There are seven main bones in the vault of the fetal skull.

1. One occipital bones
2. Two parietal bones
3. Two frontal bones
4. Two temporal bone



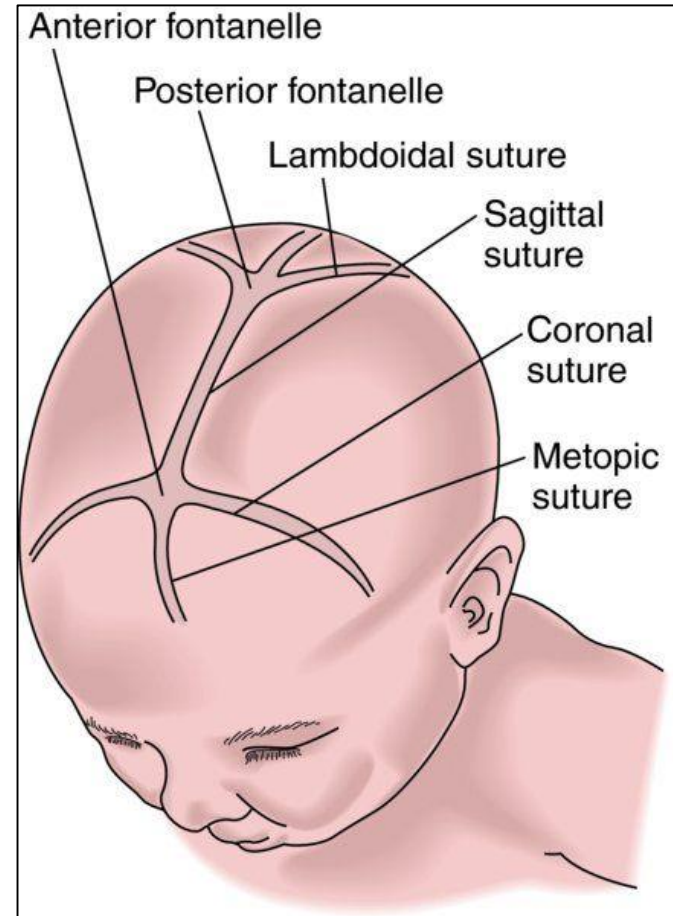
Sutures

- Is an area of membrane between the skull bones where ossification has not been completed.
- Overlap during the process of molding at the time of birth.
- There are four important types of sutures.

Sutures...

- 4 important type of suture

1. Lambdoidal suture
2. Sagittal suture
3. Frontal sutures
4. Coronal suture



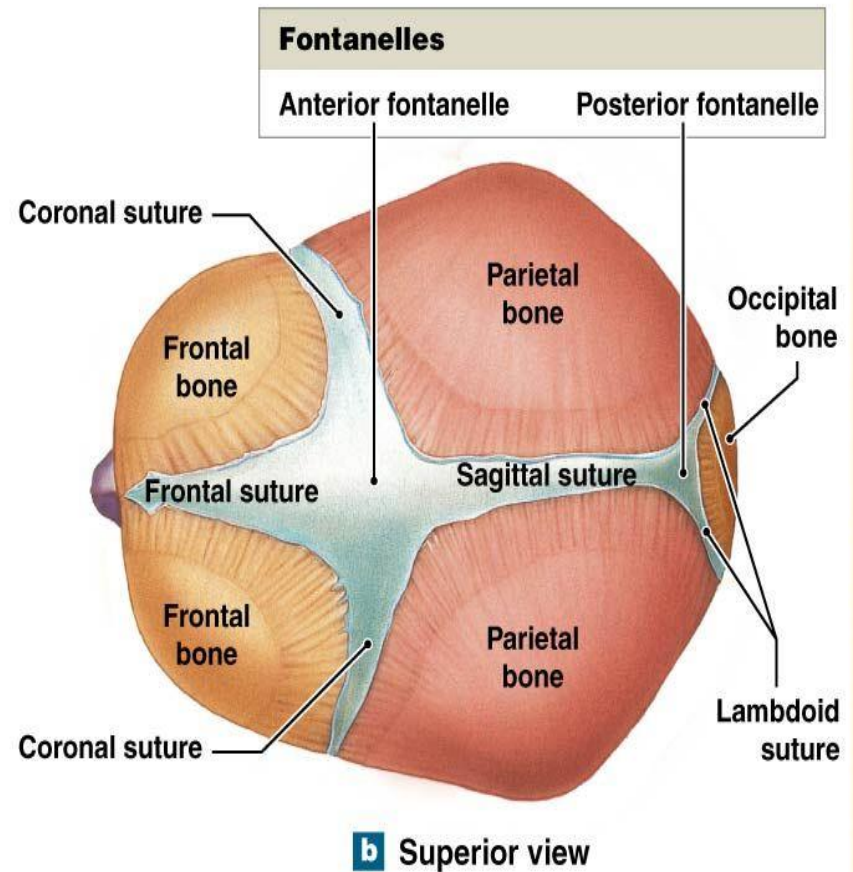
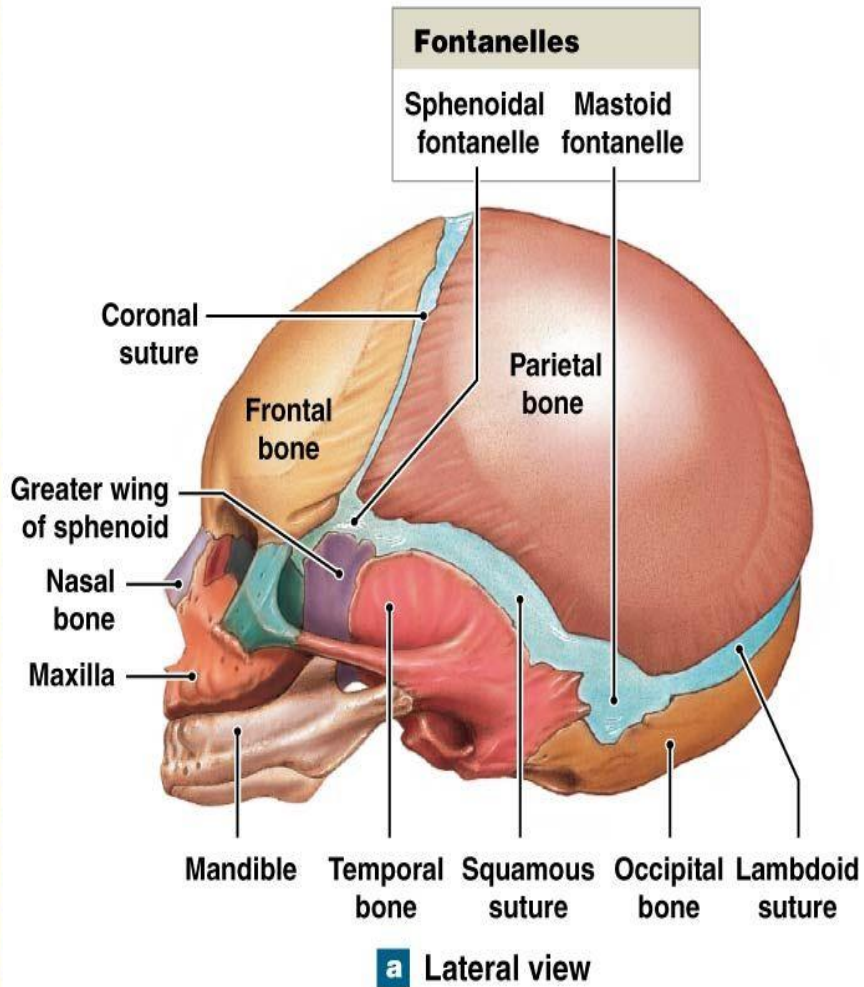
Fontanel

- Are areas where two or more suture meet.
- There are 6 fontanel
- Only two are of great obstetrical importance.
 1. The anterior fontanel or bregma
 2. Posterior fontanel

Fontanels..

Fontanels type		Description
1	The anterior fontanel or bregma	• Formed by sagittal, coronal and frontal sutures meet. • It is diamond in shape. • It is much longer than posterior fontanel. • Pulsation of cerebral vessels can be felt through it. • Normally closes at 18 months of age
2	Posterior fontanel	• The junction of the lambdoidal and sagittal sutures. • It is very small and triangular in shape • It is normally closes by 6 weeks of age

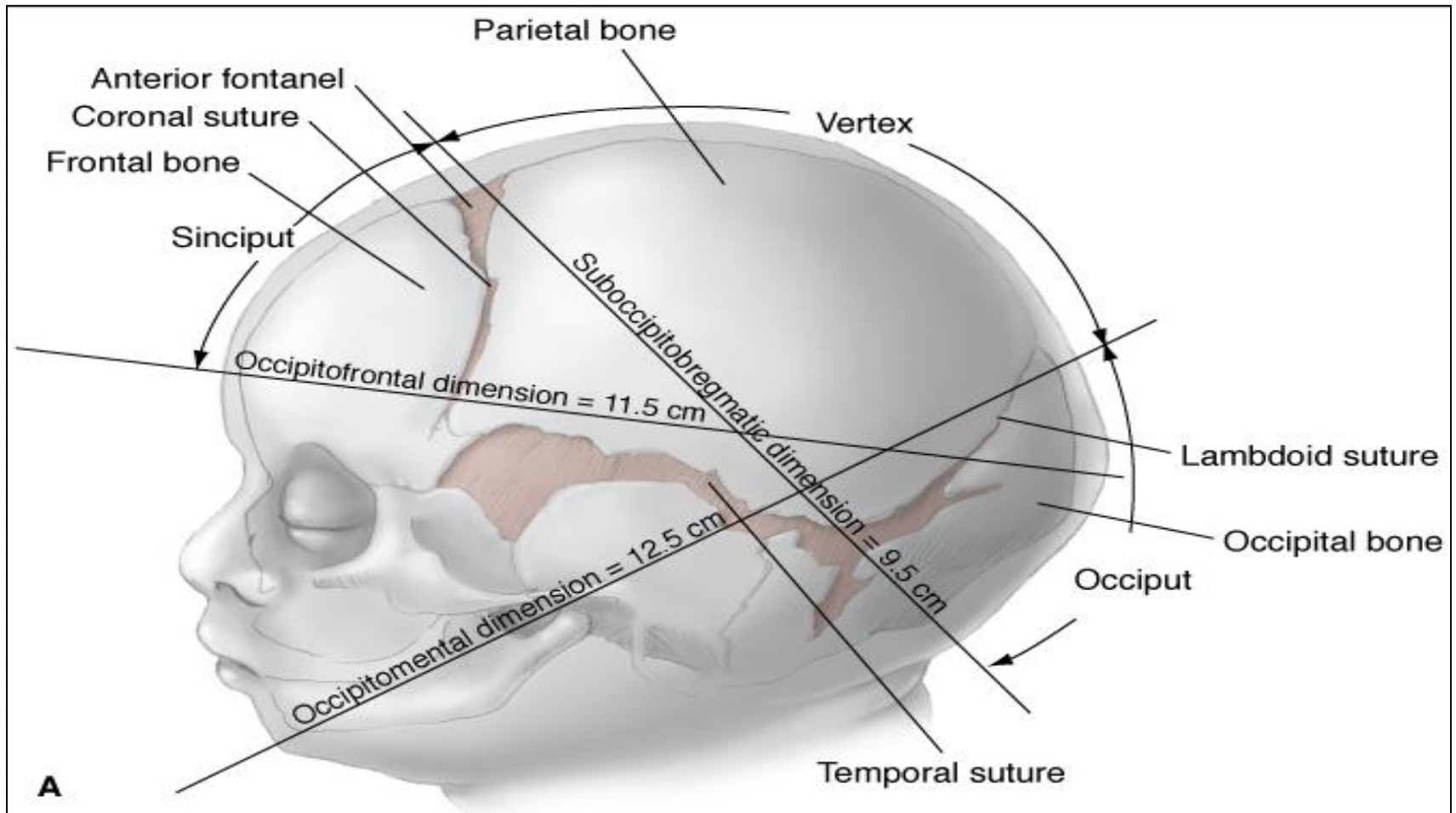
Fontanelles..



The regions of the skull

1	Vertex	• Is bounded by the anterior and posterior fontanel and parietal eminences.
2	Occiput	• lies between the foramen magnum and posterior fontanel.
3	Sinciput (brow)	• From the anterior fontanel and coronal suture to the orbital ridge
4	Face	• Extends from the orbital ridges and the root of the nose to the junctions of the chin and neck. • Between the eye brow is known as the glabella.

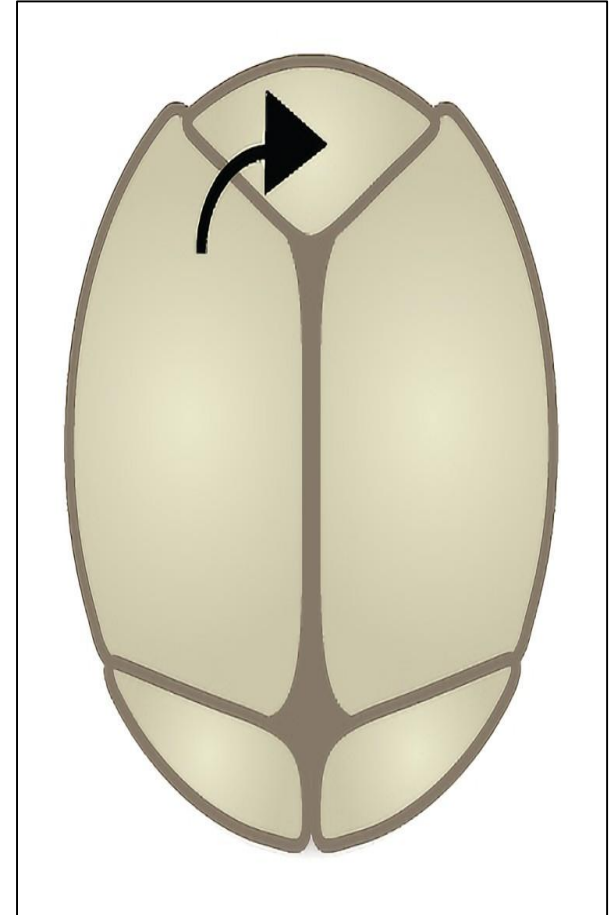
The Regions of the skull..



Source: Cunningham FG, Leveno KJ, Bloom SL, Hauth JC, Rouse DJ, Spong CY:
Williams Obstetrics, 23rd Edition: <http://www.accessmedicine.com>
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Moulding

- The overlapping of one fetal bone on the other to reduce the presenting part diameter.
- Value:
 - ⇒ 0 : separated
 - ⇒ +1: sutures apposed
 - ⇒ +2: sutures overlapped but reducible
 - ⇒ +3: None reducible sutures overlapped



Unit –II Physiology of pregnancy

Diagnoses pregnancy

Physiologic change during pregnancy

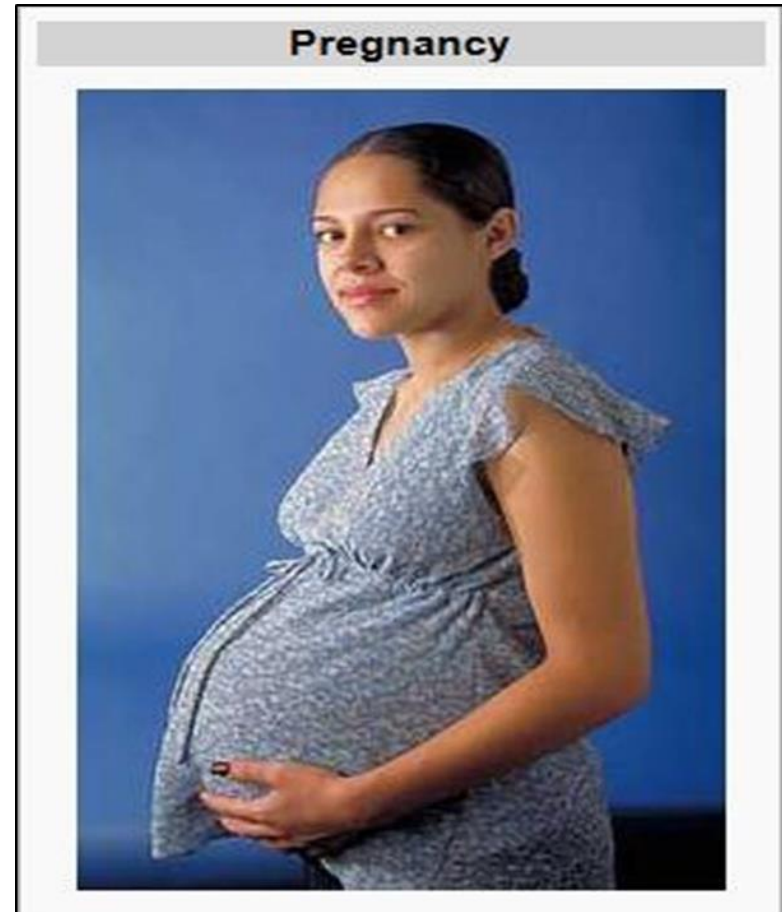
Antenatal care

Objective

- At the end of this course, the student able to:
 - ⇒ Diagnoses pregnancy
 - ⇒ Describe physiologic change during pregnancy
 - ⇒ Discuss on antenatal care

Pregnancy

- Also known as gravidity or gestation
- Is the time during which one or more offspring develops inside a woman uterus or outside the uterus.



Diagnosis of pregnancy

- Pregnancy is usually identified when a woman presents with
 - ⇒ Symptoms and signs
 - ⇒ Urine test (hCG)
 - ⇒ Sonography

Diagnosis of pregnancy

Presumptive symptoms	Probable signs	Positive or absolute signs
• Amenorrhea • Frequency of micturition • Morning sickness • Fatigue • Breast changes • Skin changes • Quickening	• Abdominal enlargement • Braxton-Hicks contractions • External ballottement • Outlining the fetus • Chadwick's sign • Changes of uterus in size, shape • Softening of the cervix • Internal ballottement • Immunological test.	• Palpation of fetal parts • Perception of active fetal movements by examiner • Auscultation of fetal heart sounds • Ultrasound evidence of embryo as early as 6th week • Radiological

Physiological change during pregnancy

- Pregnancy is characterized by major adaptations in the maternal anatomy, physiology and metabolism.
- Most pregnancy related changes are prompted by stimuli provided by the fetus and placenta(hormone).
- Many gestational changes begin soon after fertilization and continue throughout pregnancy.

Physiological change...

- Almost all return to the prepregnancy state after delivery and lactation.
- Important of knowing normal physiological change
 - ⇒ To diagnose common problems of pregnancy.
 - ⇒ To manage problems of pregnancy accordingly.

Affected organ

Genital organ Breasts

Central nervous system

Gestational weight gain

Cardiovascular system

Skin

Skeletal and postural changes

Hematologic change

Respiratory system

Renal system

Alimentary tract

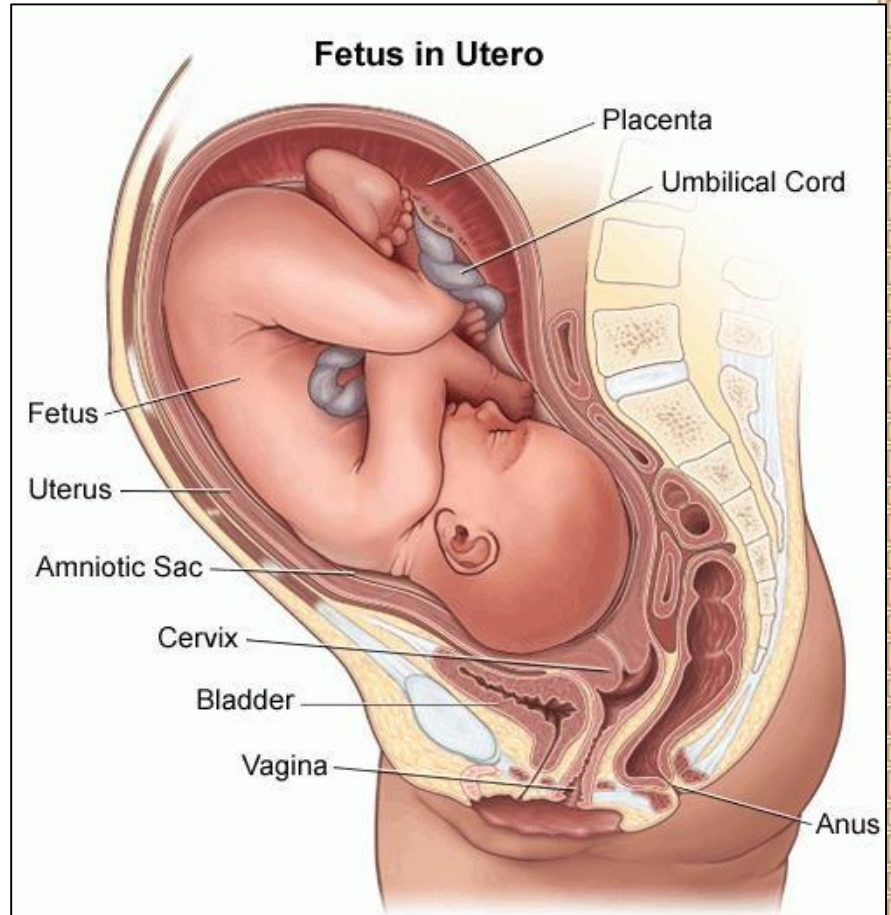
Genital organ changes

Uterine

Cervix

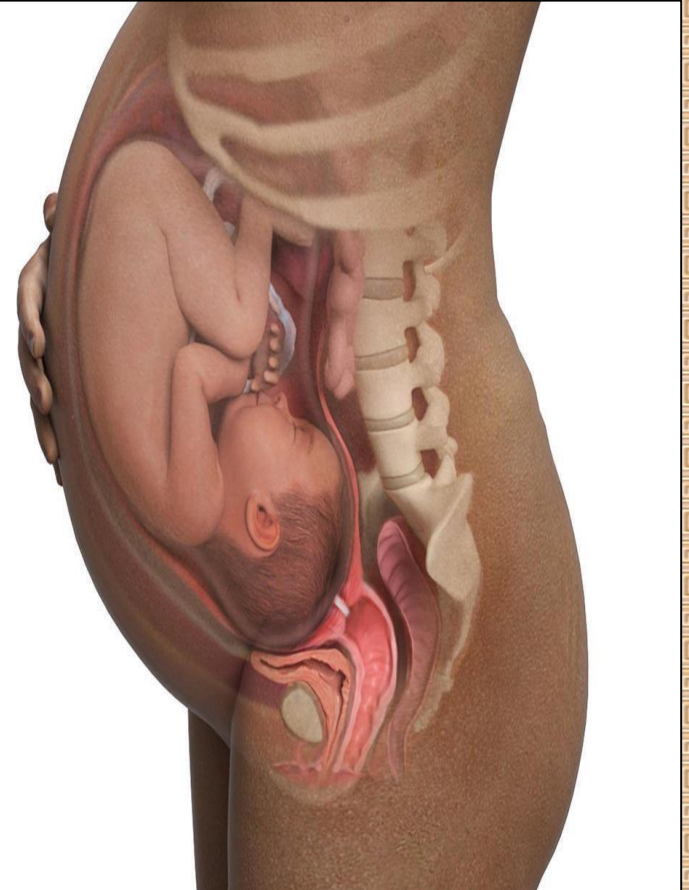
Vagina

Vulva



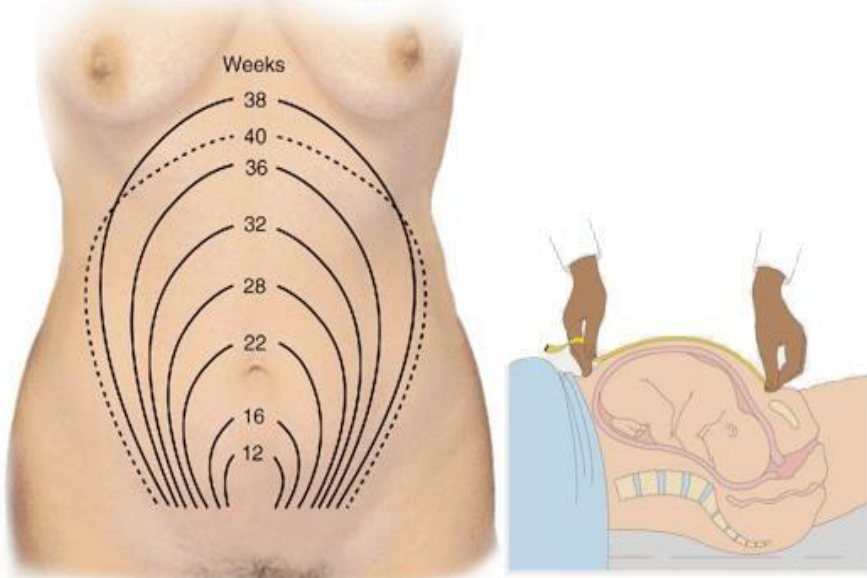
Uterus

- Become thin-walled muscular organ.
- At term weight $\Rightarrow$ 1100g
- Total volume > 5 L
- Enlargement due to stretching and marked hypertrophy of muscle cells.
- Early pregnancy, uterus contracts irregularly which become increase



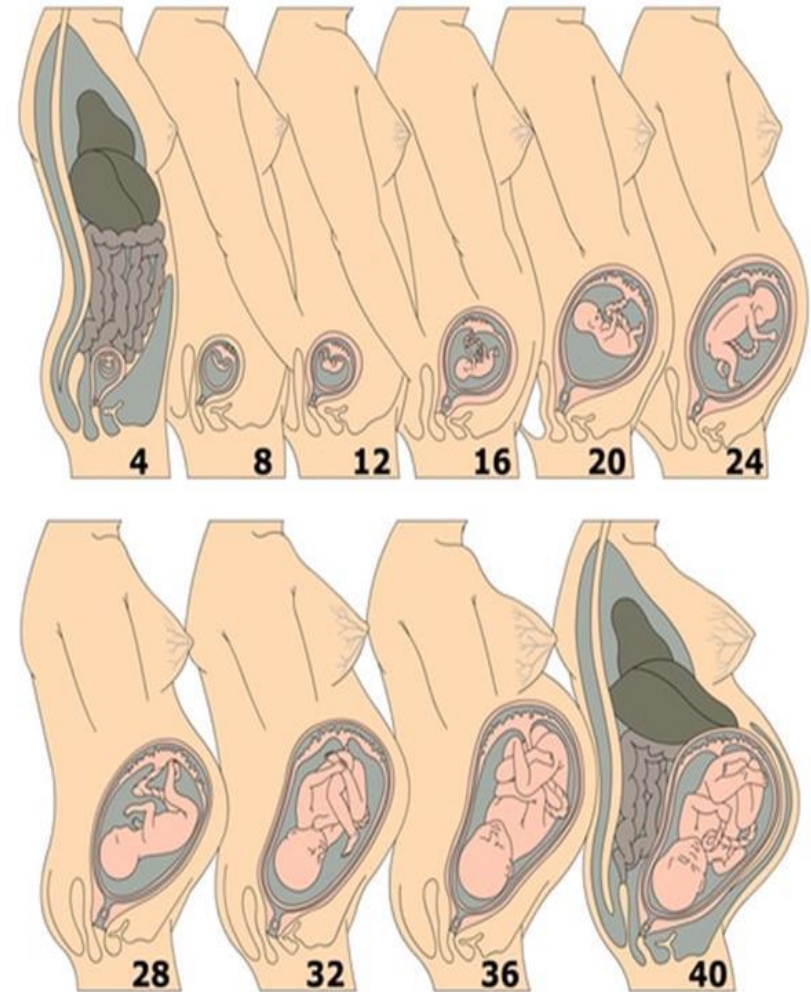
Uterus...

Fundal Height



Approximation of Fundal Height

Gestational age	Fundal Height Landmark
12 wks	Pubic symphysis
20 wks	Umbilicus
36 wks	Xiphoid process
37 - 40 wks	Regression of fundal height b/w 36 - 32 cm
Postpartum (≤ 24 hrs)	Umbilicus



Cervix

- Become softening distensible, and cyanosis(5th-6th week)
- Produces thicker, acidic copious amounts of mucus which contain IgG & IgA

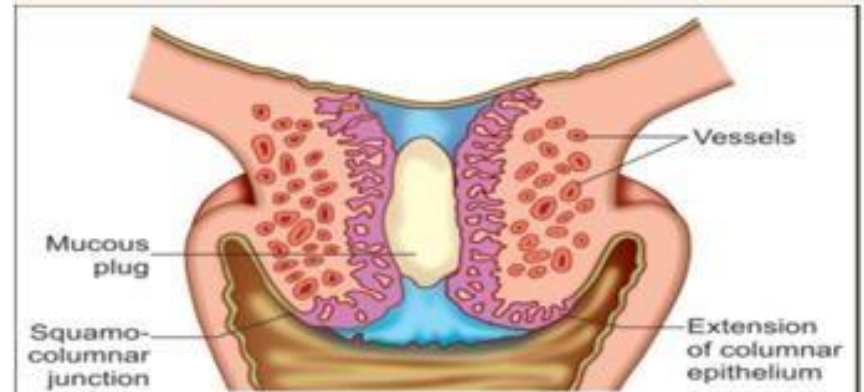
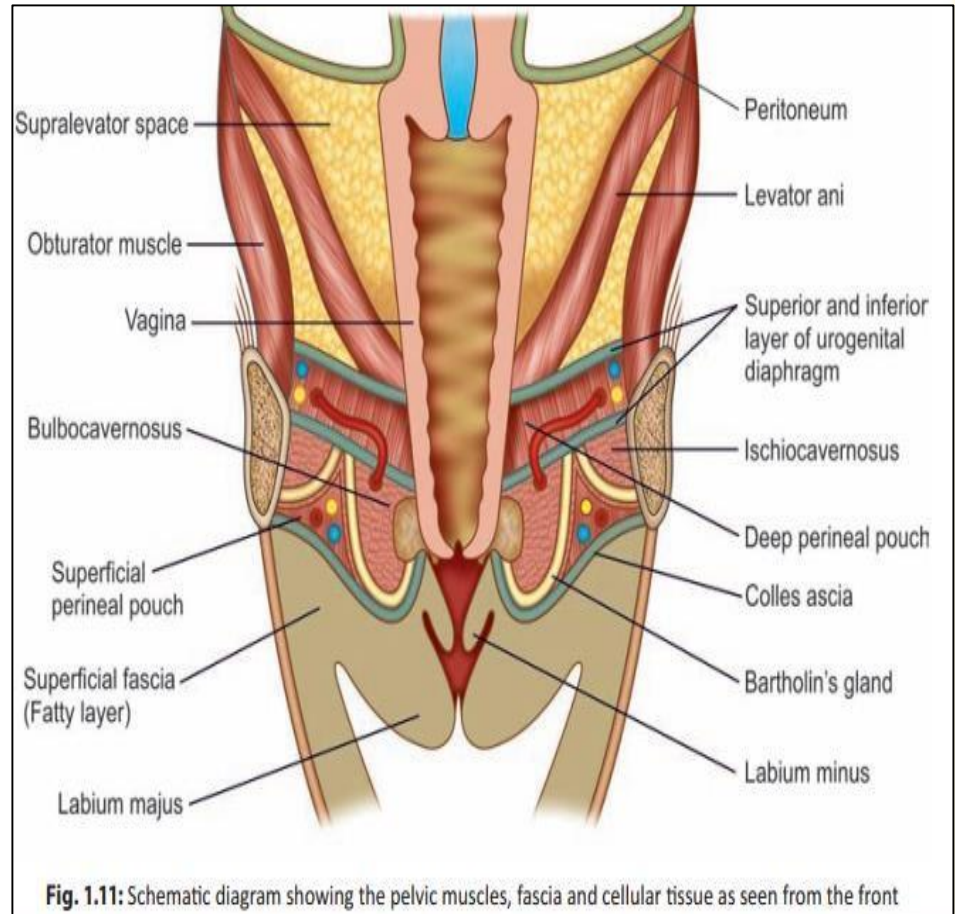


Fig. 5.4: Changes of the cervix during advanced pregnancy



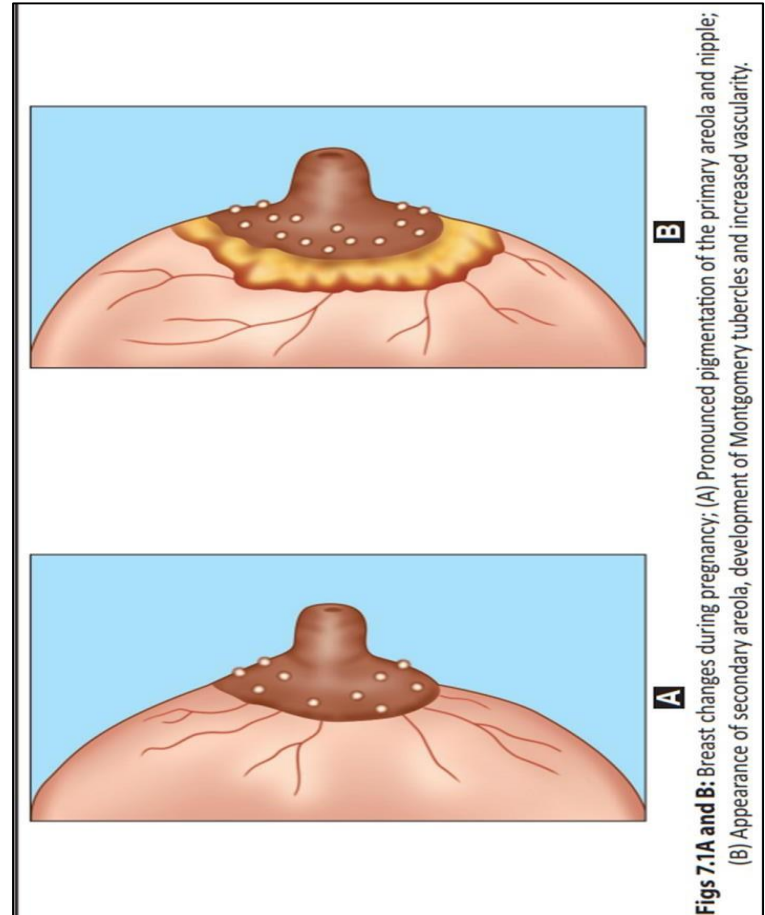
Vagina and vulva

- Muscle fibers thicken.
- Mucosa become thickness
- $\uparrow$ Glycogen $\Rightarrow$ lactic acid $\Rightarrow$ $\downarrow$ vaginal pH $\Rightarrow$ $\downarrow$ infection
- Worsening of varicose veins in the vulva



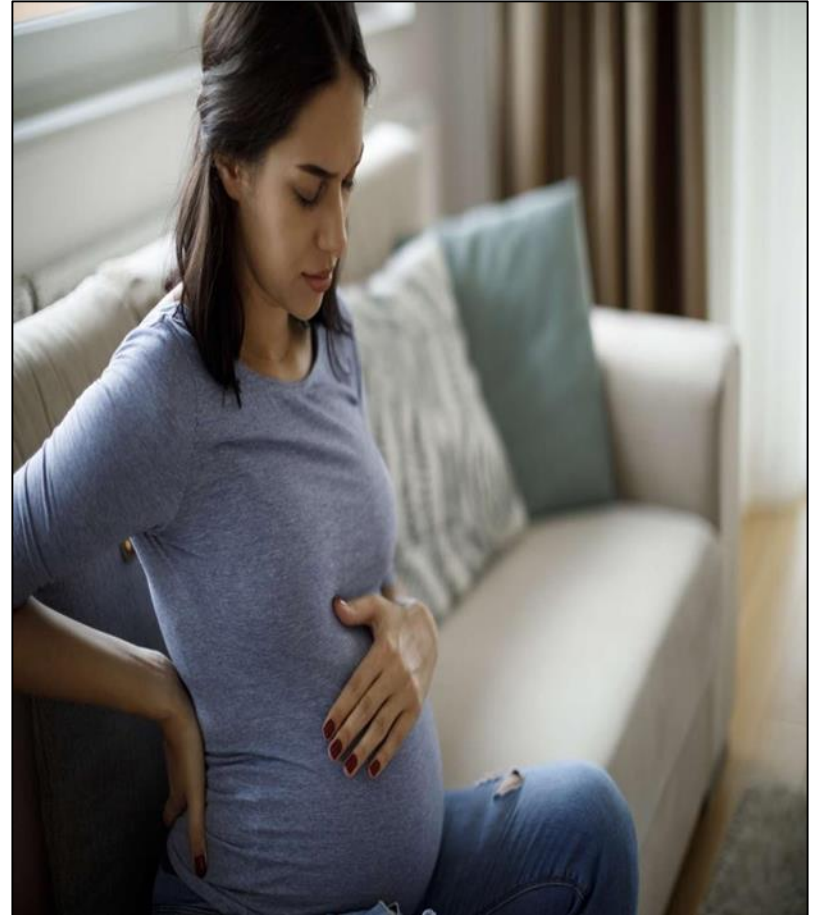
Breasts

- Enlarged by the 8th week
- Nipples become considerably ↑, deeply pigmented & erectile.
- Hypertrophied sebaceous glands called Montgomery's tubercles appears.
- Colostrum can be expressed from the nipples >10th week



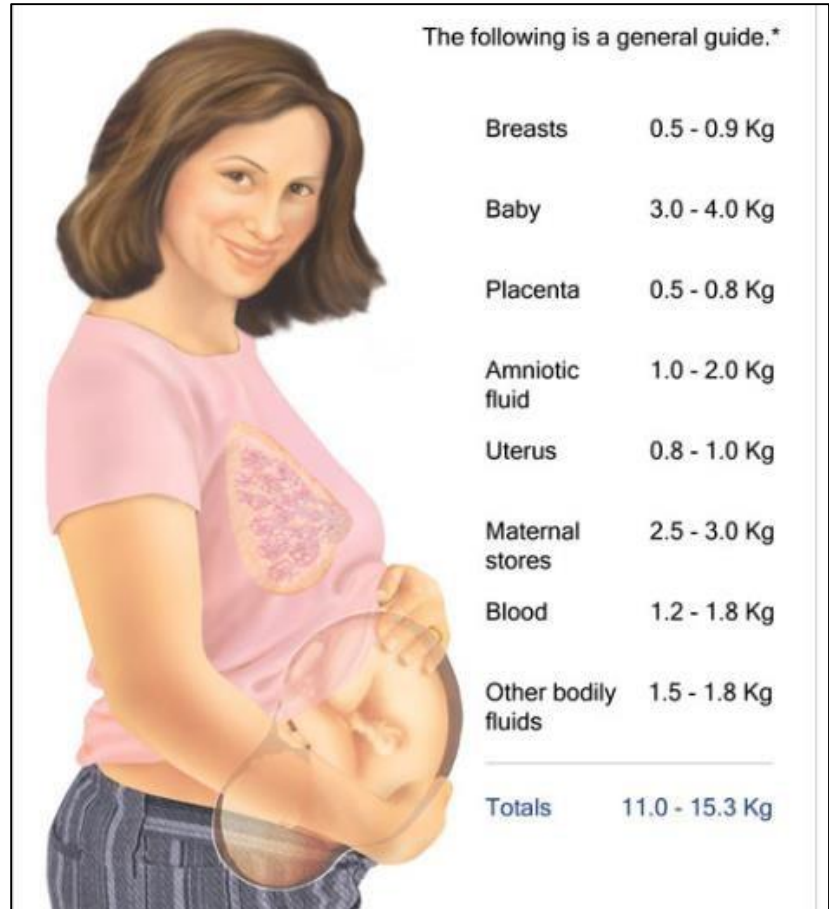
Central nervous system

- Headache is common
- Tingling of hand
- Paresthesia and sensory loss of anterolateral thigh.



Gestational weight gain

- Mean weight gains being highest in the **second trimester**.
- The total weight gain during the course of a singleton pregnancy for a healthy woman averages 11 -12.5 kg
- MUAC should be >23cm for pregnant women.



Cardiovascular system

- Heart

- ⇒ Displaces the heart upward and to the left.
- ⇒ Straightening of the left-sided heart border
- ⇒ Increased prominence of the pulmonary conus.
- ⇒ Eccentric cardiac hypertrophy is commonly noted.
- ⇒ Cardiac output increase of 30% to 50%
- ⇒ Most CO is directed to the uterus, placenta, and breasts.

Cardiovascular system...

- Blood pressure

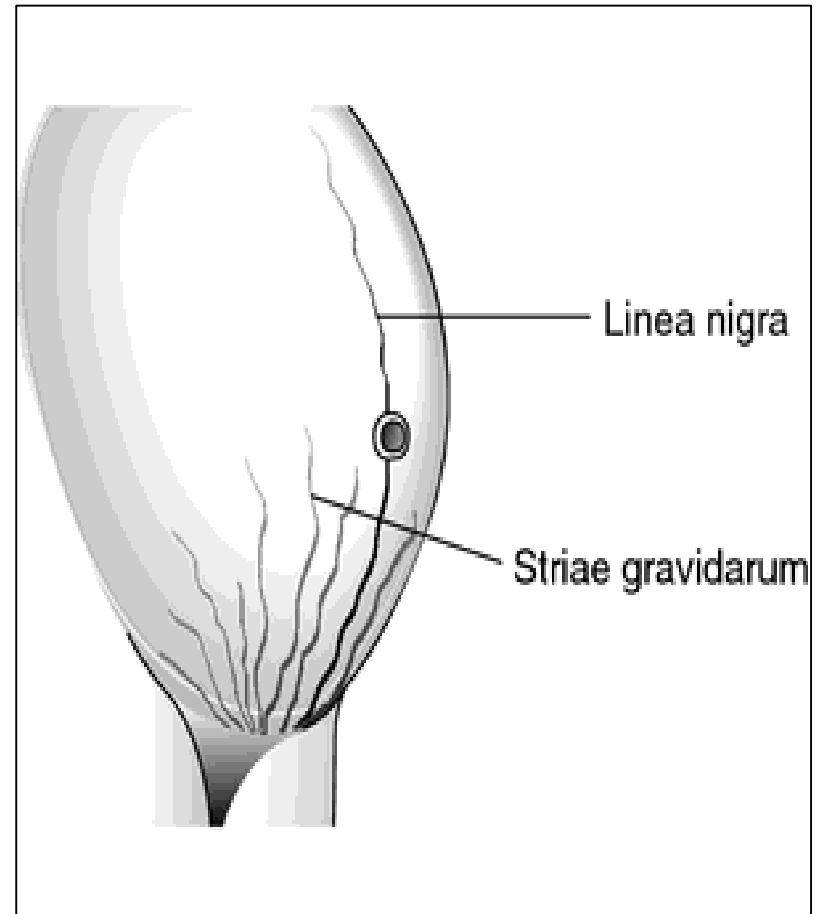
- ⇒ Arterial blood pressure ↓ during 2nd trimester & ↑ thereafter.
- ⇒ Fainting is common during 2nd trimester
- ⇒ Systemic vascular resistance decrease
 - ⇒ Smooth muscle relaxing effect of progesterone
 - ⇒ Vasodilatory effect (NO,PG,ANP).

Skin

- Hyperpigmentation in face areola, genital skin, abdomen observed 90% of women, likely due to

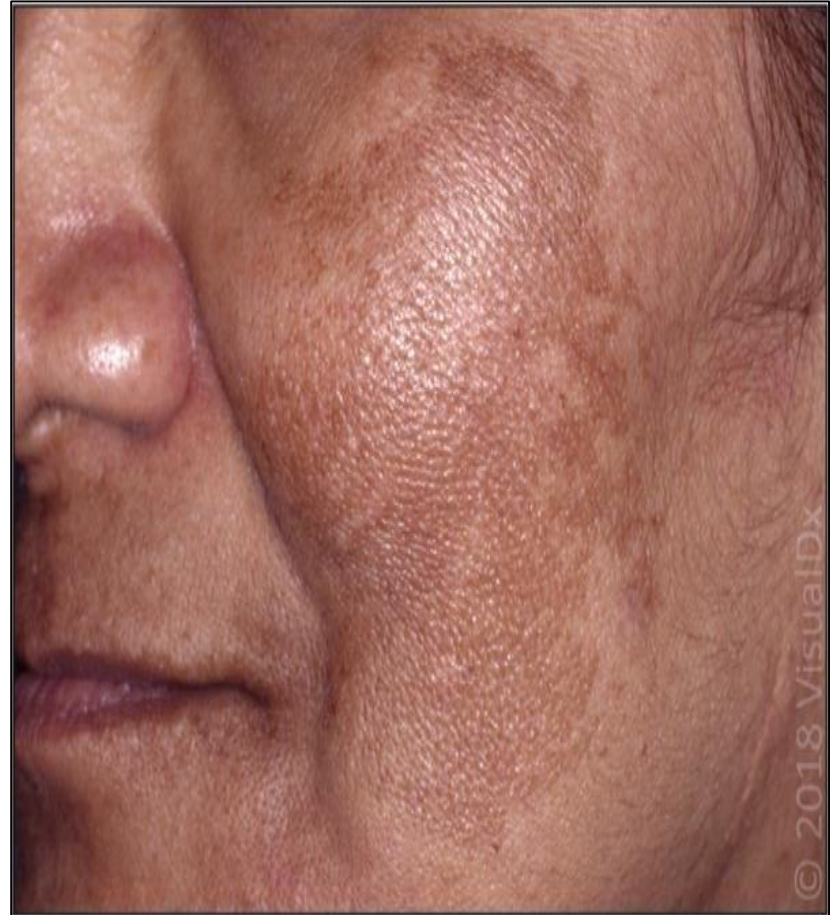
⇒ ↑ melanocyte-stimulating hormone and estrogen.

- Striae gravidarum
- Linea nigra



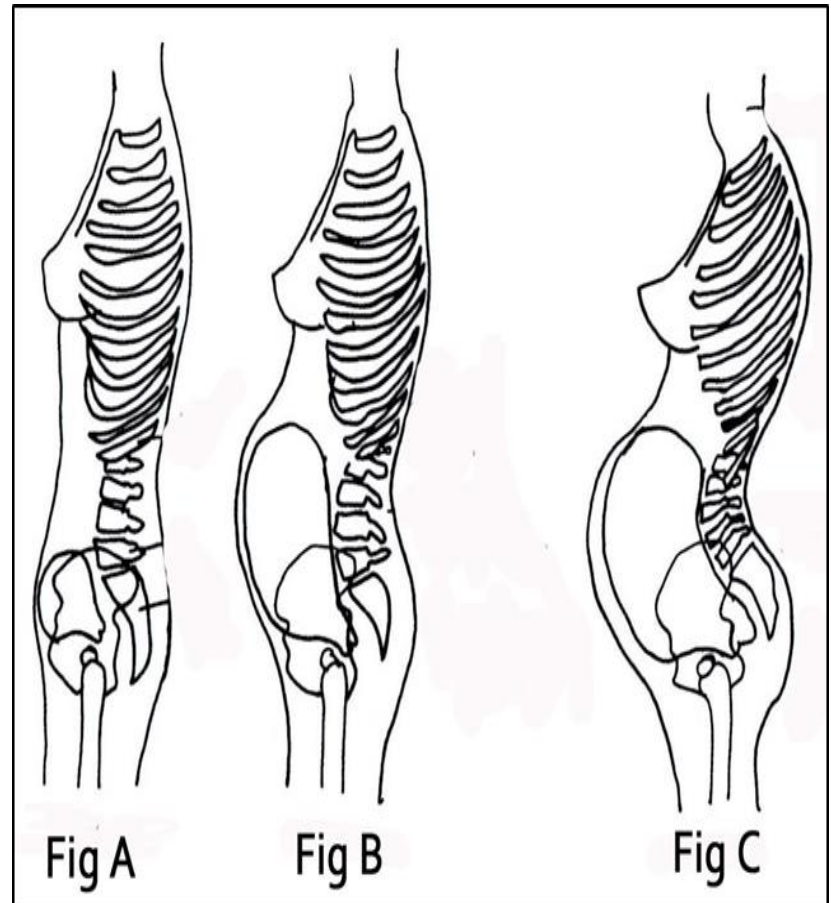
Skin...

- Cloasma or mask of pregnancy: irregular pigmentation of the face and neck.



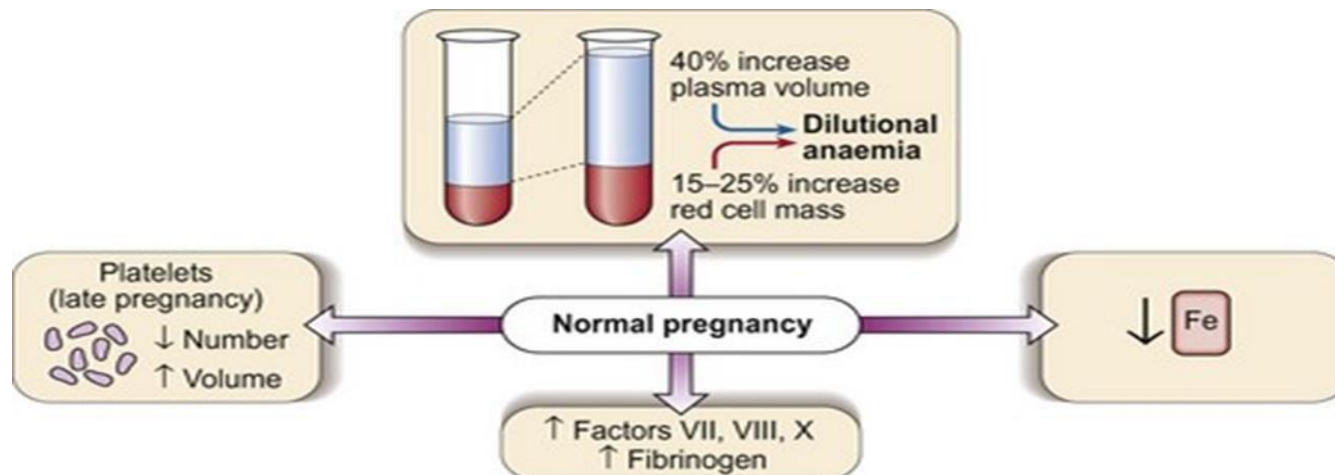
Skeletal and postural changes

- Pregnancy and lactation cause reversible bone loss
- Bone turnover increases in the third trimester
- Pregnancy results in lordosis which lead back pain.
- Increase pubic bone joint



Hematologic change

- Plasma volume increase greater 50% as compare with total red cell volume 30% $\Rightarrow$ physiologic anemia of pregnancy.
- 60mg of elemental iron is required daily.

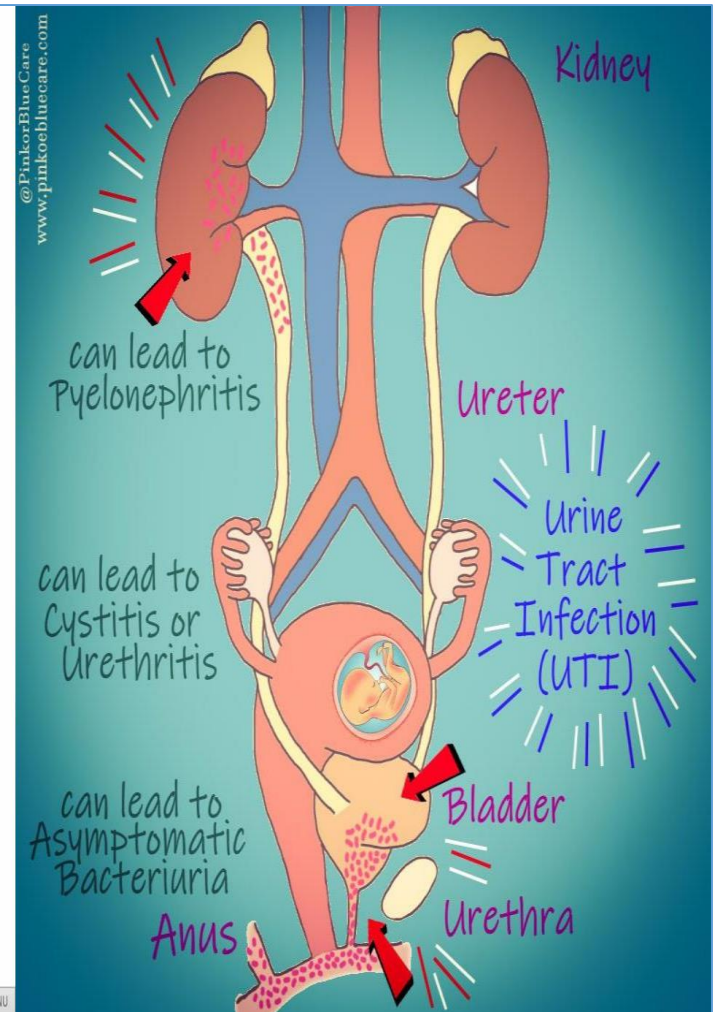
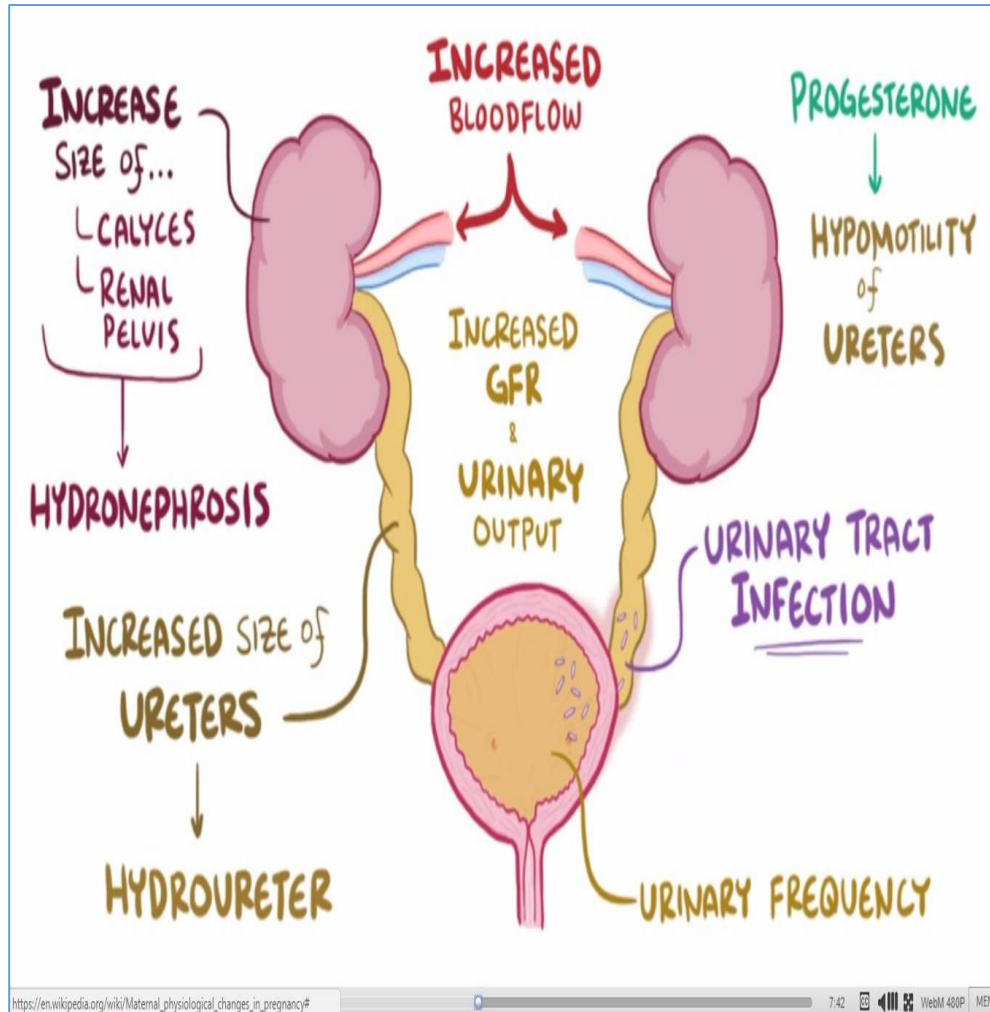


Respiratory system

- Marked nasal stuffiness and decreased nasal patency
- Chest circumference expands
- Diaphragm elevated
- Respiratory muscle function is not affected

Parameter	Change
Oxygen consumption	Increases by 20 to 50%
Minute ventilation	Increases by 50%
Tidal volume	Increases by 40%
Respiratory rate	Unchanged/Slightly Increases
PaO ₂	Increases by 10%
PaCO ₂	Decreases by 15%
HCO ₃	Decreases by 15%
FRC	Decreases by 20%

Renal system

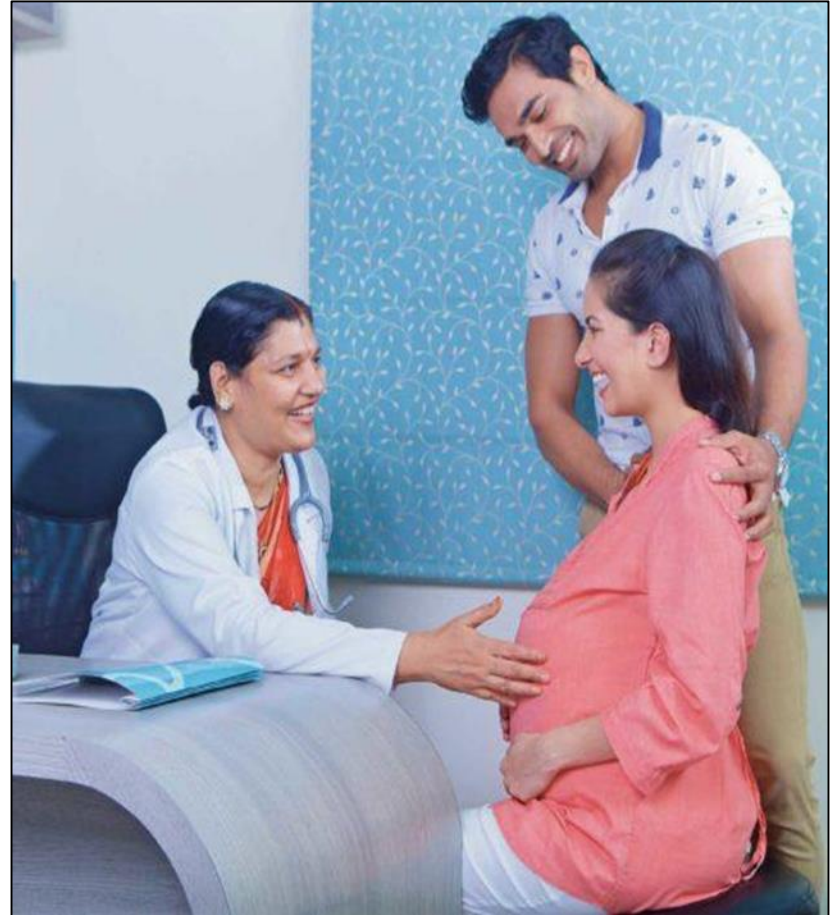


Alimentary tract

- Appetite will increase food intake
- The RDA calls for an additional 300 kcal/day
- History of pica $\Rightarrow$ poor weight gain or refractory anemia.
- Ptyalism is most often occurs
- The gums swell and may bleed $\Rightarrow$ gingivitis of pregnancy.
- Stomach and gastroesophageal (GE) sphincter tone are $\uparrow$.

Antenatal care

- A complex interventions that a pregnant woman and adolescent girl receives from skilled health care professionals in order to ensure the best health conditions for both mother and baby during pregnancy.



Components of ANC

1

- Health promotion and disease prevention

2

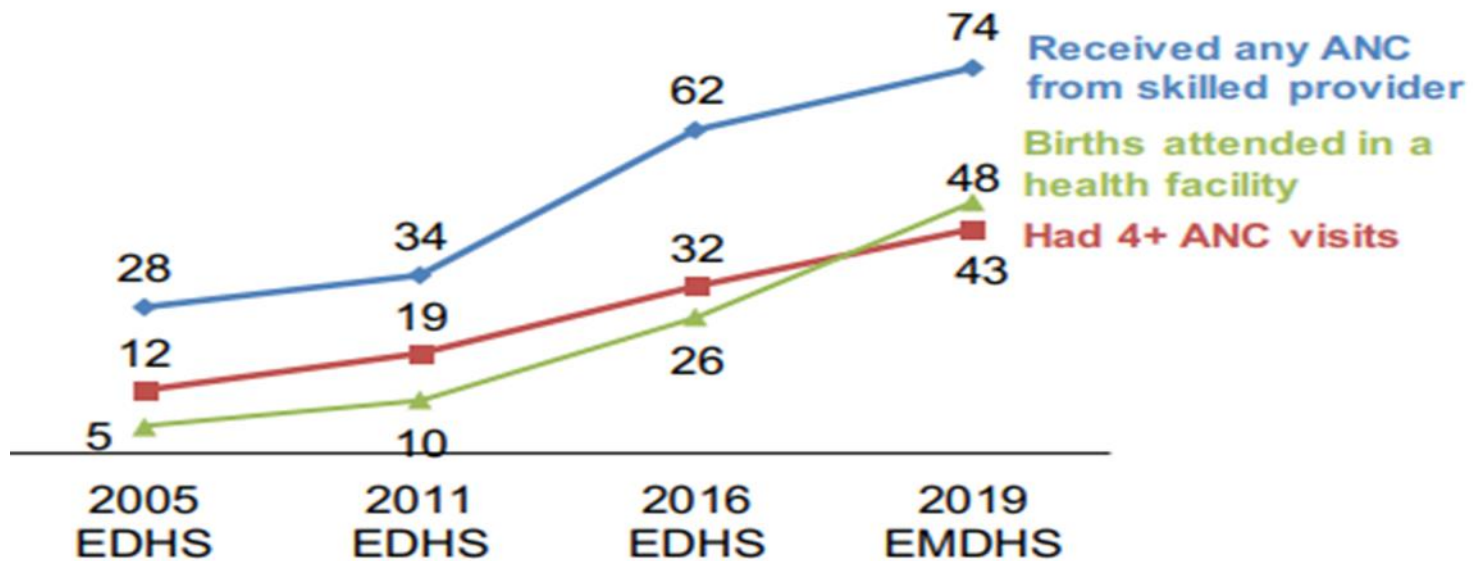
- Screening, diagnosis & management or referral for disease

3

- Birth planning and complication readiness.

Prevalence

Percentage of women age 15-49 who had a live birth in the 5 years before the survey (for the most recent birth)



ANC contacts

- Focused ANC that emphasizes
 - ⇒ Individualized care
 - ⇒ Client centered
 - ⇒ Fewer but comprehensive visits
 - ⇒ Disease detection not risk classification
 - ⇒ Care by a skilled provider

Table 1. 2016 WHO ANC model

First trimester
Contact 1: up to 12 weeks
Second trimester
Contact 2: 20 weeks Contact 3: 26 weeks
Third trimester
Contact 4: 30 weeks Contact 5: 34 weeks Contact 6: 36 weeks Contact 7: 38 weeks Contact 8: 40 weeks
Return for delivery at 41 weeks if not given birth. Note: Intermittent preventive treatment of malaria in pregnancy should be started at ≥ 13 weeks.

Purpose

- Reduces maternal and perinatal morbidity & mortality through:
 - ⇒ Screening, diagnosis & treatment of risk factors(complications)
 - ⇒ Identification of risk women
 - ⇒ Prevent and manage concurrent diseases
 - ⇒ Ensuring referral to an appropriate level of care

Important term

A nullipara is one who has never completed a pregnancy to the stage of viability. She may or may not have aborted previously.

A primipara is one who has delivered one viable child. Parity is not increased even if the fetuses are many (twins, triplets).

A multigravida is one who has previously been pregnant. She may have aborted or have delivered a viable baby.

A parturient is a women in labor.

A nulligravida is one who is not now and never has been pregnant.

A primigravida is one who is pregnant for the first time.

Multipara is one who has completed two or more pregnancies to the stage of viability or more.

A puerpera is a woman who has just given birth.

ANC contacts

During each ANC contact

Gather and interpret information



Classify the type of care (basic versus specialized)



Develop care plan



Implement care plan (take action)



Evaluate care plan.

1. History taking

1. Social history

Date: _____ ANC Reg. No: _____ Medical Record Number (MRN): _____
Name of Client: _____ Name of Facility: _____
Woreda: _____ Kebele: _____ House No: _____ Age (Years): _____
LMP: _____ / _____ / _____ EDD: _____ / _____ / _____
Gravida: _____ Para: _____ Number of children alive: _____ Marital Status: _____

2. Chief complaint

- A major symptoms for which the patient is seeking care.
- Patient may have come for ANC follow up or other complaints.
- Note the duration of each complaint

1. History taking...

3. History of present pregnancy

- Gravidity: all forms of pregnancy
- Parity is number of delivery after 28 weeks
- Abortion termination of pregnancy before 28 weeks
- Last normal menstrual period (LNMP)
- Expected date of delivery (EDD)

1. History taking...

- EDD which could be calculated by
 1. LNMP+9 months +10 days if pagume is not passed
 2. LNMP + 9 months + 5 days if pagume is passed
- Calculate gestational age(GA) in completed weeks and days as follow

$$GA = \frac{\text{Visiting date} - \text{LNMP}}{7}$$

7

1. History taking...

- Quickening:
 - ⇒ The first time the mother felt fetal movement
 - ❖ Premigravida it is around 18-20 weeks
 - ❖ Multigravida is it around 16-18 weeks of gestation
- Presence of Antenatal care elsewhere.
 - ⇒ Place and number of visits
 - ⇒ What was done and laboratory results.
- Danger symptoms of pregnancy

1. History taking...

4. Past obstetric history

- Date, month and year of gestation
- Length of gestation:
 - ⇒ Abortion (< 28 weeks)
 - ⇒ Preterm (< 37 complete weeks)
 - ⇒ Term (> 37-42 weeks)
 - ⇒ Post term (> 42 completed weeks).

History taking...

- Onset of labor (spontaneous or induced)
- Fetal presentation(vertex, face,brow,breech or shoulder)
- Duration of labour
- Mode of delivery (spontaneous ,operative,C/S)
- Fetal out come (alive or dead, sex, weight of the new born, malformation and current condition)
- Post partum complication-haemorrhage and Breast feeding

1. History taking...

5. Gynaecology history

- Family planning methods – use, type, duration, side effect
- Sexual history- assess risk of STI and HIV/AIDS
- Gynaecology operations- female genital mutilation, laparotomy, dilatation and curettage, evacuation and curettage, manual vacuum aspiration
- Menstrual history

1. History taking...

6. Past medical and surgical history:

- History of DM, hypertension, hypo or hyper thyroidism...
- Blood transfusion(haemolytic disease of the new born).
- Drugs because of risk of teratogenicity or allergic reaction
- Maternal infection- TORCH syndrome.

1. History taking...

7. Personal, family and social history

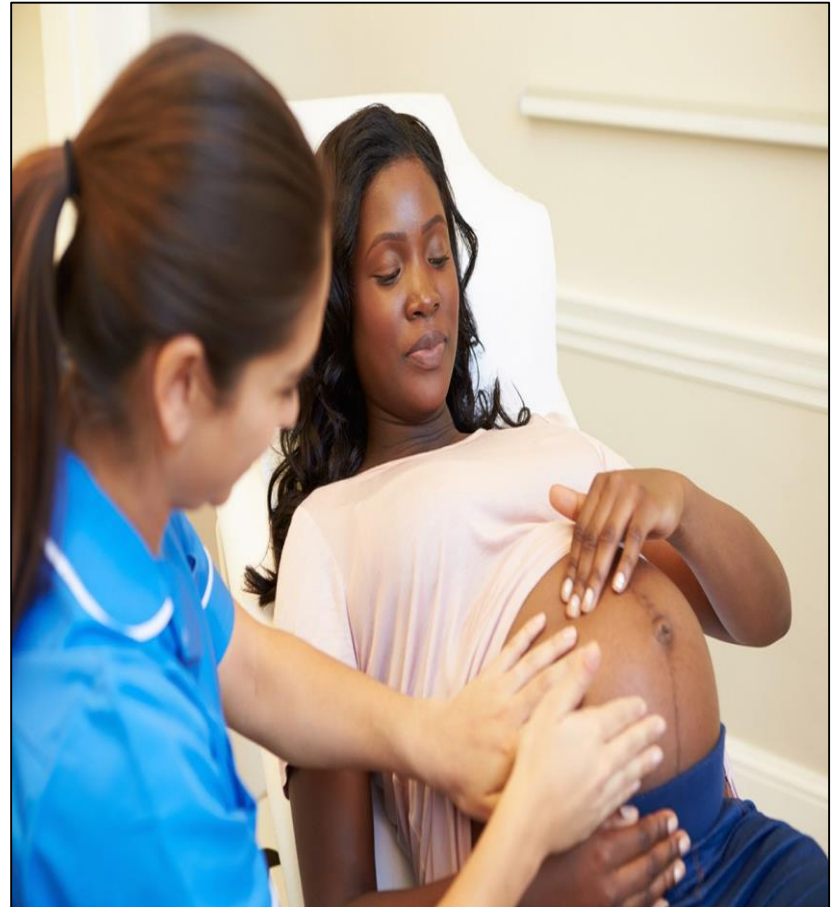
- Habit like alcohol, smoking and illicit drug use
- Occupation- exposure to radiation,halothane, chemical factory
- Income- low socio economic status(preeclampsia, preterm...)
- Family history- DM, hypertension, multiple pregnancy, genetic disorder

2. Physical examination

- Examination must be done in a private room in the presence of an assistant preferably female
- Proper explanation must be offered to the patient before, during and after the examination
- Bladder should be emptied and the patient properly positioned on the couch

2. Physical examination....

- Warm hand and instruments must be used
- Adequate light
- Appropriate gloves and swabs should be prepared.
- Always keep eye contact through out the examination.



2. Physical examination....

1. General appearance

- Apparent state of health.
- Is the pregnant mother
 - ⇒ Sick looking
 - ⇒ Acutely sick looking
 - ⇒ Chronically ill looking



2. Physical examination....

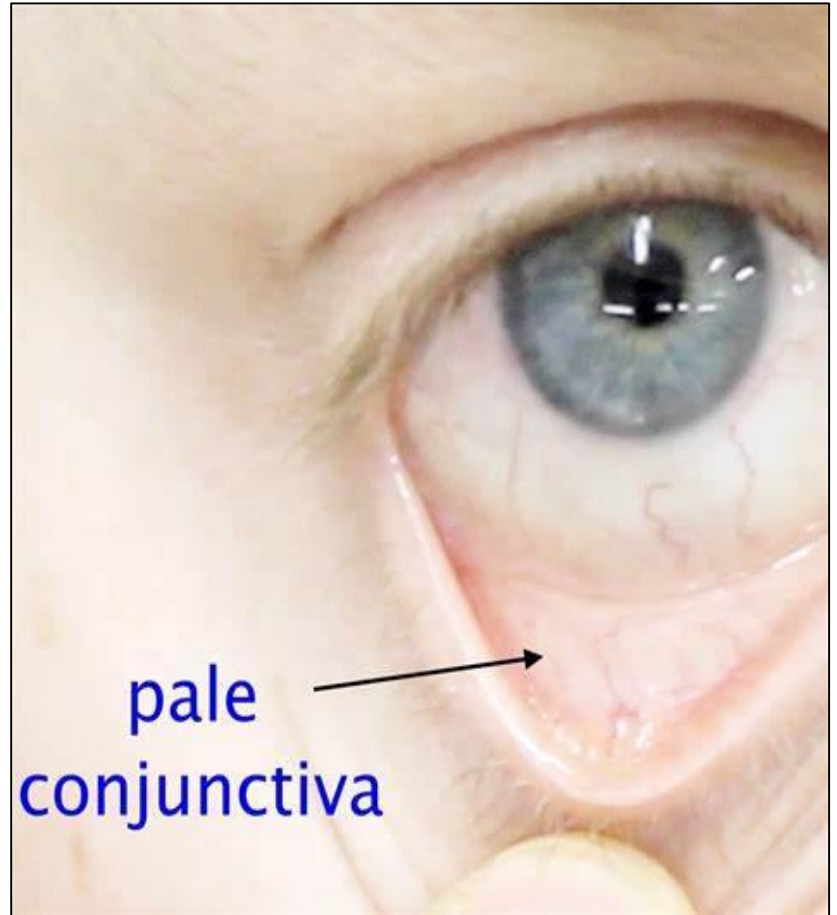
2. Vital signs and anthropometric measurements

- Blood pressure(measured in left lateral e or sitting position).
- Pulse rate- increase 10-15 beat/ minute in pregnancy
- Respiratory rate- increases 1-4 breath/ minute
- Temperature
- Weight- increment of more than 1kg/week is abnormal
- Height- less than 150 centimetres could be constitutional

2. Physical examination....

3.HEENT

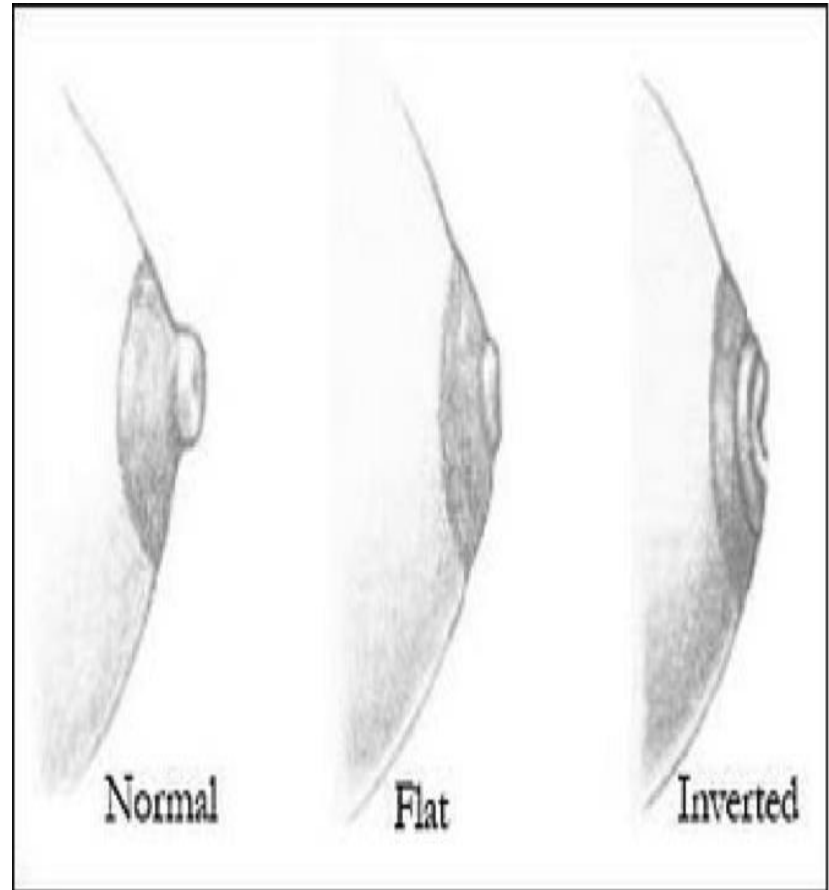
- Emphasis on conjunctiva, sclera, teeth, and buccal mucus membrane to look for pallor, jaundice, mucosal congestion and dental carries



2. Physical examination....

4. Lymphoglandular system:

- Thyroid gland for hyper or hypo thyroidism sings
- Breast for nipple reaction, pigmentation,lumps,discharge, colour change



2. Physical examination....

5. Respiratory and cardiovascular system

- Normal chest auscultation
- Point of maximal impulse (PMI) deviation to left
- S3 gallop may be heard
- Functional systolic murmur may be heard

2. Physical examination....

4. Abdomen:

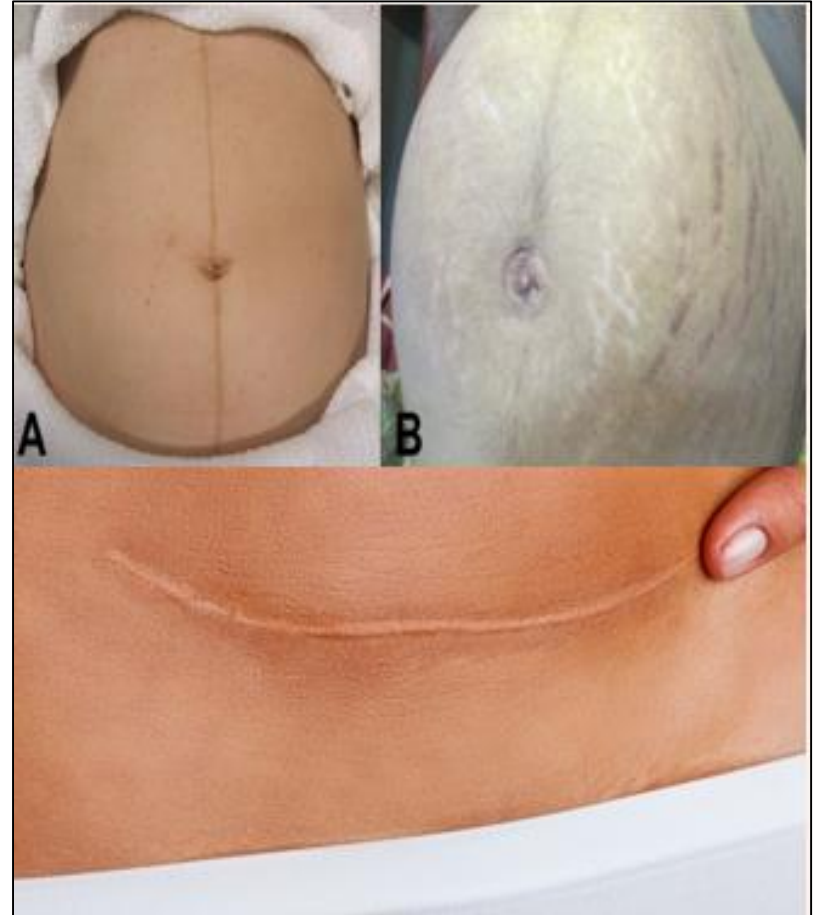
- Aims of abdominal examination
 - ⇒ To observe signs of pregnancy
 - ⇒ To assess fetal size & growth
 - ⇒ To assess fetal health.
 - ⇒ Localization of fetal parts



2. Physical examination....

A. Inspection

- Shape
- Striae gravidarum
- Surgical or non surgical scar
- Size
- Skin



2. Physical examination....

B. Palpation

- Superficial palpation⇒rigidity, tenderness, superficial mass
- Deep palpation⇒Mass, organomegally...
- Obstetric palpation or Leopold maneuver

2. Physical examination....

A. The first leopard maneuver or fundal palpation

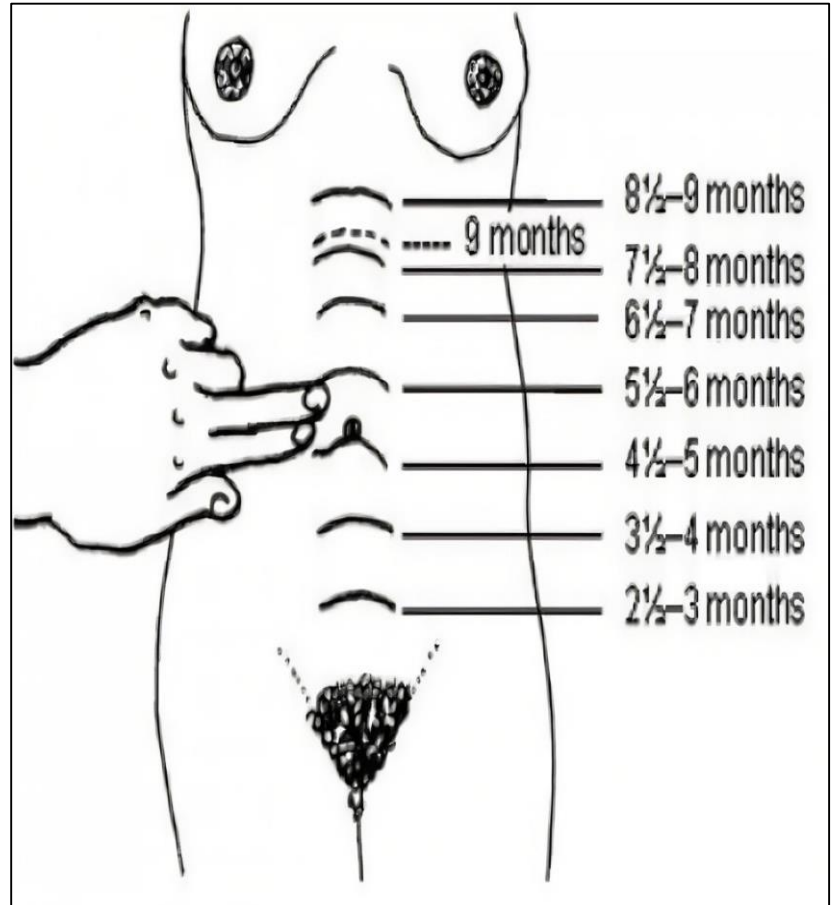
I. Fundal height measurement

- First correct for asymmetry before measurement.
- Then use one of the following methods
 - ⇒ Finger method
 - ⇒ Tape meter measurement

2. Physical examination....

A. Finger method

- 1 finger above umbilicus=2wks
- 1 finger below umbilicus=1wks
- Uterus felt at symphysis pubis corresponds to 12 weeks.
- At umbilicus it is 20 weeks and at xiphysternum it is 38 weeks.



2. Physical examination....

B. Tape measurement

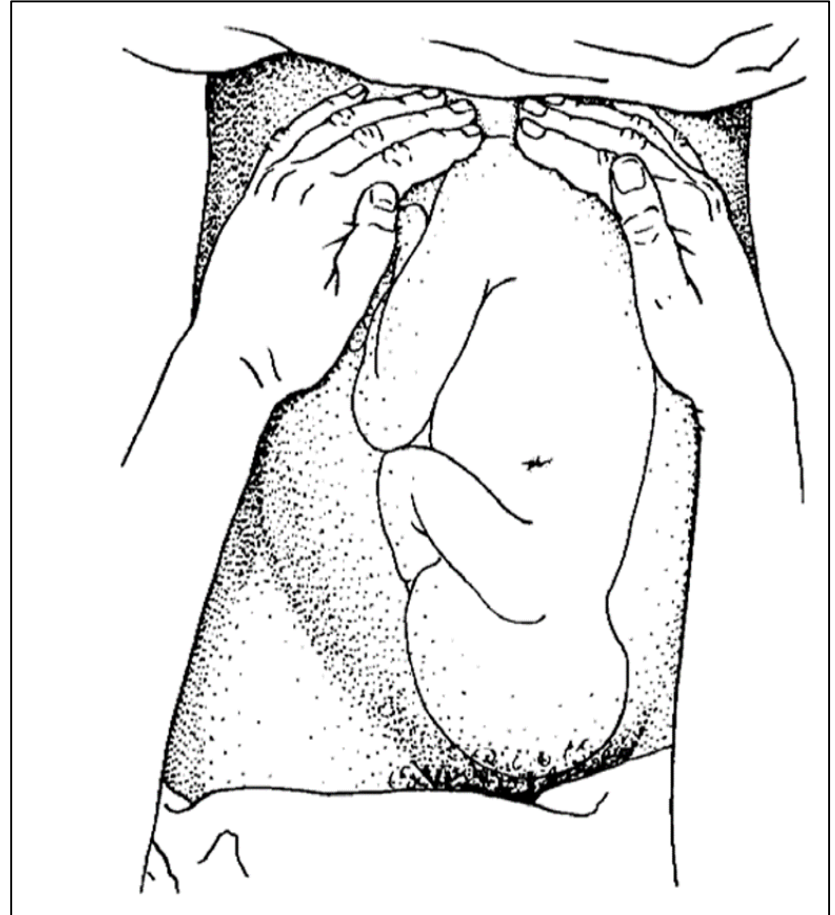
- Symphysis to fundal height in centimetre with tape meter between 18-34 weeks is accurate to within two weeks of actual gestational age.



2. Physical examination....

II. Determine what occupies the fundus.

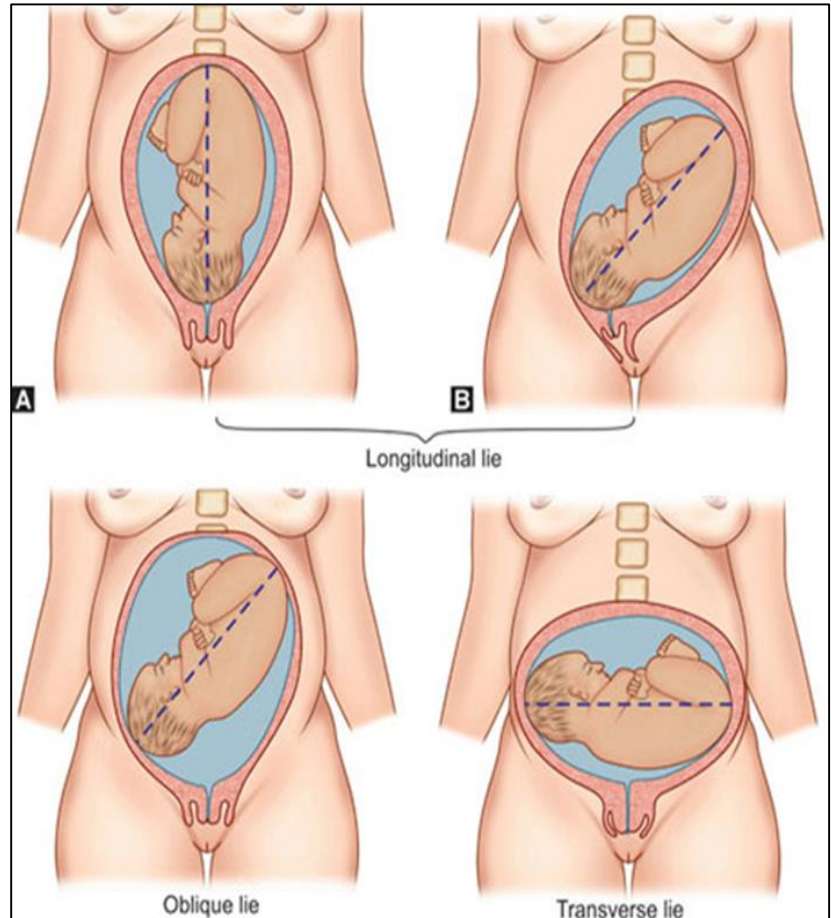
- If soft, irregular, bulky mass is found it is breech.
- If hard, round, smooth ballotable mass is found, it is the head.



2. Physical examination....

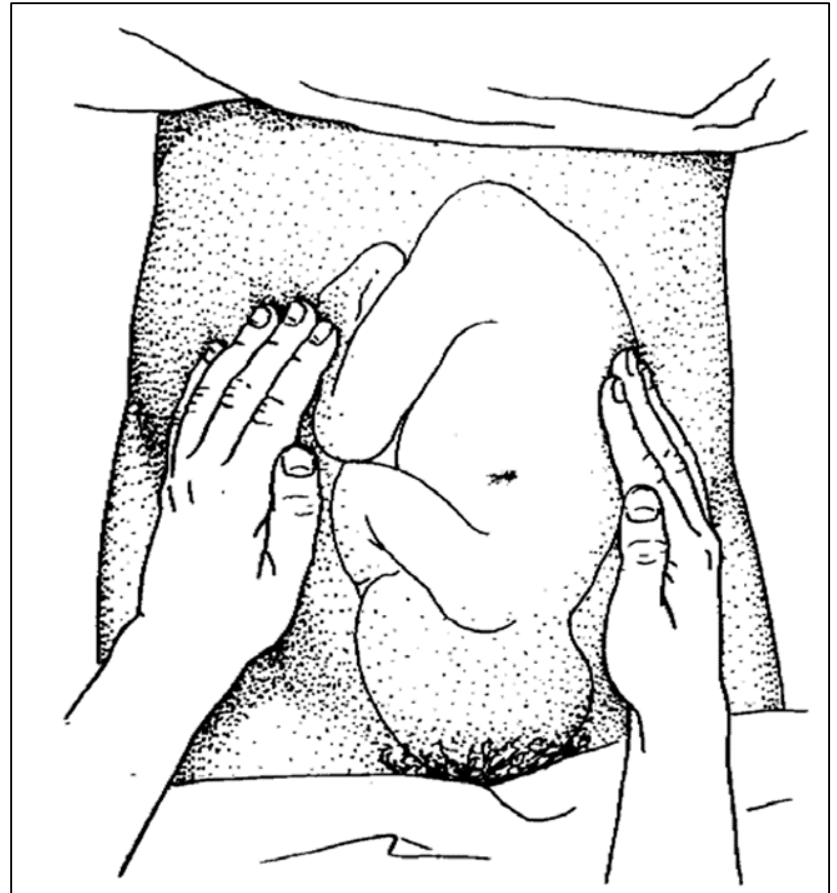
B. The 2ND Leopold maneuver or lateral palpation

- Determines the lie of the fetus which could be
 - ⇒ Longitudinal
 - ⇒ Transverse
 - ⇒ Oblique lie



2. Physical examination....

- The back of the fetus is linear, rigid and smooth in outline.
- The extrimites are felt as small irregular and bulky masses.
- The fetal heart beat is best heard on the back side.



2. Physical examination....

C. Third Leopold maneuver or pelvic palpation

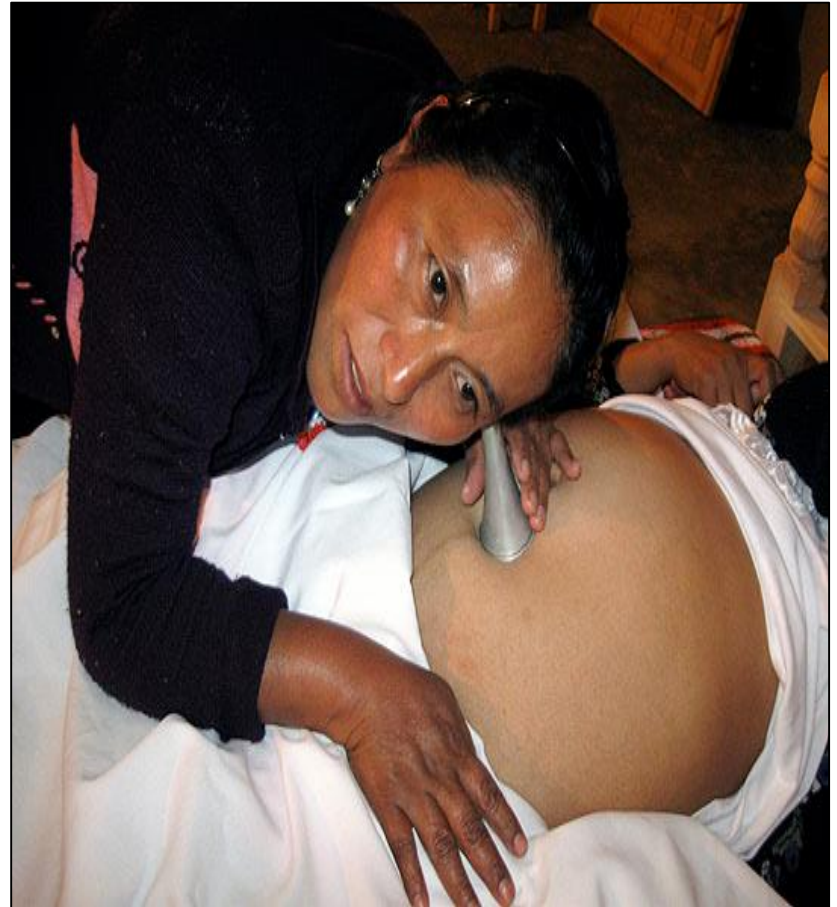
- Determine presentation.
 - ⇒ Head (cephalic)
 - ⇒ Breech
 - ⇒ Shoulder



2. Physical examination....

C. Auscultation(fetal heart beat)

- First heard in the back side at 16-18 weeks in multipara and 18-20 weeks in primigravida
 - ⇒ Breech⇒above umbilicus.
 - ⇒ Cephalic⇒ below umbilicus
 - ⇒ Occiputo posterior⇒flanks.



2. Physical examination....

5. Genitourinary System

- Costovertebral and suprapubic tenderness.
- Pelvic examination- Had better to be done with indication

2. Physical examination....

6. Integumentary system

- Check rash
- Check hemorrhage
- Check pregnancy
- Check palmar pallor in anemic mother



2. Physical examination....

8. Extremities

- Dependent edema(pretibial, ankle and pedal), seen in 80% of normal pregnancy
- Pathological edema (non dependent), involves the face, fingers or the whole body
- Check dilated vessel



2. Physical examination....

9. Central Nerves System

- As general physical examination
- Starting from consciousness
- All cranial nerves could be assessed.



History and physical examination summery

	Contents of Care		Eight schedule of ANC contacts (weeks of gestation)							
			1 st	2 nd	3 rd	4 th	5 th	6 th	7 th	8 th
			12	20	26	30	34	36	38	40
History	Date of contact									
	Gestational age									
	Present Pregnancy History (complaint)									
	Past Pregnancy History									
	Medical History and Surgical History									
	Mental Health									
	Family/Social History									
Physical Examination	General Appearance									
	Blood pressure									
	Weight									
	Pallor									
	Breast									
	Chest									
	Abdominal examination	Fundal height (wks)								
		Fetal heart beat								
		Presentation								
Pelvic assessment (as required/indicated)										
Investigations	Ultrasound (up to 24 weeks of gestation)									
	Haemoglobin									
	Blood group, RH									
	RPR/VDRL									
	HIV (PITC)									
	HBsAg									
	Urine test									
	Screening for active TB <i>for HIV positives</i>									
	Indirect coomb's test <i>for RH negatives</i>									
	75 gm oral glucose test <i>(for those at risk)</i>									

Classify the type of ANC

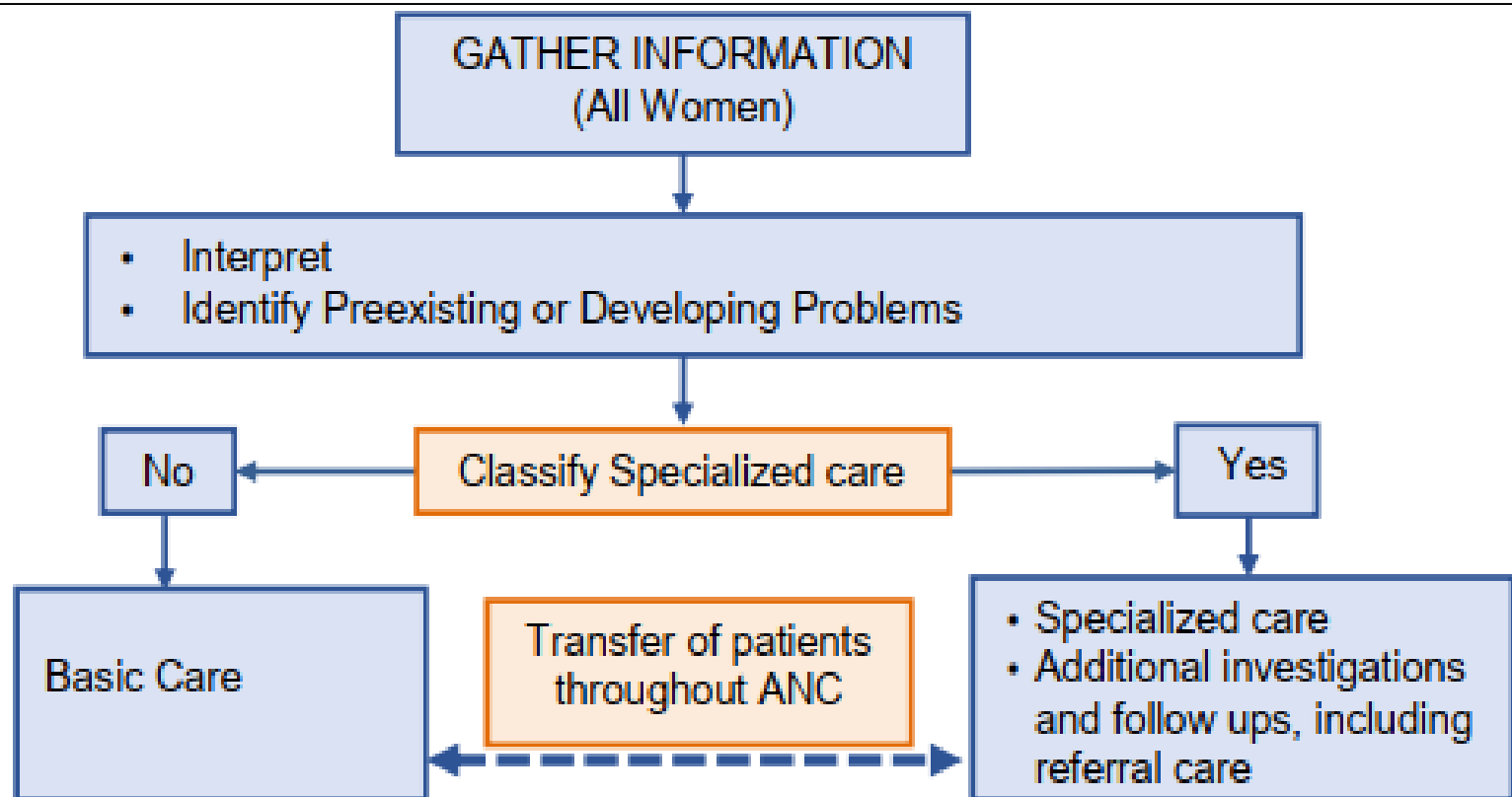


Figure 1. ANC classification diagram.

Annex 5. ANC INFORMATION GATHERING AND CLASSIFYING FORM.

Date: _____ ANC Reg. No: _____ Medical Record Number (MRN): _____

Name of Client: _____ Name of Facility: _____

Woreda: _____ Kebele: _____ House No: _____ Age (Years): _____

LMP: _____ / _____ / _____ EDD: _____ / _____ / _____

Gravida: _____ Para: _____ Number of children alive: _____ Marital Status: _____

INSTRUCTIONS to Fill Classifying form: Answer all of the following questions by placing a cross mark in the corresponding box.

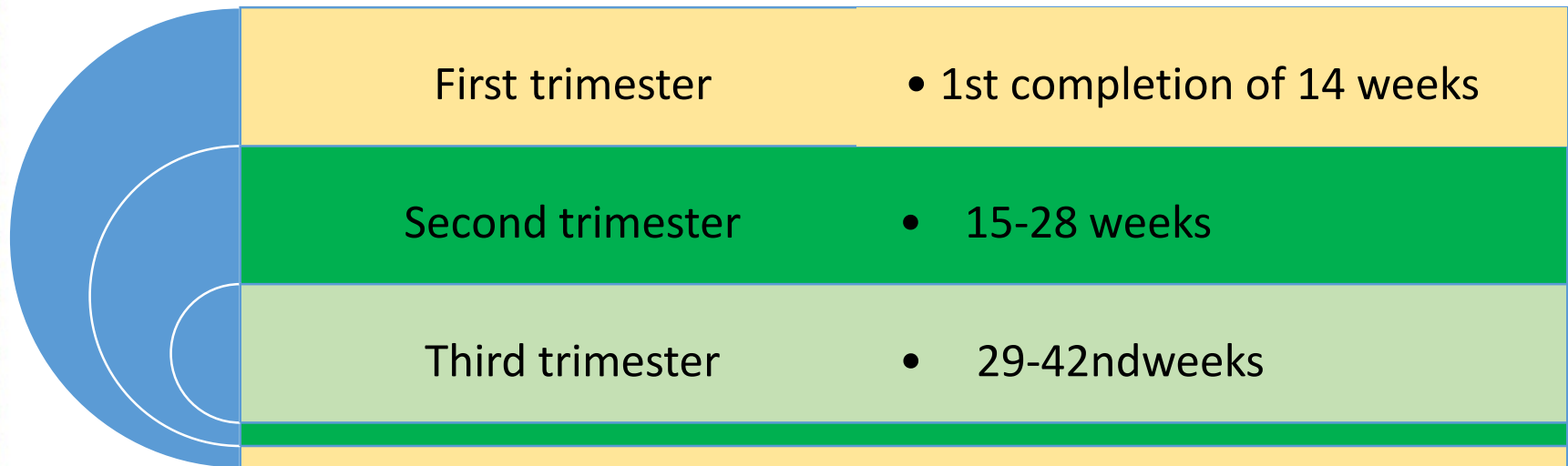
OBSTETRIC HISTORY	No	Yes
1. Previous stillbirth or neonatal loss?		
2. History of 3 or more consecutive spontaneous abortions?		
3. Birth weight of last baby > 4000g		
4. Last pregnancy: hospital admission for hypertension or pre-eclampsia/eclampsia?		
5. Previous surgery on reproductive tract? (Myomectomy, removal of septum, fistula repair, cone biopsy, CS, repaired rupture, cervical cerclage)		
CURRENT PREGNANCY	No	Yes
6. Diagnosed or suspected multiple pregnancy?		
7. Age less than 16 years?		
8. Age more than 40 years?		
9. Isoimmunization Rh (-) in current or in previous pregnancy		
10. Vaginal bleeding		
11. Pelvic mass		
12. Systolic blood pressure $\geq$ 140 mmHg and/or diastolic pressure $\geq$ 90 mmHg,		
GENERAL MEDICAL	No	Yes
13. Diabetes mellitus		
14. Renal disease		
15. Cardiac disease		
16. Chronic Hypertension		
17. Known 'substance' abuse (including heavy alcohol drinking, Smoking)		
18. Any other severe medical disease or condition TB, HIV, Ca, DVT...		

Method of estimation of gestational age

Clinical		• Size of the uterus • Palpation of fetal parts • Auscultation of FHR	
Last normal menstrual period (LNMP)		• Expected date of delivery (EDD)calculated by 1. LNMP+9 months +10 days if pagume is not passed 2. LNMP + 9 months + 5 days if pagume is passed • Calculate gestational age(GA) $GA = \frac{\text{Visiting date} - \text{LNMP}}{7}$	
Investigation records	Ultrasound	• 1st trimester	• Gestational sac • Yolk sac • Crown rump length & cardiac activity
		• 2nd & 3rd trimester	• Biparietal diameter (BPD) • Head circumference (HC) • Abdominal circumference (AC) • Femur length (FL)
	hcG	• Will be positive 8-9 days of fertilization	

Pregnancy duration

- Calculated from first day of LNMP is close to 294 days(42 weeks).
- A pregnancy is divided in to three equal trimesters



Maternal health education

Nutrition supplements

Weight gain

Immunizations

Lifestyle modification

Breast feeding

Family planning

Danger signs of pregnancy

Minor disorder in pregnancy

ADVICE DURING PREGNANCY



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Nutrition

- Dietary modifications should be started early
- 300 kcal per day increase in calories above nonpregnant levels
- Multiple gestation- 600 kcal over that of the nonpregnant woman.
- Pregnant mother needs 1000mg iron
- All fertile women to take at least 4mg of folic acid per day and 5 mg/day for risk offspring with neural tube defects
- ACOG-caffeine consumption (<200 mg/day=2cup)

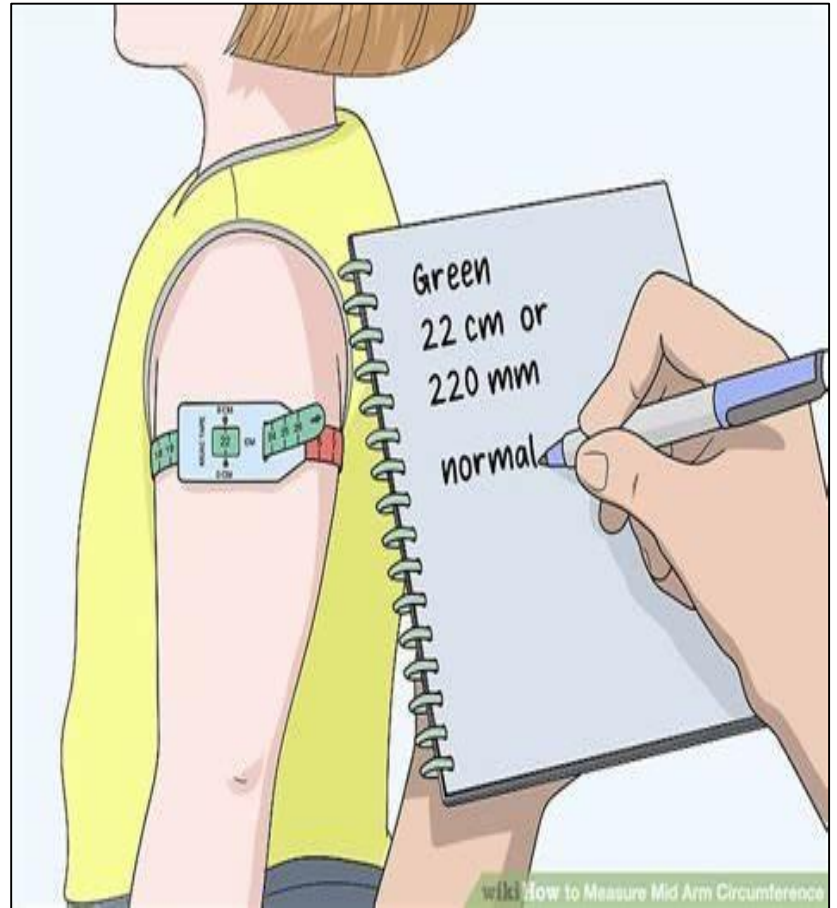
Nutrition...

- Iron requirements for pregnancy and the puerperium

	Function	Requirement
	Increased red blood cell mass	450 mg
	Fetus and placenta	360 mg
	Vaginal delivery	190 mg
	Lactation	1 mg/day

Weight gain

- The total weight gain during the course of a singleton pregnancy for a healthy woman averages 11 -12.5 kg
- MUAC should be $>23\text{cm}$ for pregnant women.



Immunizations

- Give only toxoids or killed bacteria or viruses
 - ⇒ Tetanus, diphtheria, influenza, HBV vaccine
- At least TT1 and TT2 vaccine should be offered
- Avoid attenuated live-virus vaccines (Measles, rubella vaccine)

Visit	Interval	Antigen	Protection duration
1	0 (as early as possible)	TT1	No protection
2	At least 4 weeks after TT1	TT2	Three years
3	At least 6 months after TT2	TT3	Five years
4	At least 1 year after TT3 if not pregnancy	TT4	ten years
5	At least 1 year after TT4 if not pregnancy	TT5	For thirty years (reproductive life)

Lifestyle modification

- Drug

- ⇒ Alcohol intake of any dose during pregnancy can never be safe since alcohol leads fetal alcohol syndrome
- ⇒ Heroin, cocaine, marijuana and cigarette smoking should be avoided
- ⇒ Any drug administered will cross the placenta and reach the fetus and should be avoided as much as possible.
- ⇒ Asymptomatic bacteriuria give amoxicillin 500 mg PO TID for 7 days.

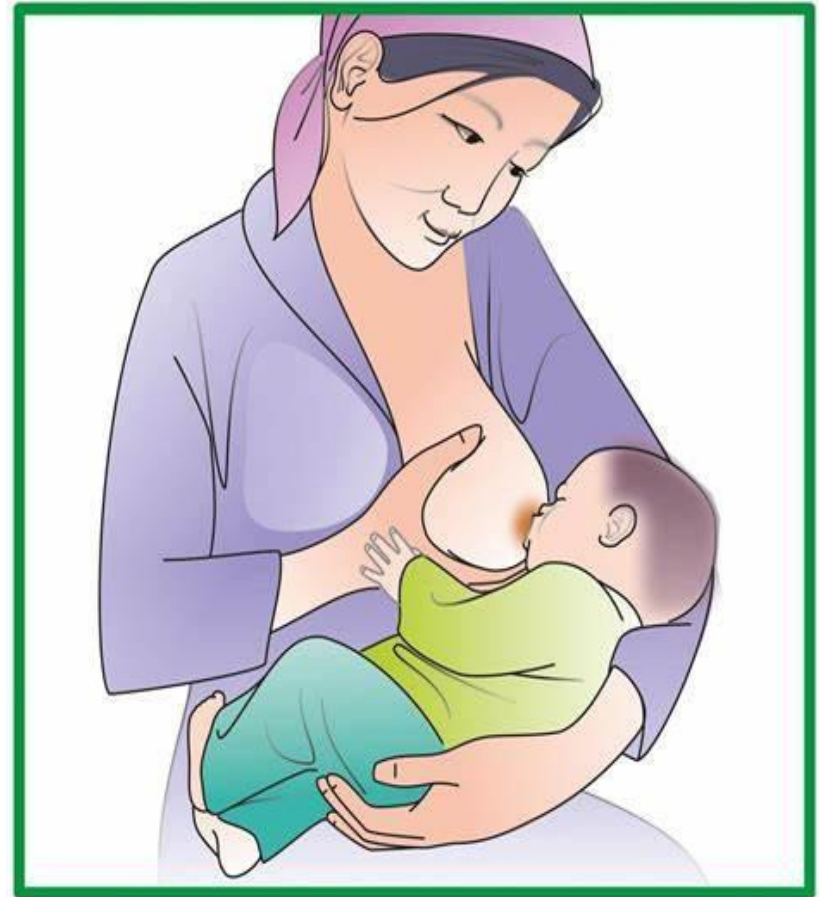
Lifestyle modification...

- Work

- ⇒ With on risk routine daily life may continue to work without interruption until the onset of labor.
- ⇒ Restrict exercise in placenta previa, cervical incompetence, hypertension, premature labor and multiple gestation
- ⇒ Travel in early and last 6 weeks of pregnancy should be restricted.
- ⇒ A pregnant woman should move around every 2 hours to prevent

Breastfeeding

- Teaching the mother the proper way of breast feeding
- Advice balanced diet
- Increase their daily caloric intake by 300 to 500 kcal above prepregnancy levels and consume 1200 mg/day of calcium.



Family planning

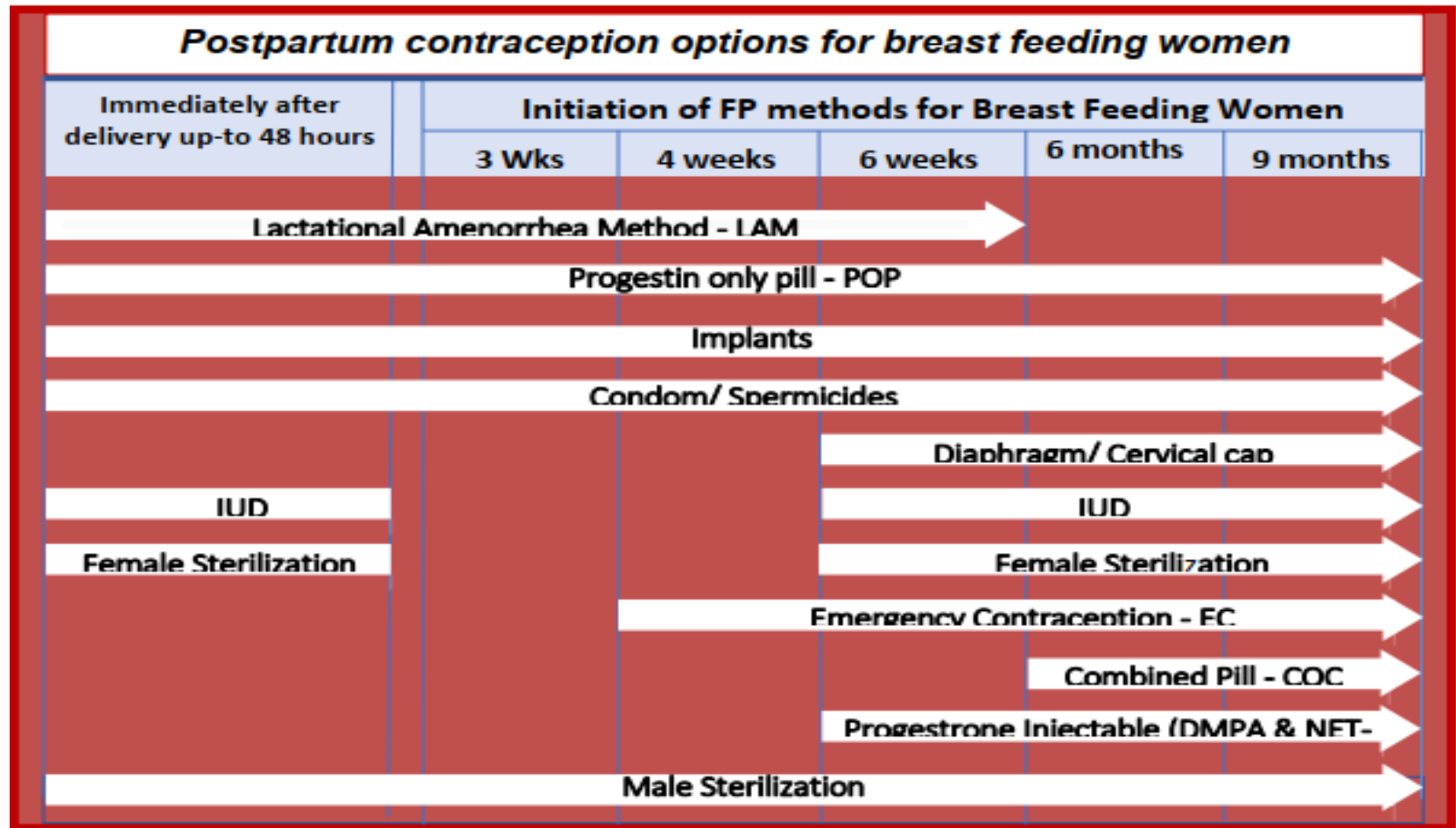


Figure 30. *Post partum contraception options for breast feeding women.*

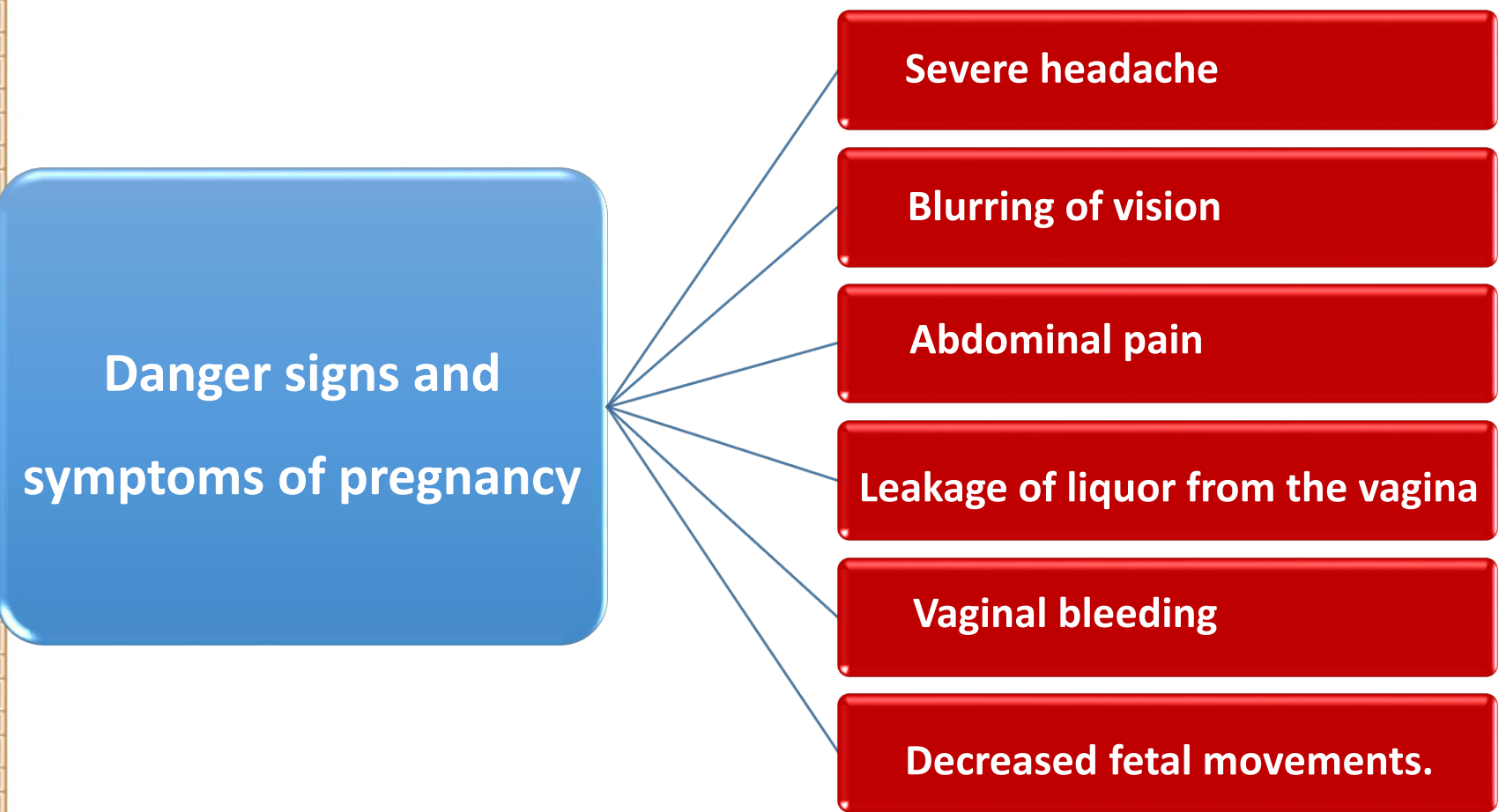
Family planning....

Postpartum contraception options for non-breastfeeding women					
Immediately after delivery up-to 48 hours	Initiation of FP methods for Non-Breastfeeding Women				
	3 Wks	4 weeks	6 weeks	6 months	9 months
	Progestin only pill - POP				
	Implants				
	Condom/ Spermicides				
			Diaphragm/ Cervical cap		
IUD			IUD		
Female Sterilization			Female Sterilization		
		Emergency Contraception - EC			
	Combined Pill - COC				
	Progestrone Injectable (DMPA & NET-ET)				
	Male Sterilization				

Figure 31. *Post partum contraception options for non-breast feeding women.*

Danger signs

Danger signs and symptoms of pregnancy



```
graph LR; A[Danger signs and symptoms of pregnancy] --- B[Severe headache]; A --- C[Blurring of vision]; A --- D[Abdominal pain]; A --- E[Leakage of liquor from the vagina]; A --- F[Vaginal bleeding]; A --- G[Decreased fetal movements.]
```

Severe headache

Blurring of vision

Abdominal pain

Leakage of liquor from the vagina

Vaginal bleeding

Decreased fetal movements.

Minor disorder in pregnancy

Minor disorder	Management
Nausea and vomiting	• Ginger, chamomile, vitamin B6
Heartburn	• Avoidance of large, fatty meals • Cessation of smoking • Raising the head of the bed to sleep • Antacid preparations
Leg cramps	• Avoid long journey and standing
Low back and pelvic pain	• Regular exercise throughout pregnancy • Physiotherapy, support
Constipation	• High fiber diet, regular bowel habit • Adequate fluid intake
Varicose veins and edema	• Compression stockings • Leg elevation and water immersion
Urinary complain	• Adequate fluid intake, do not postpones urination

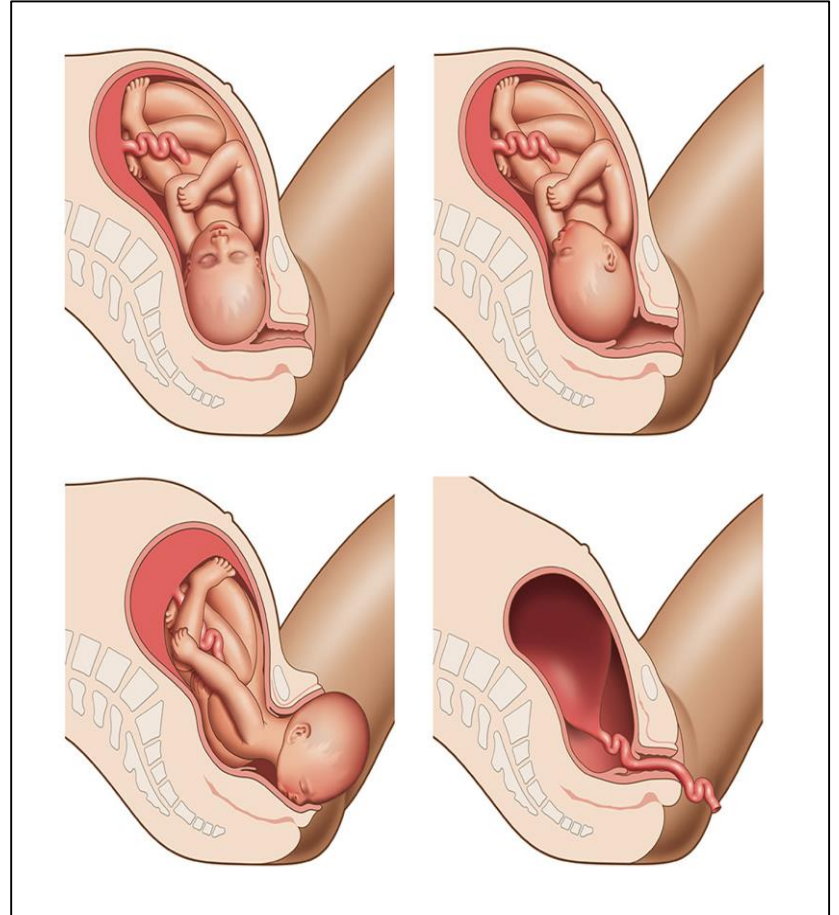
Unit-III :Physiology of normal labor

- Objective

- ⇒ Define labor and delivery
- ⇒ Explain the cause and mechanism of labor
- ⇒ Discuss on each stage of labor
- ⇒ Explain immediate new born care
- ⇒ Define episiotomy ,indication ,type and complication

Labor

- A process where regular uterine contraction results in progressive dilatation and effacement which ends in the delivery of the fetus, placenta and membranes.
- Classified:
 1. Normal
 2. Abnormal



Delivery

- A process by which the fetus, placenta and membranes are expelled through the birth canal.
- It can occur before term or at term



Normal labour

- Normal labour occurs at term and is spontaneous in onset with the fetus presenting by the vertex.
- The process should be completed with acceptable time with in 24 hours vaginally with no complications arise.
- Can be
 1. True
 2. False

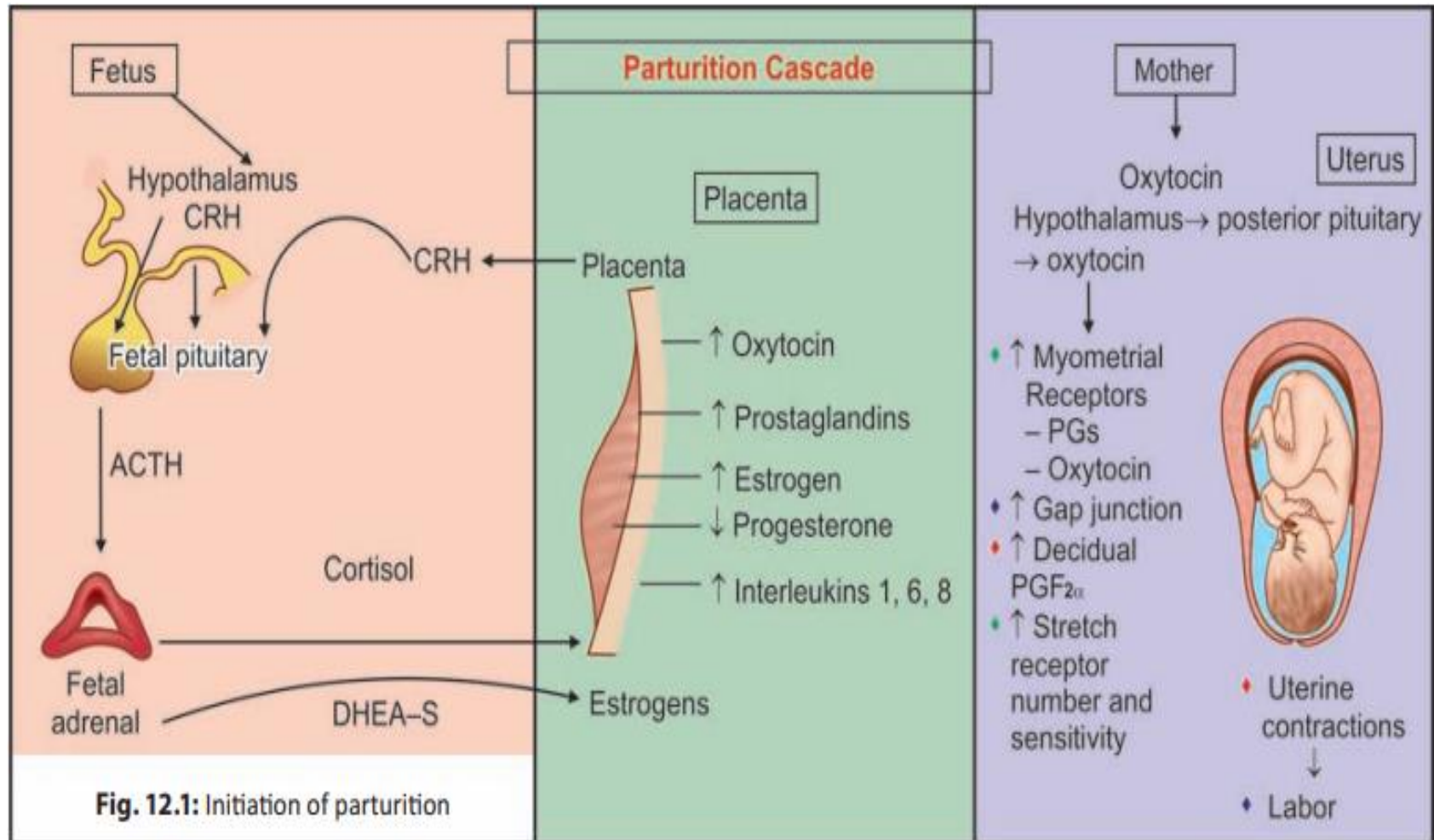
Normal labour...

True labor pain	False labor pain
• Regular contraction	• Irregular contraction
• Contraction ↑ progressively	• Contraction not progressive
• Pain over lower abdomen & back	• Lower abdomen
• Dilatation & effacement of cervix	• No effect on cervix
• Pain not relieved by sedatives & antispasmodics	• Pain relieved by analgesics
• There is show	• No show

Normal labour...

- Condition for normal labor
 - ⇒ Parturient without any risk(eclampsia...)
 - ⇒ Begins spontaneously
 - ⇒ At term(37th - 42th weeks)
 - ⇒ Vertex presentation
 - ⇒ Lasting with 24hours
 - ⇒ Fetus is alive
 - ⇒ Spontaneous delivery
 - ⇒ With out maternal& fetal complication
 - ⇒ Uncomplicated puerperium

Cause of the onset of labor



Diagnostic criteria

- Regular, rhythmic uterine contractions (≥ 2 contractions in 10 minutes) with one or more of:
 1. Rupture of the membranes.
 2. Cervical dilatation of 4 centimeters.
 3. Cervical effacement of $\geq 80\%$.
 4. Bloody show

Factors affecting labor process(4 p's)

Passage(birth canal)

Passenger(the fetus)

Power(force of contraction)

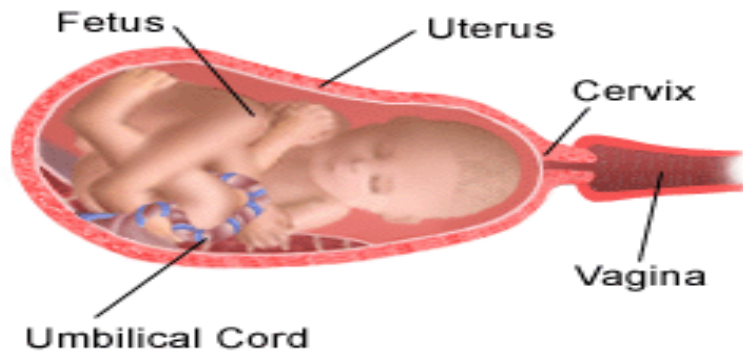
Psychosocial considerations(psyche)

Position of the mother

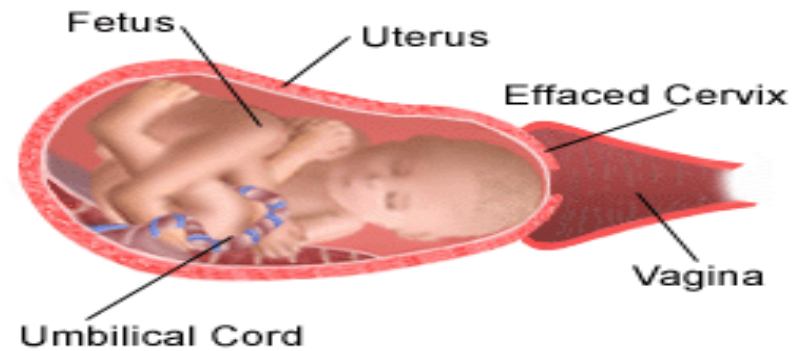
Stages of labor

Stage 1

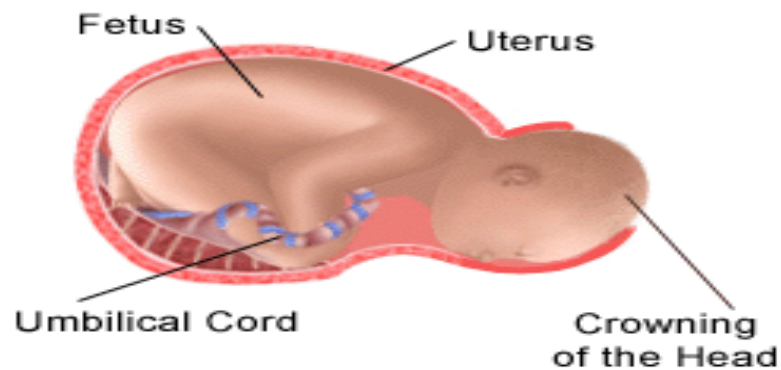
Initial (Latent) Phase



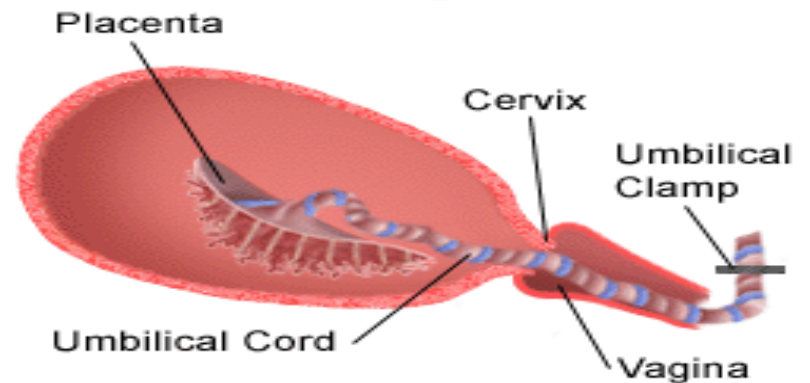
Active Phase



Stage 2



Stage 3



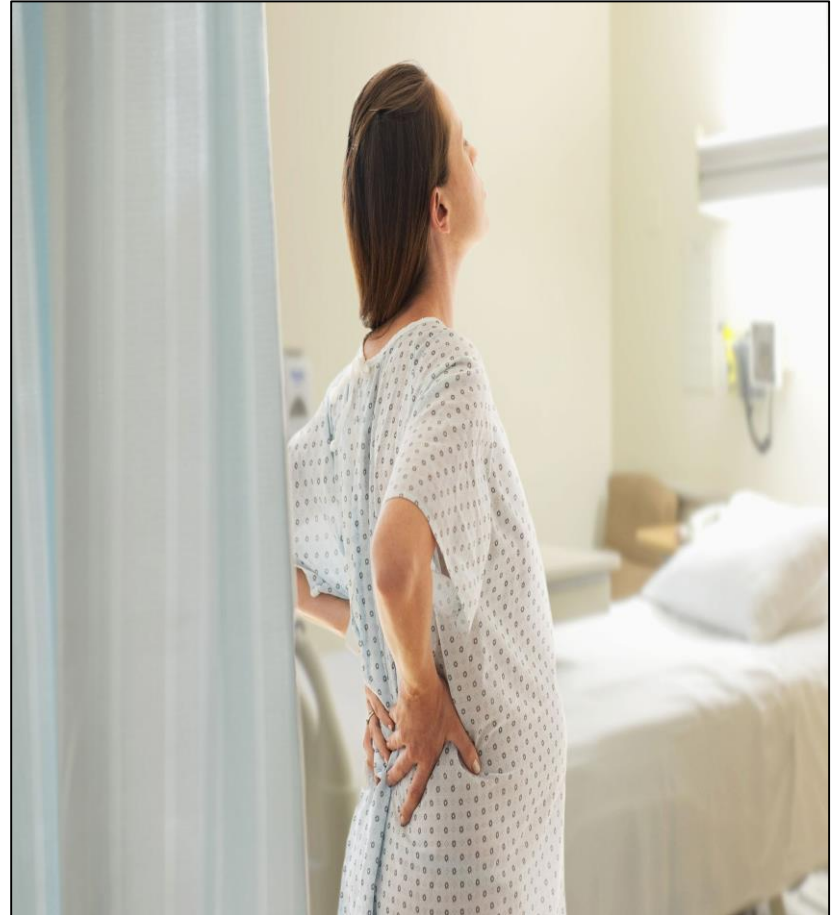
First stage of labor

- From onset of true labor to complete dilatation of cervix.
- It takes about 10-20 hours in primigravida and about 6-8 hours in multipara.
- Has two phases
 1. latent phase⇒ Onset true labor to 4 cm dilatation
 2. Active phase⇒ From 5cm-10cm cervical dilatation

First stage of labor....

A. Latent phase

- From onset of true labor to 4 cm dilatation of the cervix
- Is slow takes about 8 hours in primigravida and 4 hours in multigravida.



First stage of labor....

Name _____ Age _____ MRN _____											
Admission diagnosis _____ Status of membranes _____											
If Ruptured duration in hours _____ Admission date _____ Time _____											
Date	Time	BP	PR	RR	T ⁰	FHR	Uterine contraction	Cervical dilatation & effacement	Colour of liquor	Remark	Sign

- Uterine contraction, FHR & PR every 1 hour.
- BP every 4 hours if not indicated more frequently.
- Cervical condition every 4 hours but should be done after spontaneous rupture of membranes, in the presence of abnormal FHR or before giving analgesia.

- Latent phase of first stage of labor follow up chart.

Physiological process of labor

1. Fundal dominance
2. Contraction with retraction of uterine muscles
3. Formation of upper and lower uterine segment
4. Development of the retraction ring
5. Effacement and dilatation
6. Formation of fore water and hind water
7. Rupture of membrane
8. Show

Management of 1st stage of labor

- Is the care given through out the 1st stage of labour

A. Admission procedure

- Well coming the mother
- On arrival
 - ⇒ Greet the mother
 - ⇒ Introduce your self
 - ⇒ Inform relative to wait



Management of 1st stage of labor...

B. Admission criteria

1. Active first stage of labor
(cervical dilation is ≥ 4 cm)
2. All women in latent first stage of labor with known risk such that hypertensive ,previous scare, cardiac disease...
3. Ruptured membranes



Labor assessment

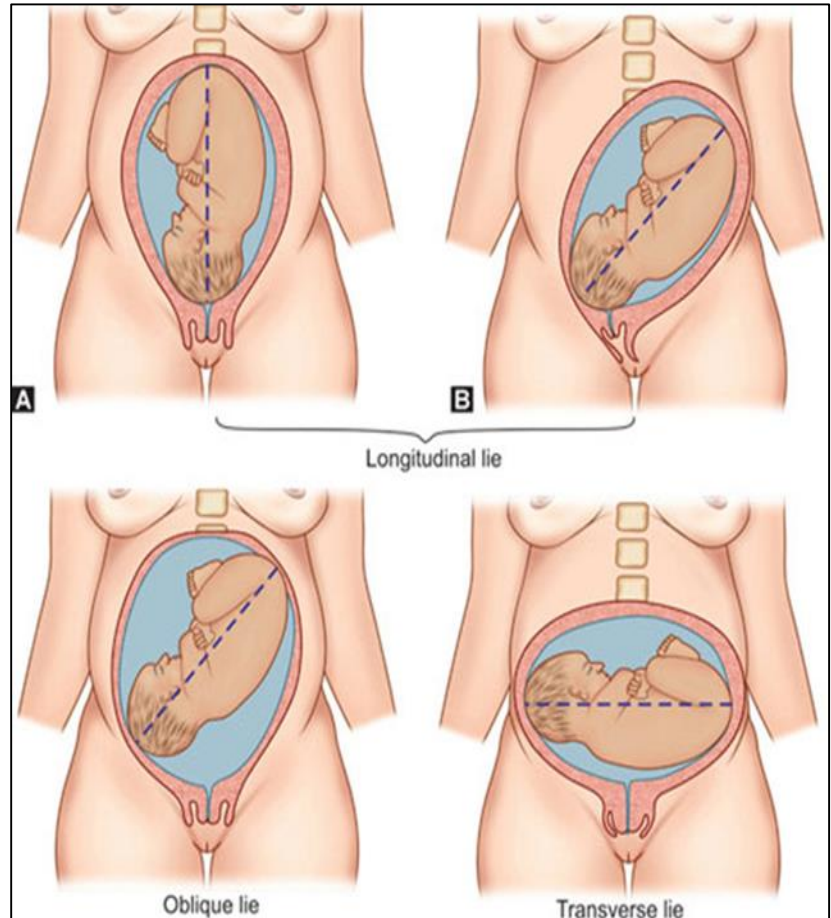
History taking	Physical examination	Investigation
• Gravidity, Parity • LMP, gestational age • History of use of OCP • On set of labour(time) • Passage of liquor • History of menstrual cycle • Previous delivery, c/s, instrumental delivery, prolonged labor • Ask If any danger sign • Blood group and Rh factor	• General appearance • Vital sign • Rapid HEENT – scleral and conjunctiva • LGS-examine breast • Assessment of chest – look for CVS & RS problems • Abdominal examination ⇒ Fundal height ⇒ Fetal presentation(lie) ⇒ Engagement ⇒ Auscultation FHB ⇒ Contraction	• Blood group and Rh • Hematocrit • Urine analysis • VDRL • HIV testing • HBS Ag • For those with ANC record, update Hematocrit and urine analysis.

Physical examination...

- Lie

⇒ Is the relationship of the long axis (spine) of the fetus to the long axis of the mother's uterus

⇒ The normal lie in longitudinal.



Physical examination...

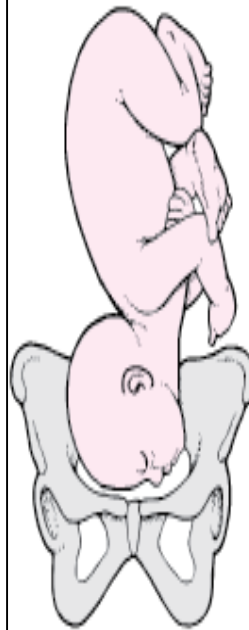
Presentation

Normal Delivery

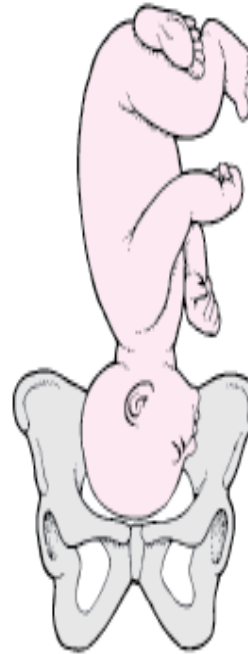


Head First Facing Backwards

Face



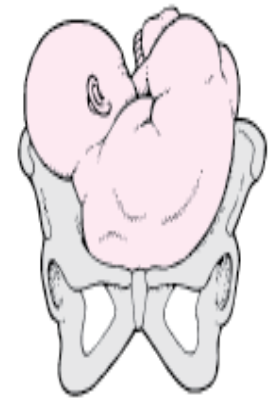
Brow



Breech



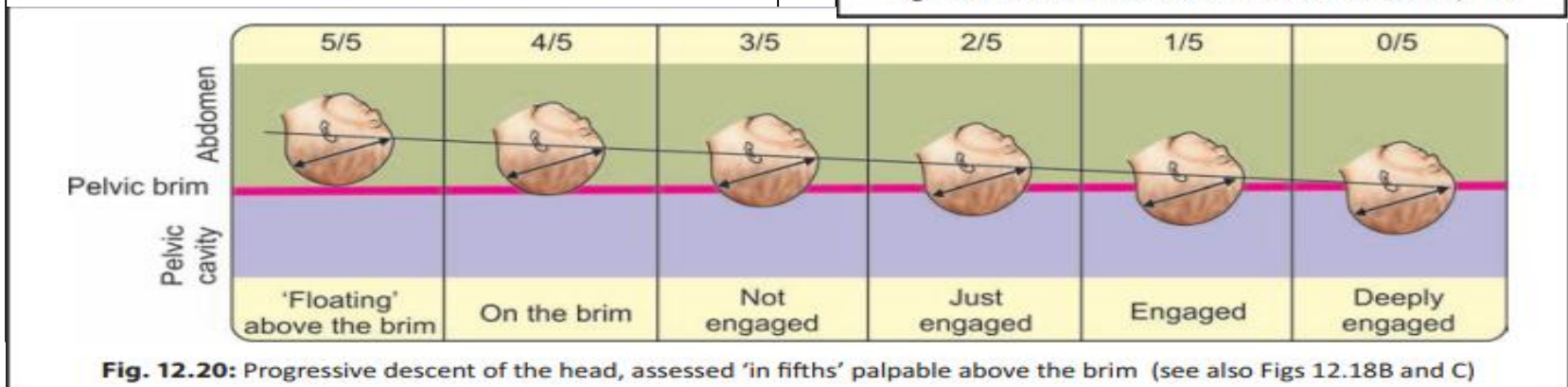
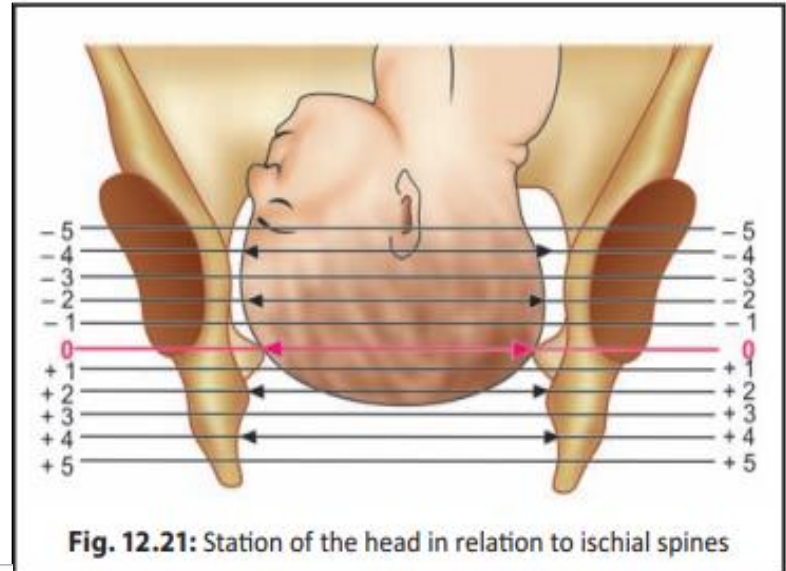
Shoulder



Abnormal Presentations

Physical examination...

- Descent
 - ⇒ Gradual movement of presenting part from engagement up to delivery of the fetus.



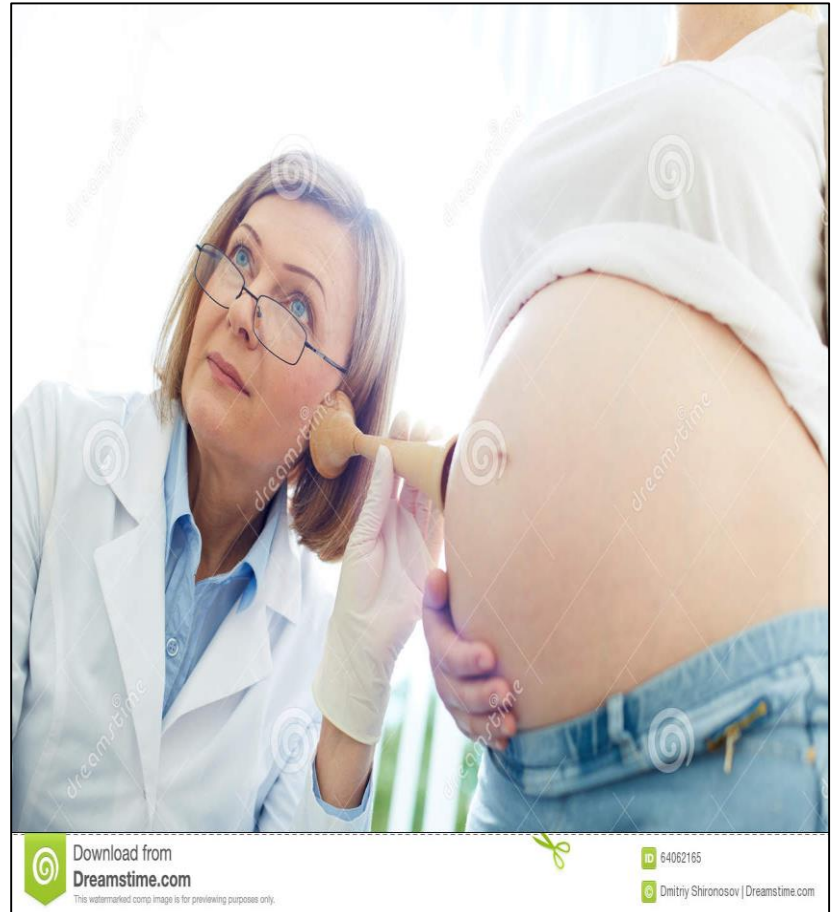
Physical examination...

- Uterine contraction
 - ⇒ Frequency in 10 minutes every 30 minutes
 - ⇒ Regularity
 - ⇒ Duration
 1. Mild(0-20 sec)
 2. Moderate(20-40 sec)
 3. Strong(40-60 sec)



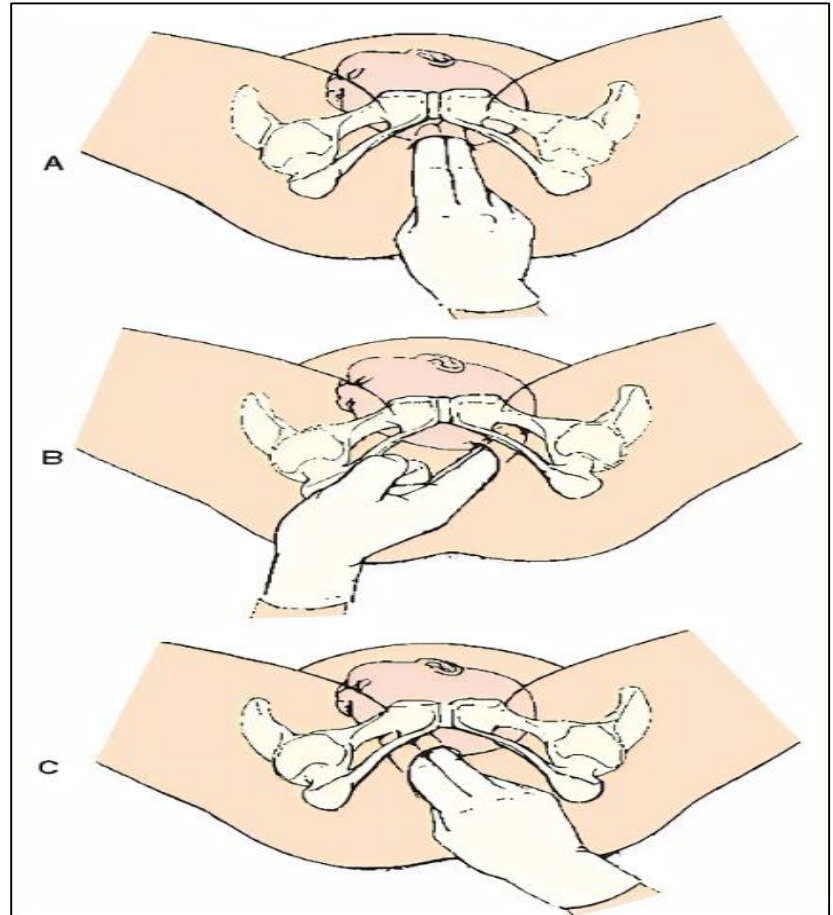
Physical examination...

- Fetal heart beat(FHB)
 - ⇒ Heard below umbilicus on cephalic presentation
 - ⇒ Heard above umbilicus on breech presentation
 - ⇒ Normal value:100-160 beat/minute.

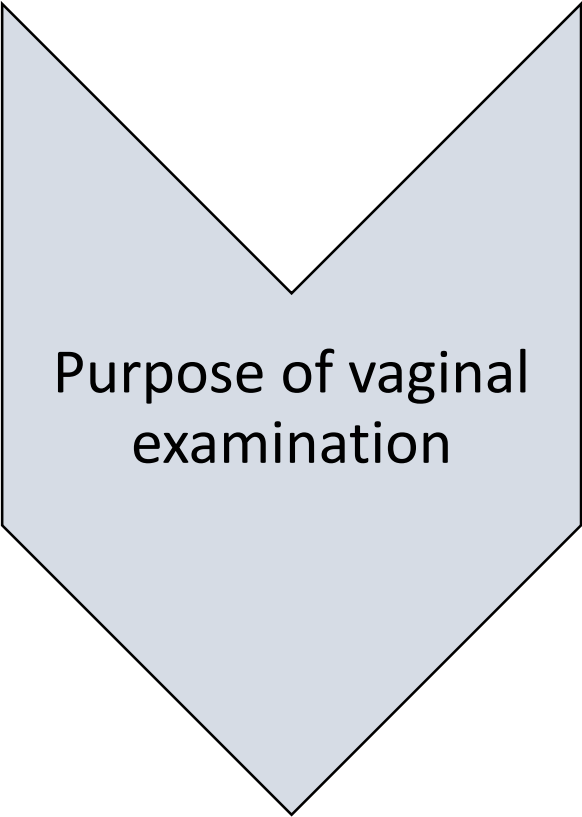


Physical examination...

- Pervaginal examination (PV)
 - ⇒ Vagina is not sterile cavity but the uterus is sterile
 - ⇒ Increase the risk of intra uterine infection if it carelessly perform



Physical examination...

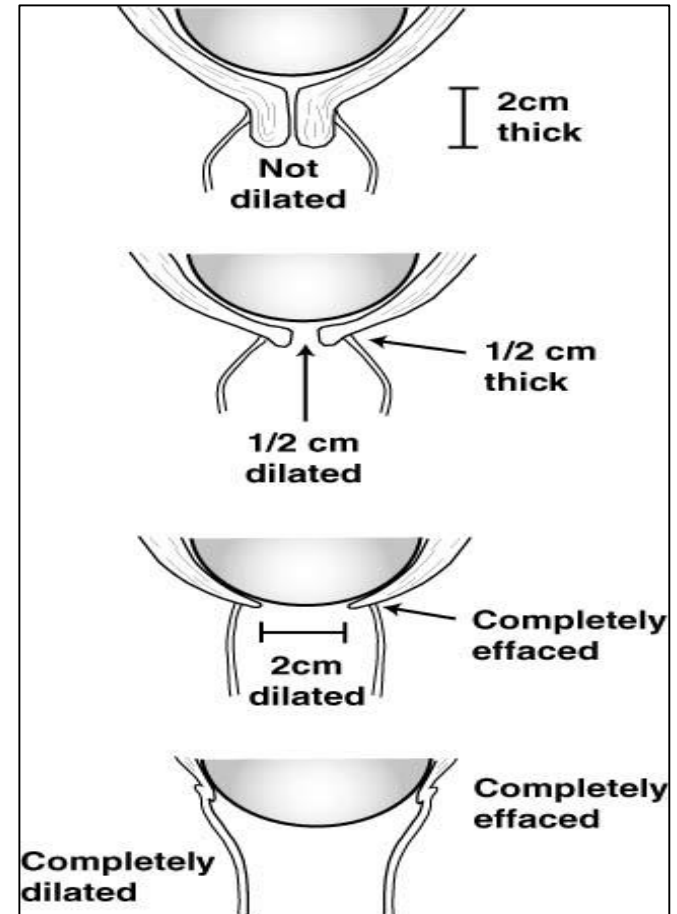


Purpose of vaginal examination

- Pelvic adequacy
- Presentation and presenting part
- Station
- Position
- Degree of molding and presence of caput
- Membrane /intact or ruptured(fetal distress)
- Cervical dilation & effacement
- Vaginal consistency, warmth

Physical examination...

- Assessment of cervical dilatation
 - ⇒ 1 finger = 1.5 cm
 - ⇒ 2 finger = 3cm
 - ⇒ 3 finger = 5-6 cm
 - ⇒ If posterior cervical rim absent = 7cm
 - ⇒ If left cervical rim absent = 8cm
 - ⇒ If right cervical rim absent = 9
 - ⇒ If no cervical rim at all = 10cm



Physical examination...

- Cervical effacement:
 - ⇒ Shortening & thinning of the cervix during labor.
 - ⇒ Can be 0 – 100%
 - 1. No any shortening and softening of the cervix ⇒ 0
 - 2. Cervix absent and totally in to lower uterine segment ⇒ 100

Cervical Effacement & Dilation (Borramiento y Dilatación Cervical)

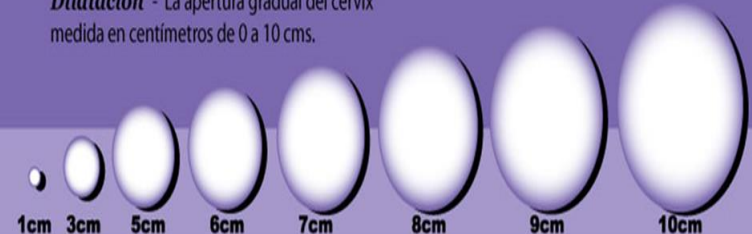
Effacement - the gradual thinning, shortening and drawing up of the cervix measured in percentages from 0 to 100%.

Borramiento - el adelgazamiento, acortamiento y encogimiento gradual del cervix medido en porcentajes del 0 al 100 %.



Dilation - the gradual opening of the cervix measured in centimeters from 0 to 10 cms.

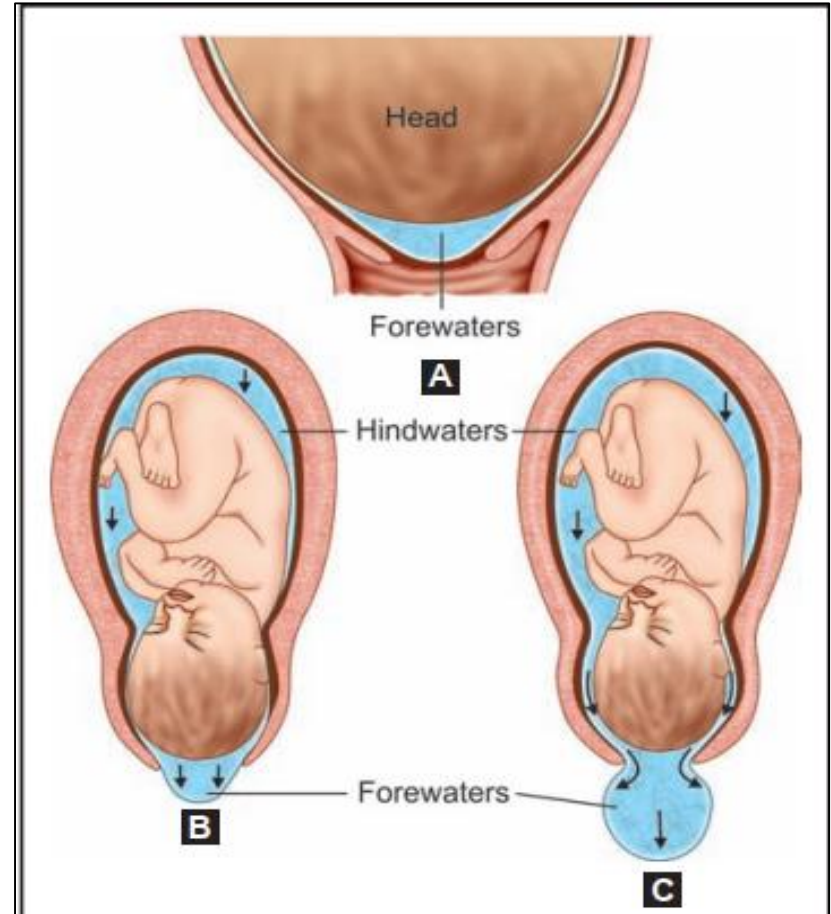
Dilatación - La apertura gradual del cérvix medida en centímetros de 0 a 10 cms.



First stage of labor....

B. Active phase

- From 4-10 cm/full dilatation of the cervix
- It is time of partograph



Partograph

- Is the graphic recording of the progress of labor and the salient condition of the mother and the fetus.
- Use of partograph:
 1. Assessment of fetal well being
 2. Assessment of maternal well being
 3. Assessment of progress of labor

Partograph...

- Components

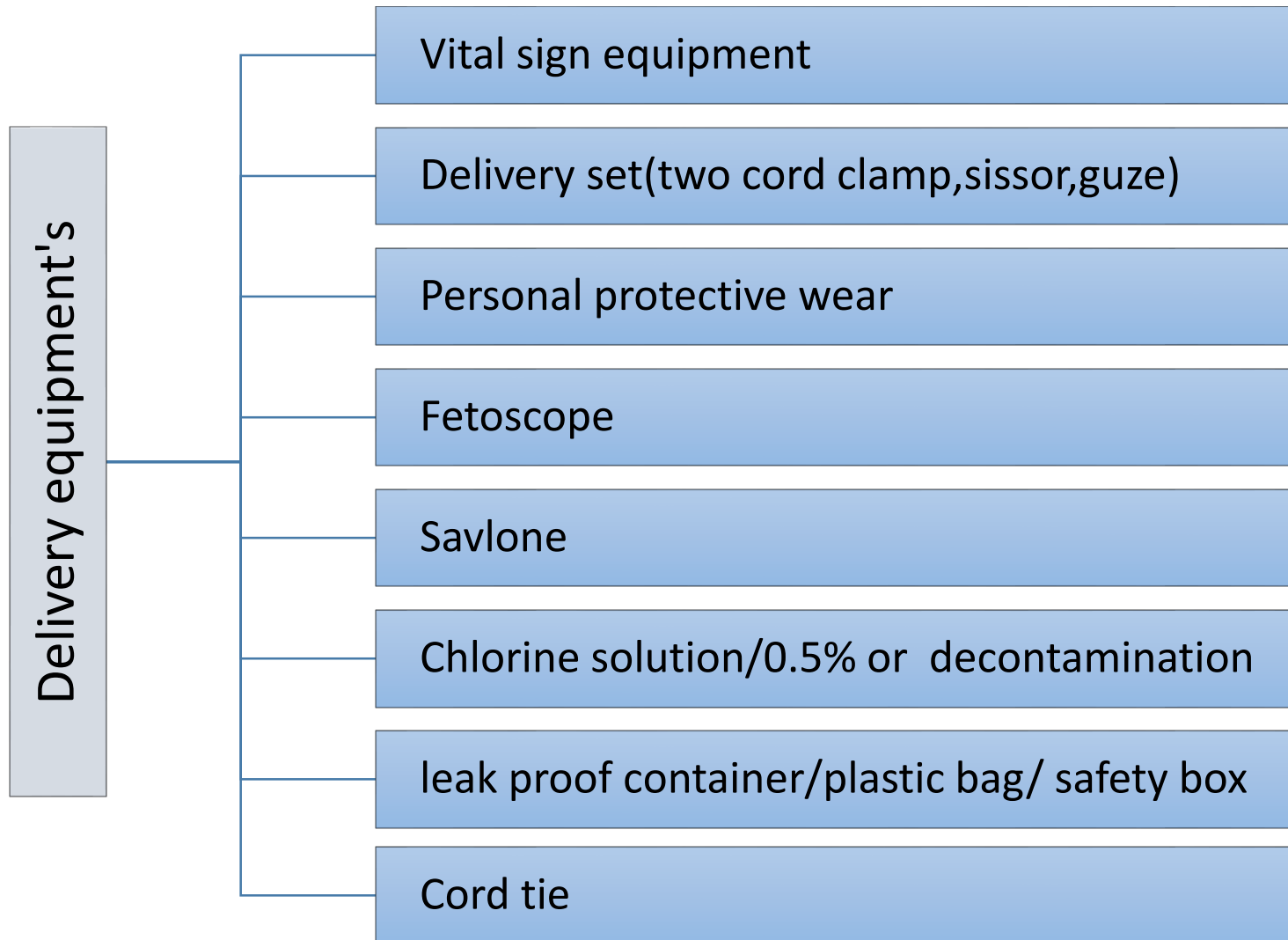
1. Fetal component
2. Maternal component
3. Labor component
4. Drugs/ medical components

PARTOGRAPH																									
Name		Gravida						Para						Hospital no.											
Date of admission				Time of admission								Ruptured membranes												hours	
Fetal heart rate		180 170 160 150 140 130 120 110 100																							
Liquor Moulding		10 9 8 7 6 5 4 3 2 1 0																							
Cervix (cm) [Plot X]		10 9 8 7 6 5 4 3 2 1 0																							
Descent of head [Plot O]		10 9 8 7 6 5 4 3 2 1 0																							
Time		Hours 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24																							
Contractions per 10 mins		5 4 3 2 1 0																							
Oxytocin U/L drops/min		5 4 3 2 1 0																							
Drugs given and IV fluids																									
Pulse and BP		180 170 160 150 140 130 120 110 100 90 80 70 60																							
Temp °C		40 39 38 37 36 35 34 33 32 31 30																							
Urine { protein, acetone, volume																									

Monitoring in active first stage of labor

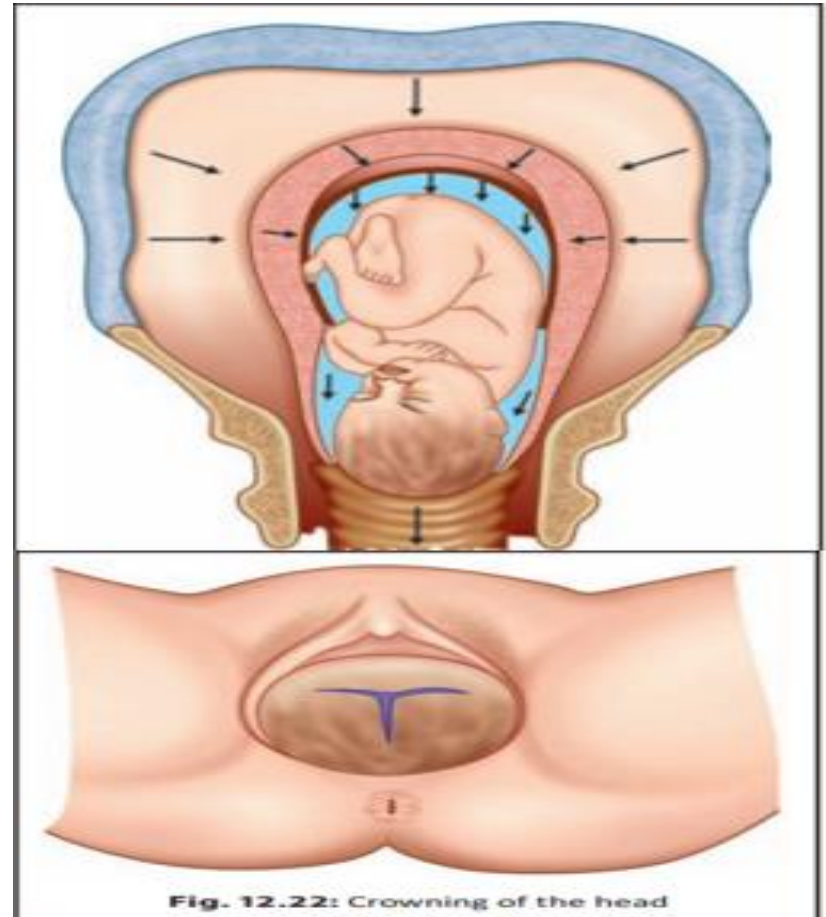
Fetal well -being	Progress of labor	Maternal wellbeing
• FHR(100-160BPM) ⇒ Every 30 min for low-risk ⇒ Every 15 min for high risk • Status of liquor (Grading of meconium) • Grading of Molding	• Uterine contraction every 30 minutes • Descent of fetal head: • Cervical Dilatation ⇒ Every 4 hours	• Vital signs: ⇒ PR - half hourly (30'). ⇒ Temperature and BP - every 4 hourly • Monitor urine output. • Maternal position • Nutrition- oral intake: • Companionship in labor: • Pain management:

Prepare the necessary equipment's



Second stage of labor

- From 10cm of cervical dilatation to delivery of fetus
- Lasts about 2 hours
- Signs of second stage:
 - ⇒ No cervix felt around head
 - ⇒ Much stronger contractions
 - ⇒ Urge to push
 - ⇒ Head crown



Mechanism of the second stage

- Is the series of movements of the fetus in its passage through the birth canal.
- Described in relation to a vertex presentation.
- The changes in position of fetal head during its passage through birth canal

Mechanism of the second stage

Cardinal movement

Engagement



Descent



Flexion



Internal rotation

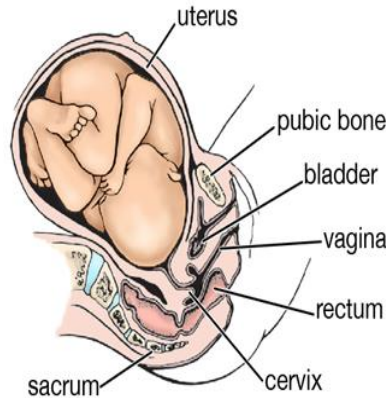


Extension



External rotation or restitution

Onset of labour



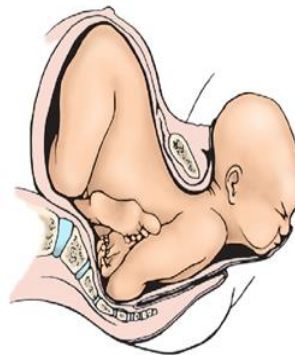
Flexion



Internal rotation of head



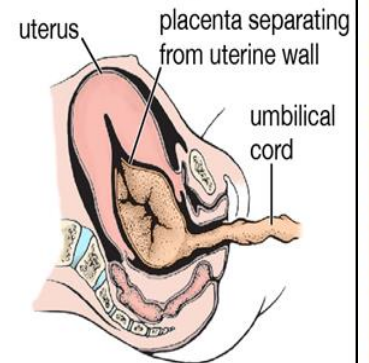
Extension



External rotation of head



Uterus immediately after birth



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Management of the second stage

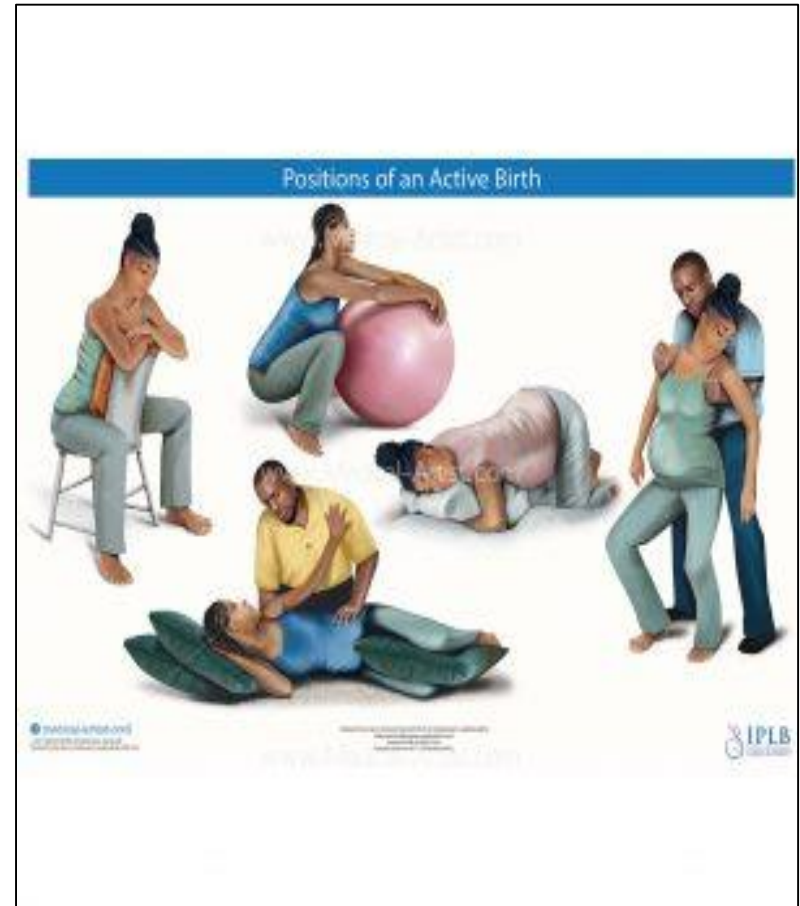
- Maternal

- ⇒ General condition: fatigue, pain, and state of hydration.
- ⇒ BP monitoring: every 1hour (if indicated more frequently)
- ⇒ PR, temperature, and RR: every 30 minutes.
- ⇒ Evaluate for the presence of the urge to push and /or effort.
- ⇒ Position the mother(Left lateral position)
- ⇒ Encouraged to empty her bladder before delivery.

Position in labor and childbirth

- Encourage any non-supine position:

- ⇒ Side lying
- ⇒ Squatting
- ⇒ Hands and knees
- ⇒ Semi-sitting
- ⇒ Sitting



Management of the 2nd stage...

• Pain control	
Psycho-prophylaxis	Pharmacological
• Breathing and relaxation techniques • Warm showers or baths • Massage • Warm or cold compresses • Changing positions • Listening to music	• Analgesics • Epidural anesthesia • Spinal; anesthesia

Management of the second stage...

- Fetal and labor
 - ⇒ Every 15 min for low-risk pregnancy.
 - ⇒ Every 5 min for high-risk pregnancy
 - ⇒ Evaluate the status of liquor
 - ⇒ Evaluate the degree of descent and / or station every 1 hour.
 - ⇒ Look for extent of caput and degree of molding

Preparation for delivery

- General

- ⇒ Notify staff that delivery is imminent.
- ⇒ Move the woman to the delivery room
- ⇒ Ready delivery and newborn care equipment
- ⇒ Pre-warmed neonatal corner for neonatal care
- ⇒ Position the mother to semi-sitting (back up and leg down)
- ⇒ Dressed and gloved appropriately

Assistance of spontaneous deliver

- Goal

- ⇒ Reduction of maternal trauma
- ⇒ Prevention of fetal injury
- ⇒ Initial support of the newborn



Assistance of spontaneous deliver...

- Delivery of the head
 - ⇒ Assist extension of the head.
 - ⇒ Check for cord around the neck disentangle it from around the head.
 - ⇒ After delivery of the head: wipe the mouth, oro-pharynx first, nose and eye.

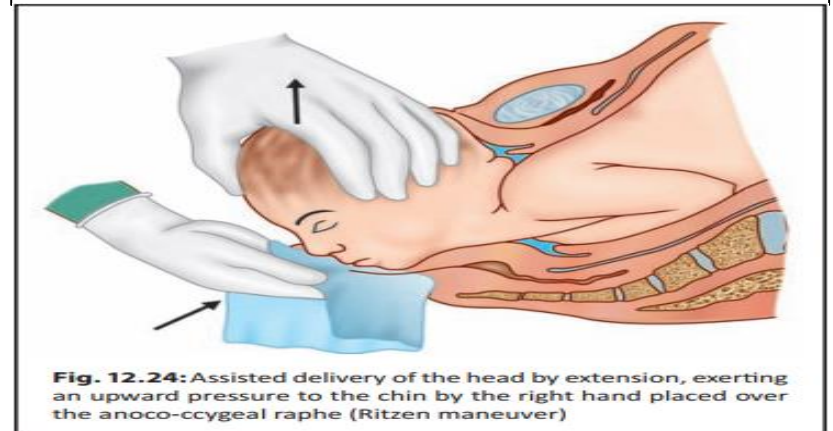
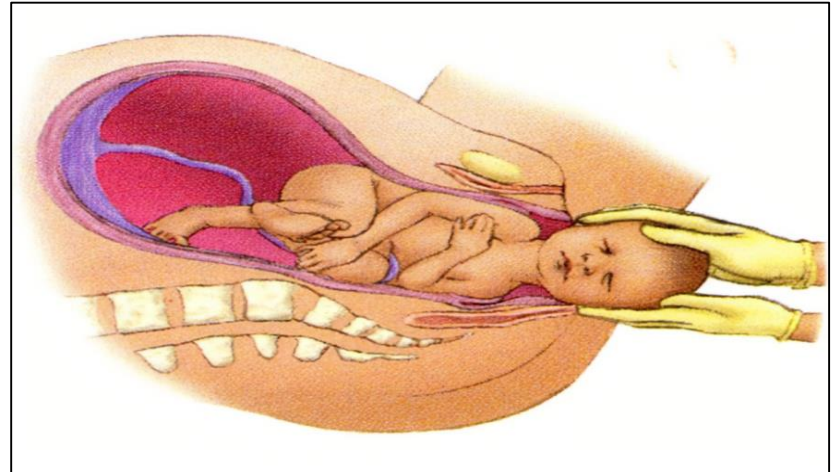
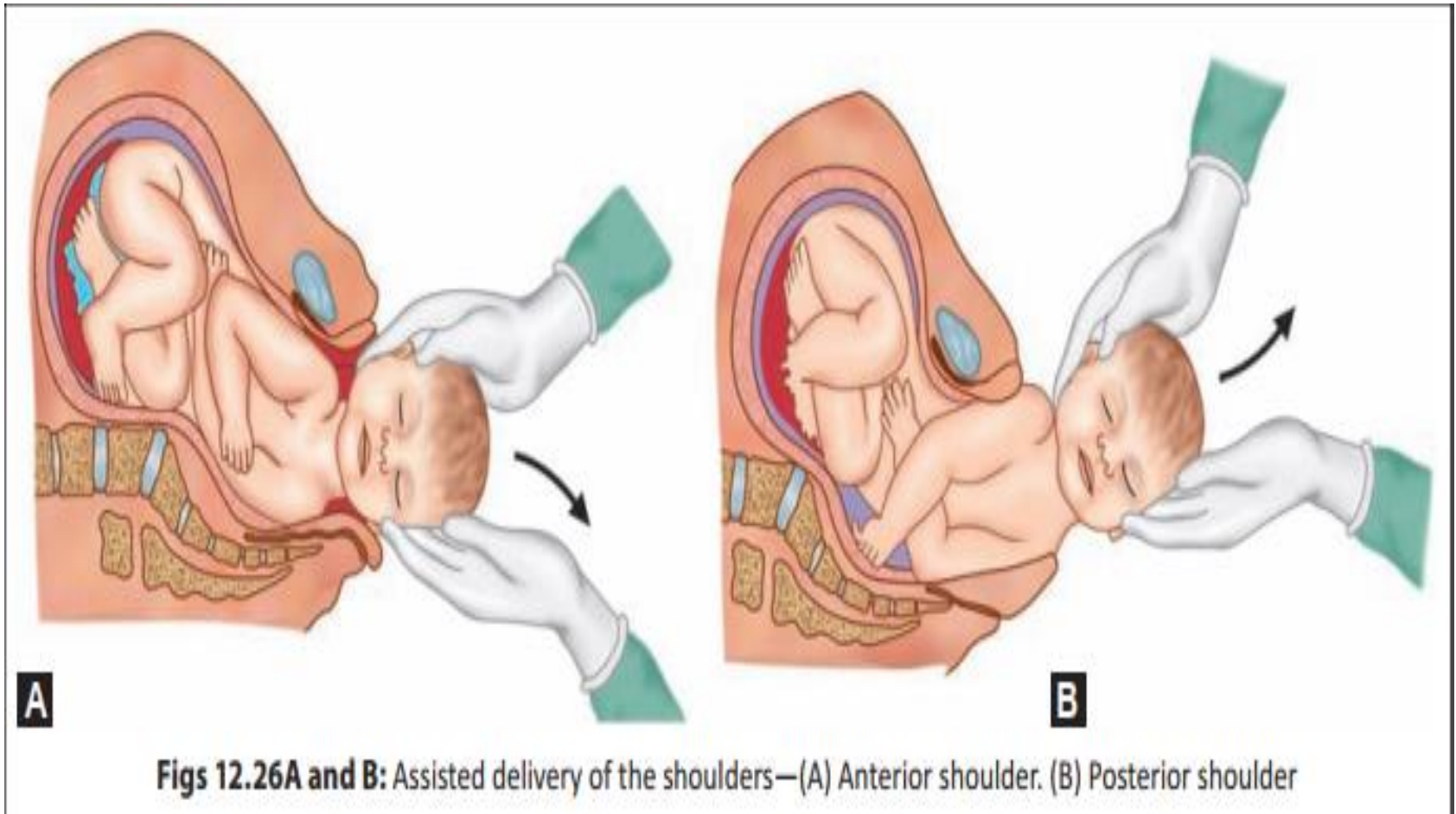


Fig. 12.24: Assisted delivery of the head by extension, exerting an upward pressure to the chin by the right hand placed over the anoco-cygeal raphe (Ritzen maneuver)

Assistance of spontaneous deliver...



Assistance of spontaneous deliver...

- Complete delivery of the rest of the body
- ⇒ After securing complete delivery, wipe the newborn's body dry with clean towels ,remove the wet towel and wrap them with a dry towel.

Stage 2:
Birth

① Presentation of head



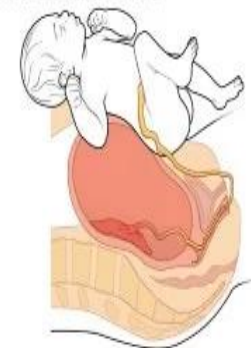
② Rotation and delivery of anterior shoulder



③ Delivery of posterior shoulder

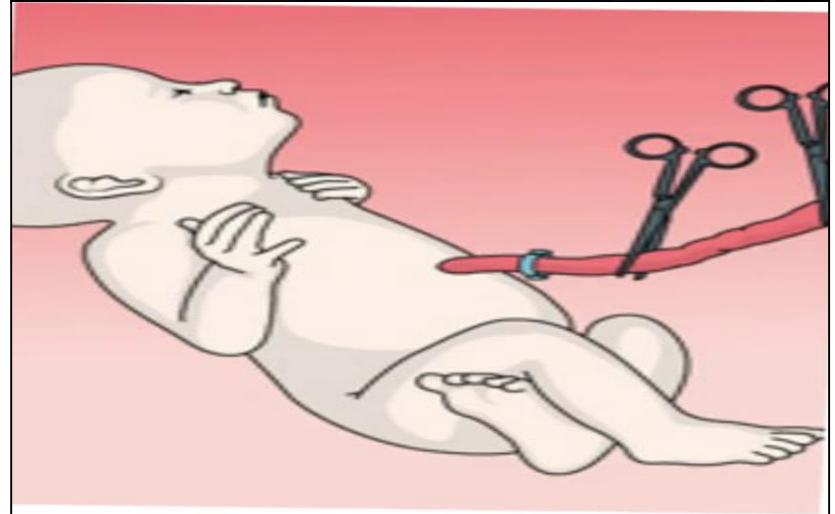


④ Delivery of lower body and umbilical cord



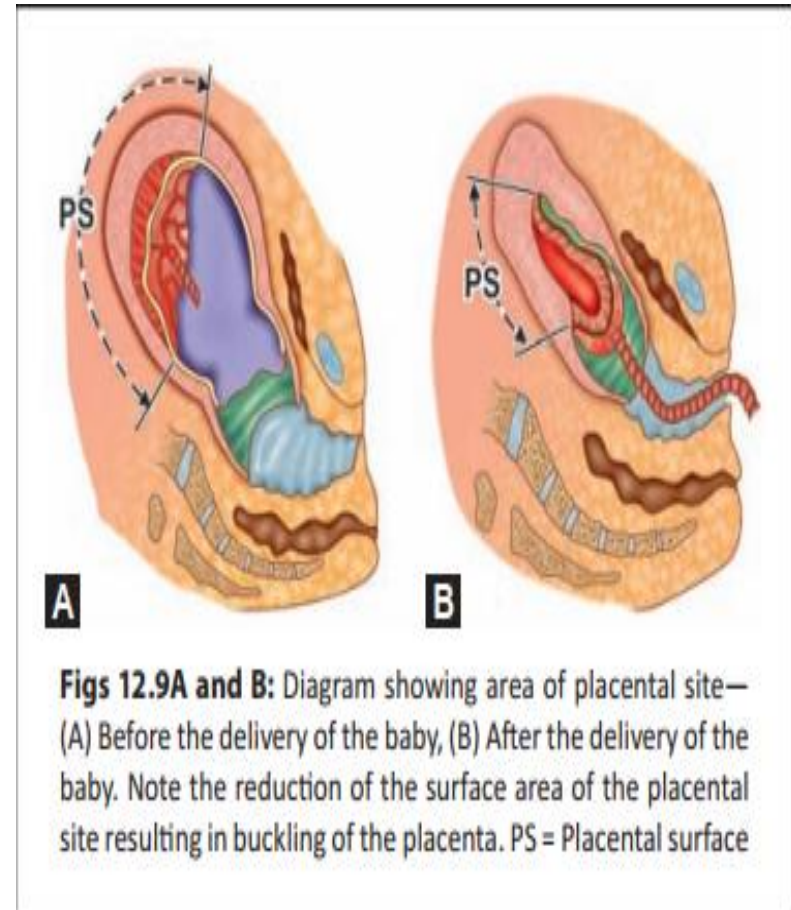
Assistance of spontaneous deliver...

- Put newborn on mother abdomen.
- Clamp after 1-3 minute or when pulsation is stop.
- Cord clamping:- 4-5 cm from fetal umbilicus
- Take cord blood if indicated for Rh negative mother



Third stage

- It is the time interval between the deliveries of the fetus up to the expulsion of the placenta
- It lasts 30 mints.



Third stage...

- Benefit of active management of third stage of labor component (AMTSL)
 - ⇒ Duration of third stage of labor will be short
 - ⇒ Less maternal blood loss
 - ⇒ Less need for oxytocin in post partum
 - ⇒ Less anemia in the post partum
- AMTSL should be offered to all women who are giving birth.

Third stage...

Active management of third stage of labor component

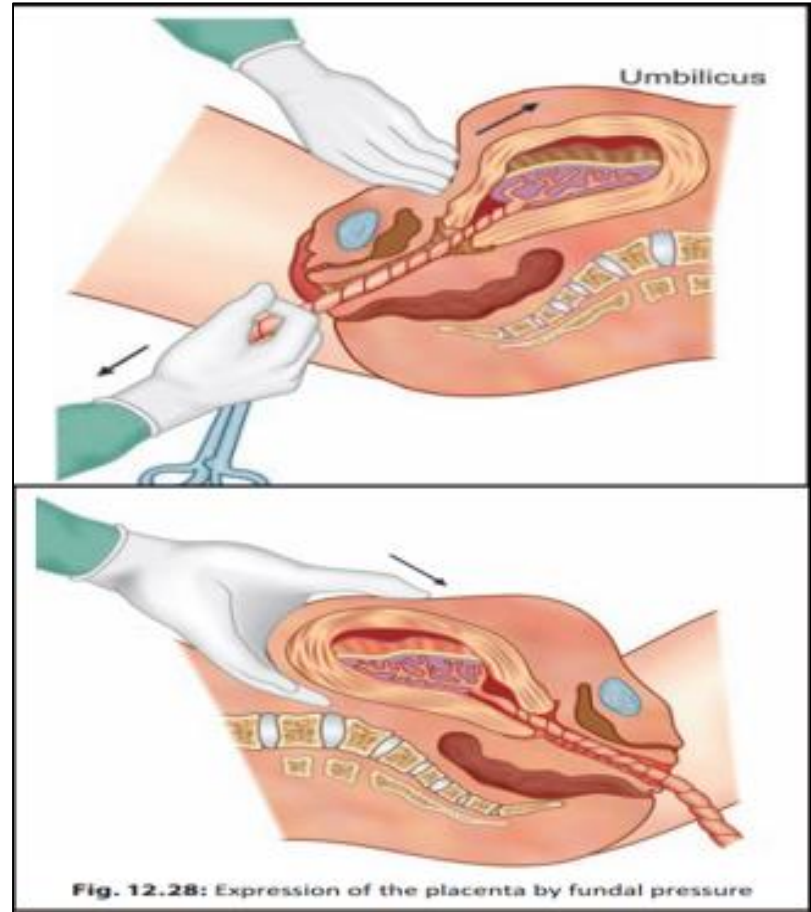
Inject uterotonic agents (oxytocin 10IM 1m)



Controlled cord traction



Uterine massage (after the delivery of the placenta)



Third stage...

- Drugs used for AMTSL
 - ⇒ Oxytocin 10 IU injected deep into thigh muscles IM
 - ⇒ Carebetocin 100 micrograms IV or IM, or
 - ⇒ Ergometrine 0.2 mg IM within 1 minute of the delivery or
 - ⇒ Misoprostol 400-600 mcg sublingual / oral / rectal

Immediate newborn care

- Normal newborn
 - ⇒ Born at term
 - ⇒ Weight 2500-4000gm
 - ⇒ Cries immediately
 - ⇒ Normal vital sign
 - ⇒ Adopt environment



APGAR score

Sign	0	1	2
Heart rate	Absent	<100BPM	>100BPM
Reflex irritability	No response	Facial grimace	Sneeze, cough
Respiratory effort	Apneic	Weak, irregular gasping	Regular
Muscle tone	Flaccid	Some flexion of arms and legs	Good flexion
Color	Blue,pale hands and blue feet	Body pink	Pink

- Total Babies with a score of
 1. 0-3 out of ten-severely depressed:
 2. Moderately depressed:4-6 out of ten
 3. Normal:>7-10

Steps of Essential Newborn Care.

Step 1: Deliver baby on to mother's abdomen or into her arms



Step 2: Dry baby's body with dry towel. Wipe eyes. Wrap with another dry one and cover head



Step 3: Assess breathing and color. If <30 breaths per minute, blue tongue, lips or trunk or if gasping then start resuscitating



Step 4: Tie the cord two fingers from abdomen and another tie two fingers from the 1st one (if no clamp). Cut the cord between the 1st and 2nd tie (clamp)



Step 5: Place the baby in skin-to-skin contact and, on the breast, to initiate breastfeeding.

Step 6: Apply Tetracycline eye ointment



Step 7: Give Vitamin K, 1mg IM on anterior mid-thigh



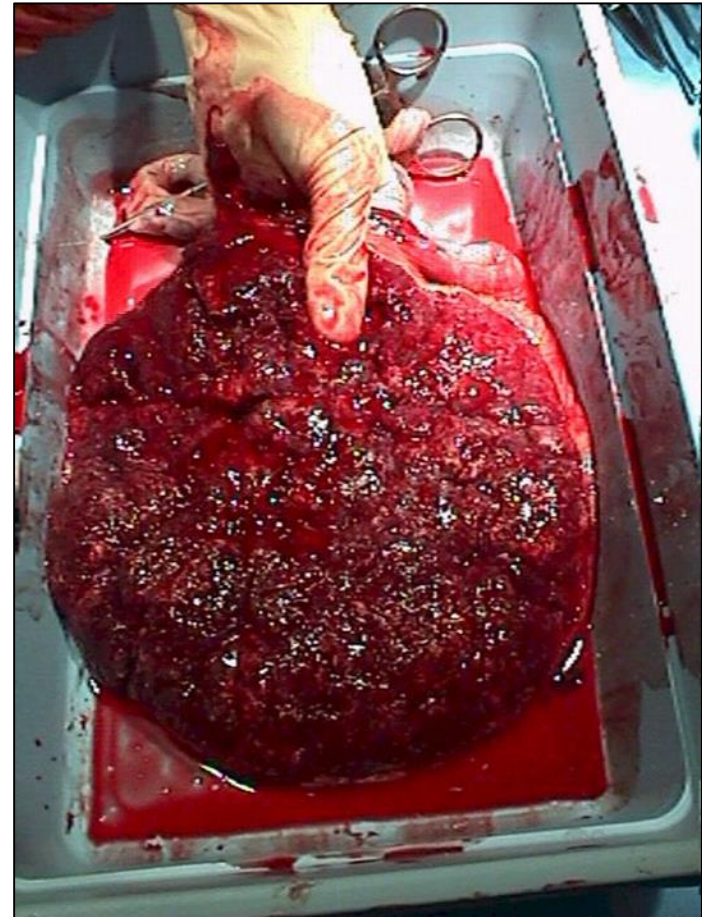
Step 8: Place the newborn's identification bands on the wrist and ankle



Step 9: Weigh baby (if <1500 gm, refer urgently)

Examination of placenta

- Check
 - ⇒ Membrane
 - ⇒ Completeness of lobes
 - ⇒ Umbilical artery and veins



Episiotomy

- A surgically planned incision on the perineum and the posterior vaginal wall during the second stage of labor.
- Designed to enlarge the vulva out let during delivery to prevent un wanted perineal tear.

Episiotomy....

- Indications of episiotomy
 - ⇒ To speed delivery if fetal distress
 - ⇒ Prior to instrumental (assisted) delivery; forceps, vacuum
 - ⇒ To minimize the risk of intracranial damage(preterm)
 - ⇒ To prevent delay due to rigid perineum
 - ⇒ To prevent tears in tight perineum

Types of episiotomy

Medio- lateral

Median

J- shaped

Lateral

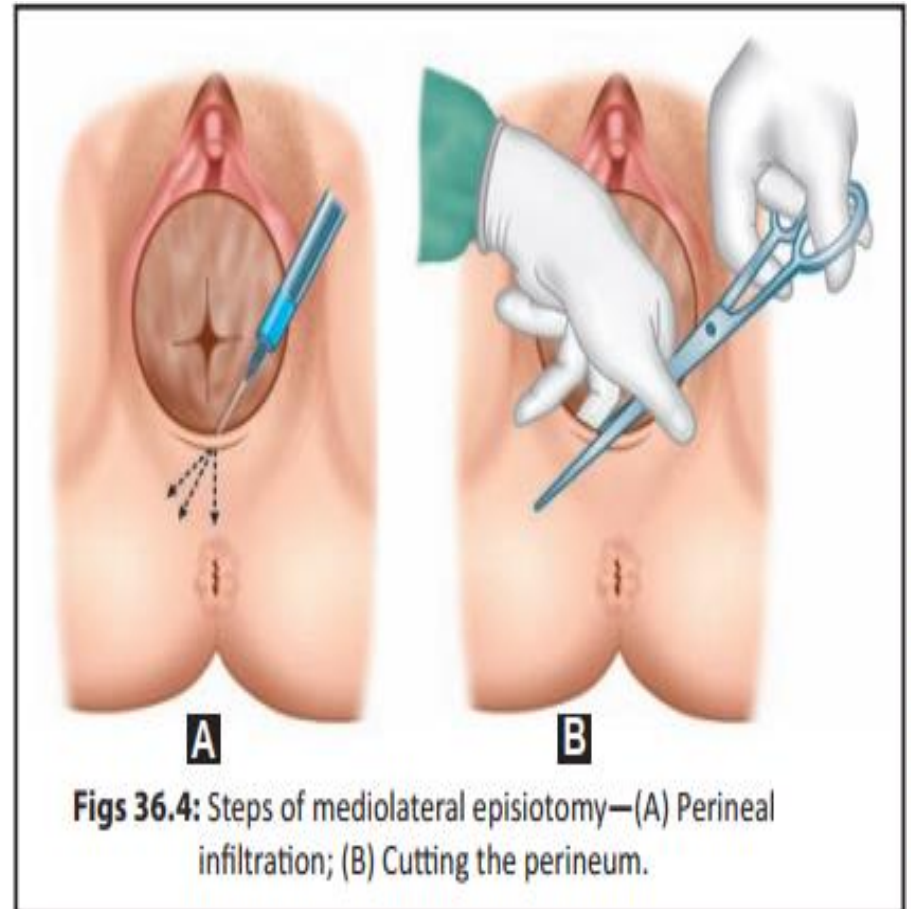


Types of episiotomy...

Characteristic		Midline	Mediolateral
Surgical		<i>Median</i>	<i>Mediolateral</i>
Faulty healing	Merits	— The muscles are not cut — Blood loss is least — Repair is easy — Postoperative comfort is maximum — Healing is superior — Wound disruption is rare — Dyspareunia is rare	— Relative safety from rectal involvement from extension — If necessary, the incision can be extended
Postoperative			
Anatomical			
Blood loss	Demerits	— Extension, if occurs, may involve the rectum — Not suitable for manipulative delivery or in abnormal presentation or position. As such, its use is selective	— Apposition of the tissues is not so good — Blood loss is little more — Postoperative discomfort is more — Relative increased incidence of wound disruption — Dyspareunia is comparatively more
Dyspareunia			
Extensions			
		Common	Uncommon

Steps of mediolateral episiotomy

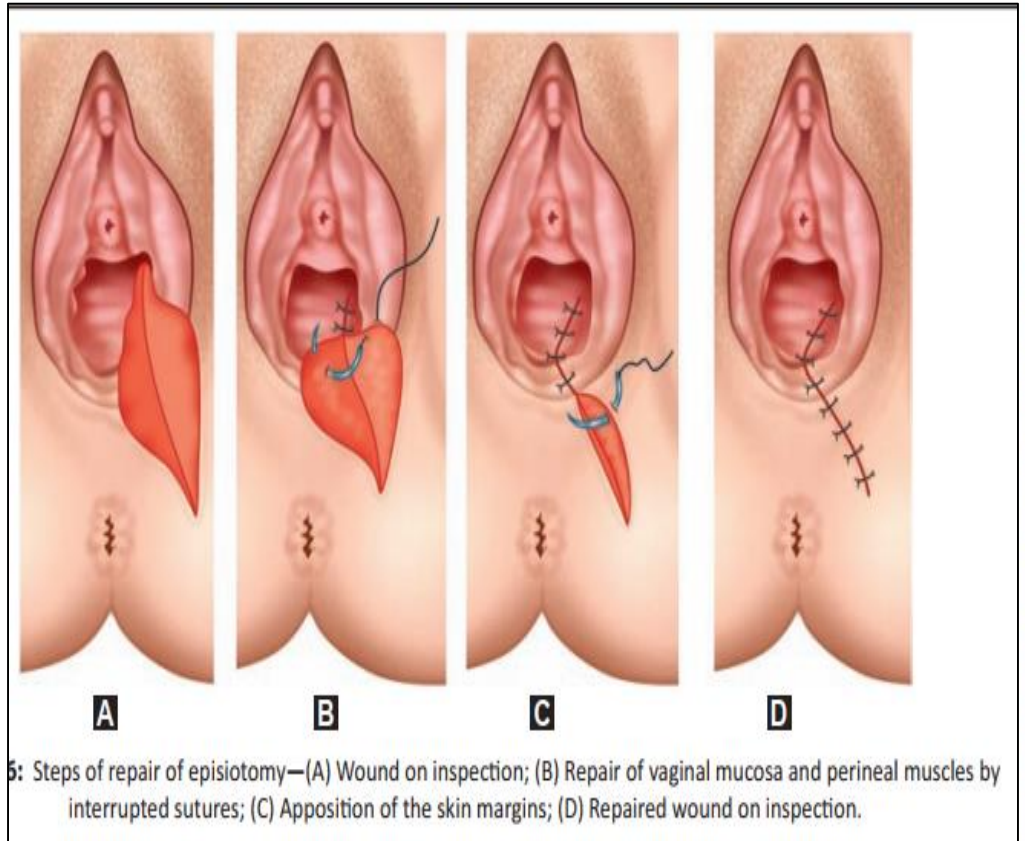
1. Swabbed with antiseptic (povidone-iodine) lotion and draped properly.
2. Local anesthesia
3. Incision



Steps of mediolateral episiotomy

4. Repair

- The repair is to be done in the following order
 - ⇒ Vaginal mucosa and submucosal tissues
 - ⇒ Perineal muscles
 - ⇒ Skin and subcutaneous tissues



Perineal lacerations

- Is a tear or spontaneous trauma in the perineal body, skin, mucosa or muscle
 - ⇒ Vaginal Tear (s)
 - ⇒ Periurethral Tear (s)
 - ⇒ Cervical Tear (s)

Perineal lacerations etiology

Over stretching of the perineum

Large baby

Instrumental delivery

Precipitated labor

Uncontrolled delivery of head

In elastic perineum

Previous scare

Elderly prim gravida

Extension of median episiotomy

Degrees of perineal tear

First degree tears	• Involve the vaginal mucosa and connective tissue.
Second degree tears	• Involve the vaginal mucosa, connective tissue and underlying muscles.
Third degree tears	• Involve complete transection of the anal sphincter.
Fourth degree tears	• Involve the rectal mucosa. • It may cause fistula of the anus

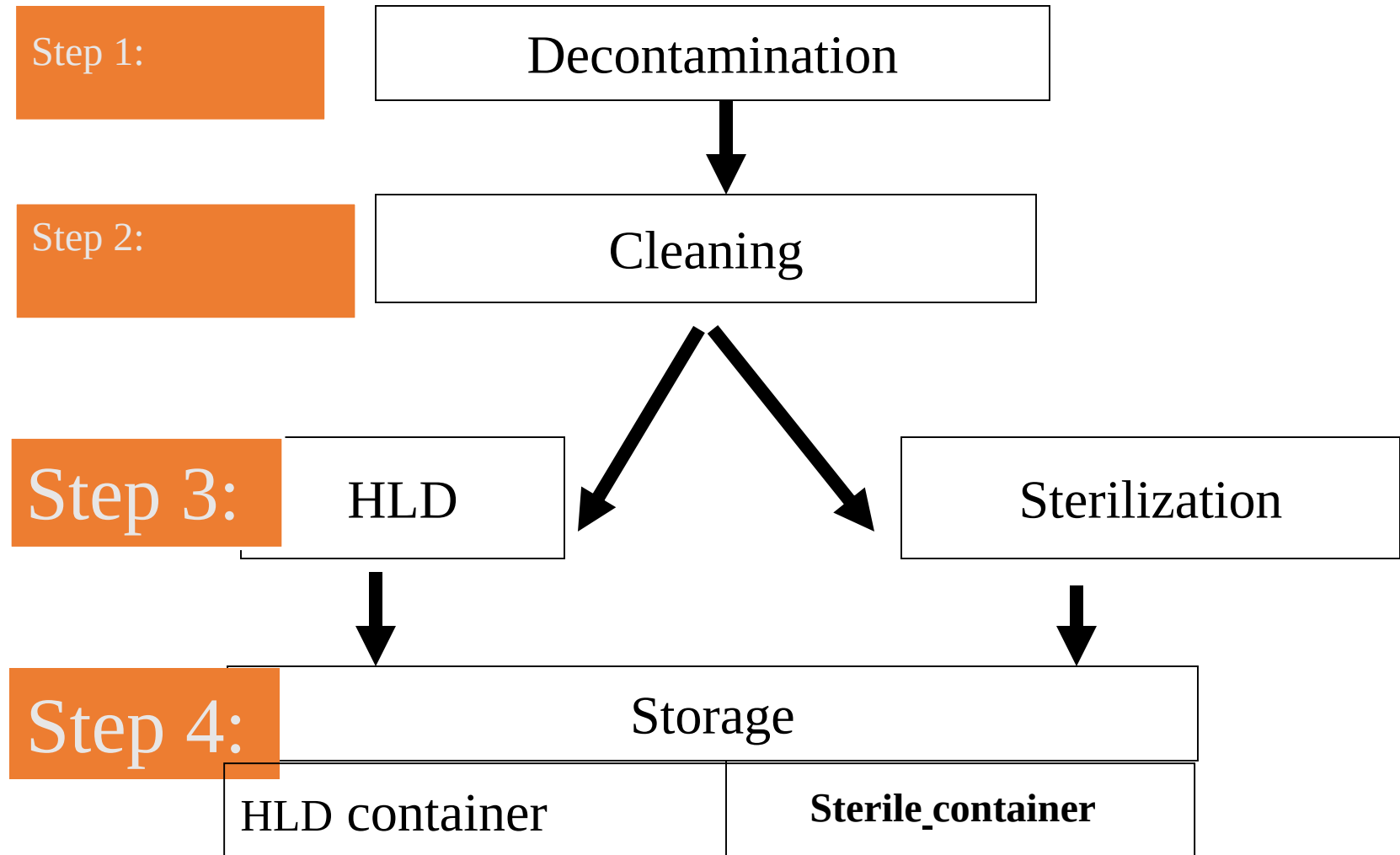
Repair of lacerations

- Is the same to episiotomy.
- But in case of 3rd and 4th degree tear the mother needs hospitalization because it repair by a physician under general anesthesia.

Complications of episiotomy and perineal tear

Immediate	Late
• Extension of the incision • Vulval hematoma • Infection • Wound dehiscence • Injury to anal sphincter	• Dyspareunia • Chance of perineal lacerations • Scar endometriosis

Infection prevention



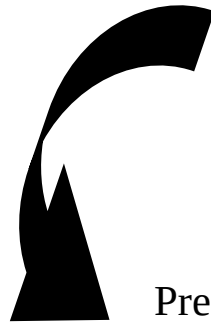
DECONTAMINATION

Soak in 0.5%
Chlorine solution for 10 minutes



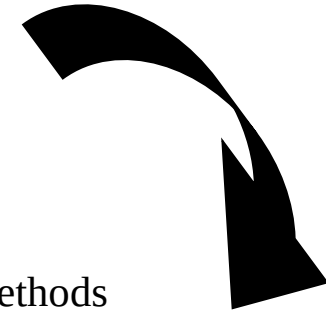
THOROUGHLY WASH AND RINSE

Wear glove and other protective
barriers(glasses, visors or goggles)



Preferred method

Acceptable methods



STERILIZATION

HIGH-LEVEL DISINFECTION (HLD)

Chemical
Soak
10-24 hours

Autoclave
106 k Pa pressure
(15 lbs./in²)
121₀C (250⁰F)
20 min. unwrapped
30 min. wrapped

Dry Heat
170⁰C
60 minutes

Boil or Steam
Lid on
20 minutes

Chemical
Soak
20 minutes



COOL
(Use immediately or store)



Unit-IV: Physiology and management of normal puerperium

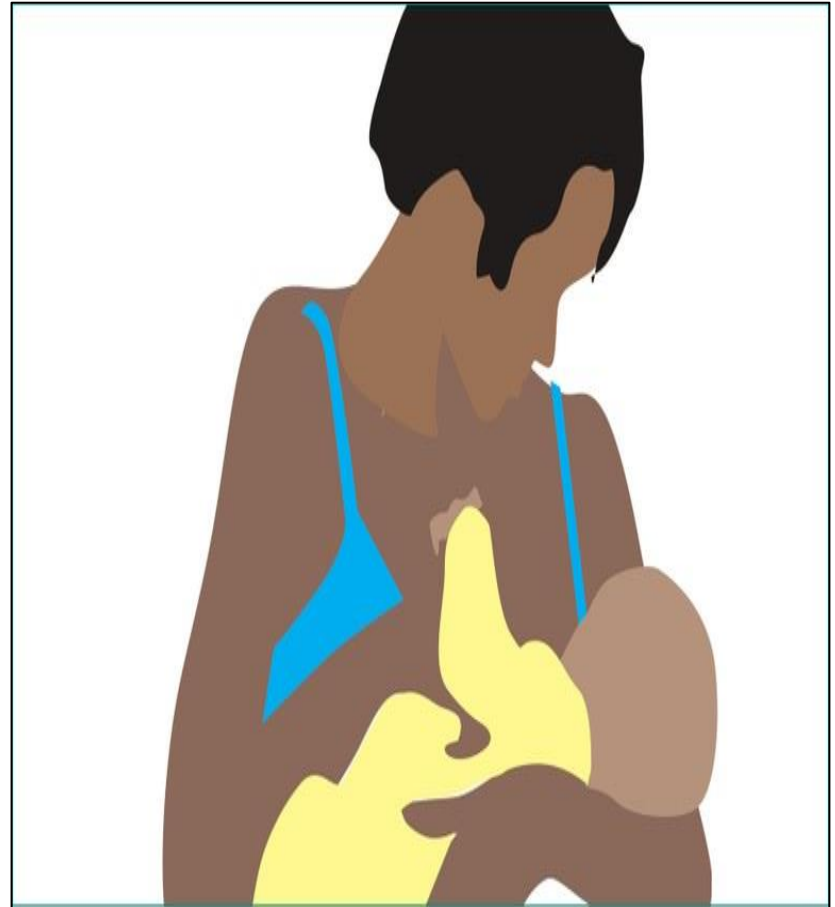
- Session objective
 - ⇒ Define normal puerperium
 - ⇒ Describe physiology and management
 - ⇒ Describe the elements postnatal care
 - ⇒ Physiology of lactation
 - ⇒ Management of breast feeding

Puerperium

- Is the period following childbirth during which the body tissues, specially the pelvic organs revert back approximately to the prepregnant state both anatomically and physiologically.
- Retrogressive changes are mostly confined to reproductive organs with exception of the mammary glands

Puerperium...

- Involution is the process whereby the genital organs revert back approximately to the state as they were before pregnancy.
- The woman is termed as a puerpera.



Duration of puerperium

- Puerperium begins as soon as the placenta is expelled and lasts for approximately 6 weeks
- The period is arbitrarily divided into
 1. Immediate – within 24 hours
 2. Early – up to 7 days
 3. Remote or remote – up to 6 weeks.

Physiology of puerperium

- During the postpartum four things are happen
 1. Affected organs return to their pre-pregnancy state.
 2. Lactation is started.
 3. Mother recuperate from the physical and emotional stress.
 4. The family adjusts to the new baby.

Uterine involution

- Is the uterus returns to its normal site, tone & position of non pregnant state.
- By the end of the Postpartum it weighs 50 gm instead of 1000 gm during pregnancy.

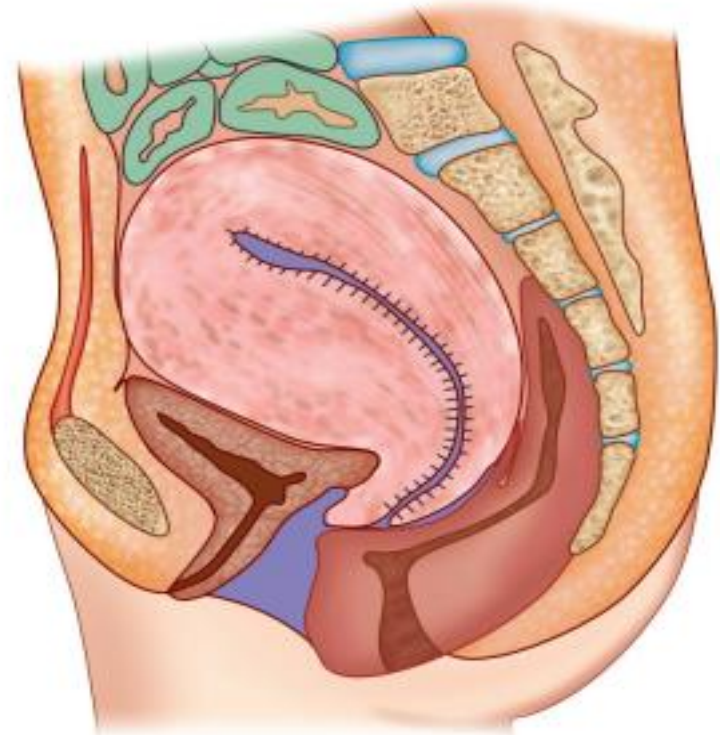


Fig. 13.1: Sagittal section showing uterus five days after delivery

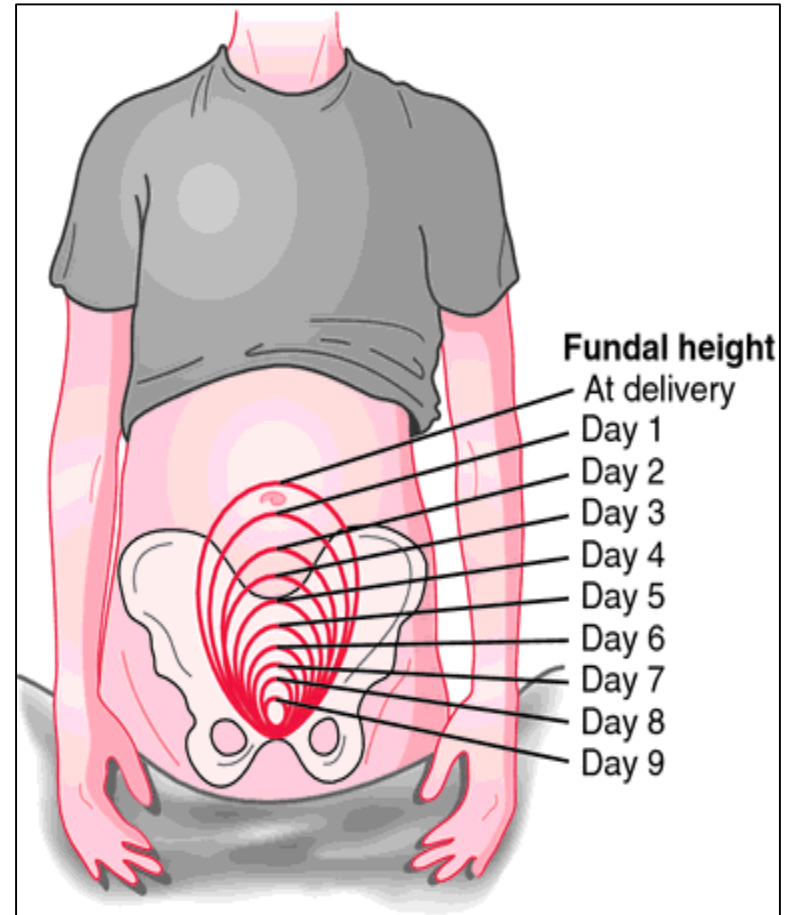
Uterine involution

- Mechanism of involution

1	Ischemia	• After the birth of the baby & placenta, the uterine muscle & blood vessels contracts so the blood circulation decreases.
2	Autolysis	• Muscle fibers are digested by proteolytic enzyme, waste product then pass in to the blood stream and are eliminated by the kidneys.
3	cast off endometrium	• Lining of the uterus is cast off and is replaced first by granular tissue and then by endometrium.

Size of the uterus

- Fundal height should be at or below the level of the umbilicus.
- Uterus should be firm, well contracted and in midline.
- Level of the fundus descends gradually at a rate of about 1 finger breadth every day.



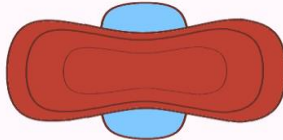
Size of the uterus....

Factors that enhance involution	Factors slow uterine involution
• Uncomplicated labor and birth. • Breast-feeding. • Early, frequent ambulation.	• Prolonged labor. • Incomplete separation and expulsion of placenta. • Previous labors. • Distended (full) bladder. • Anesthesia

Lochia

- It is the uterine discharge coming through the vagina during the first 3-4 weeks of the postpartum.
- It is alkaline in reaction
- Amount is more than the menstrual flow
- Fleshy odor.

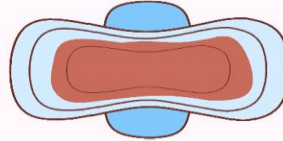
Dark Red Blood



Lochia Rubra

- ✓ From the birth until 3 - 4 days postpartum
- ✓ Flow in amounts like a heavy menstrual period.
- ✓ May have small clots
- ✓ The odor should be fleshy, like menstrual blood

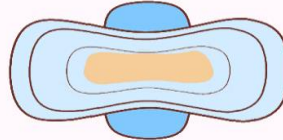
Pinkish-brownish discharge



Lochia Serosa

- ✓ From 4 to 12 days ~ 2 weeks postpartum
- ✓ Flow is moderate to small amount

Yellowish-white discharge



Lochia Alba

- ✓ From 12 days to 3 weeks postpartum
- ✓ Gradually disappearing
- ✓ Scant creamy whitish discharge

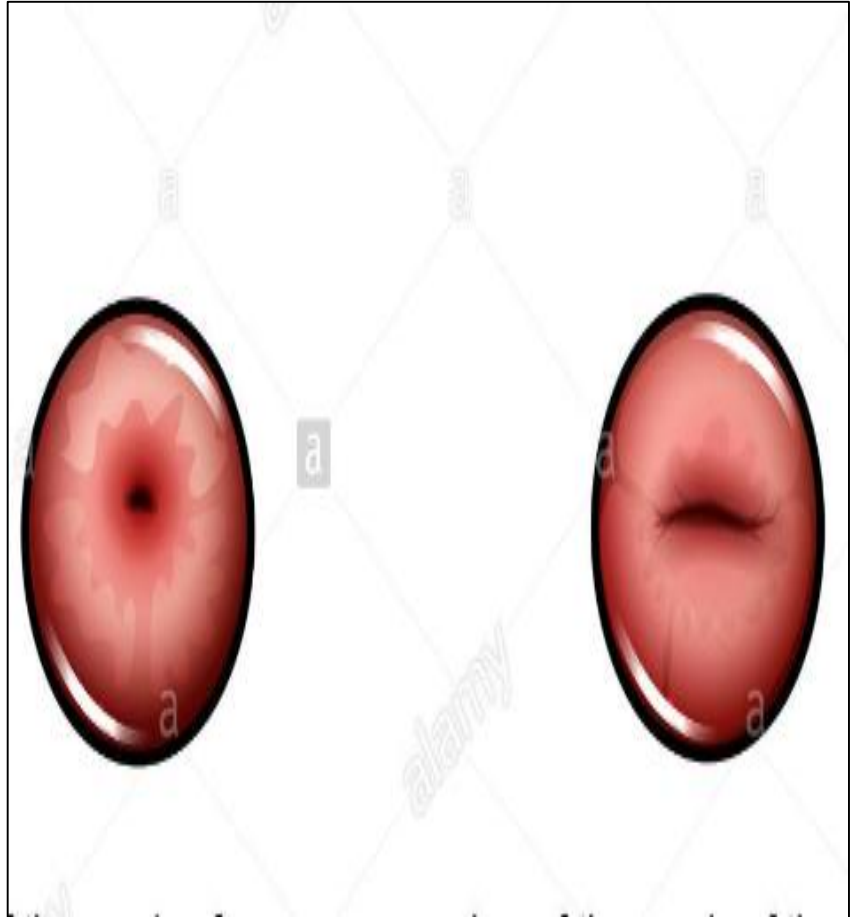
*Overactivity or stress
May increase lochia flow or change from serosa back to rubra.*

Total volume of postpartum lochial secretion is 200 to 500 mL, Can lasts from 3 to 6 weeks

Pinkorbluecare.com

Cervix

- The cervix is soft immediately after birth
- Within 2 to 3 postpartum days, shortened, become firm and regained its form.
- The cervical os closes gradually.
- Appears slit-like rather



Ovarian function

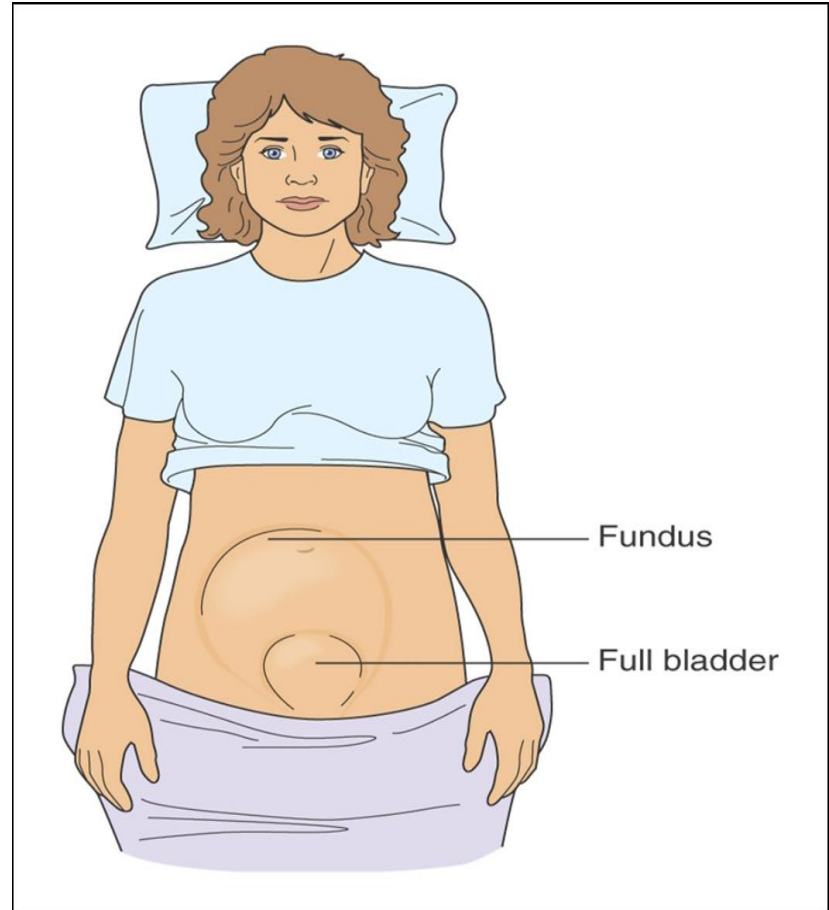
- Ovulation occurs as early as 27 days after delivery
- Mean time being ~70 to 75 days in non lactating women.
- Among those women who are breast-feeding their infants, the mean time to ovulation is about 6 months.
- The duration of an ovulation depends on
 - ⇒ The frequency of breast-feeding,
 - ⇒ The duration of each feed
 - ⇒ The proportion of supplementary feeds.

Integumentary system

- Melanin stimulating hormone levels rapidly decrease.
- Stretch marks “striae” appear purple to red
 - ⇒ Commonly on abdomen, breasts, hips
 - ⇒ Over time fade to silvery tan-white, do not disappear
- Linea nigra and other estrogen mediated skin markings gradually fade, may not disappear
- May have altered body image
- Chloasma disappears at the end of pregnancy.

The urinary tract and renal function

- The bladder increased capacity and has lost some of its muscle tone.
- Urinary retention and over distention of the bladder may cause two
 - ⇒ Urinary tract infection
 - ⇒ Post partum hemorrhage



Admission to post natal ward

- The mother and baby are transferred to the post natal ward
- Monitor mother every 15 min for the first hour; at 2, 3, and 4 hrs. then every 4 hours
- Vital sign every 30 min with in the first 2 hours.
- first BP is normal, measure the second BP within six hours
- Check uterine tone
- Check for bleeding

Postnatal care

- Include
 - ⇒ A systematic examination of the mother and the baby
 - ⇒ Appropriate advice given to the mother during puerperium
- The first postnatal examination given on discharge of the patient
- The second routine postnatal care is conducted at the end of 6th week postpartum.

Postnatal care...

Routine examination	Pelvic examination	Laboratory investigations
• Vital sign • Rrecording weight • Pallor • Tone of the abdominal muscles • Examination of the breasts.	• Cervical status • A cervical smear • Insertion of intrauterine contraceptive device may be done when desired	• hemoglobin) depending on the clinical need may be advised • Urine analysis
• Examination of the baby	• Immunization to the baby is started	

Postnatal care

- An important aspect is educational role.
- Advice the mother to care for her self and for her baby
 - ⇒ Nutrition
 - ⇒ Immunization
 - ⇒ Contraceptive
 - ⇒ Hygiene

Care of the bladder

- Bladder distention may occur within the 1st 12 hours
- Full bladder is risk for post partum hemorrhage
- Encourage the women to void
- If not possible to void (in 6 hours) intermittent catheterization

Perineal care

- For uncomplicated and satisfactorily repaired episiotomies
 - ⇒ Washing with water & soap 1 to 2/day plus after voiding & defecation
 - ⇒ Cold compresses promote healing and relieves pain
 - ⇒ Analgesics

Sexual activity

- Median time to resume intercourse is **6 to 7 weeks.**
- Couples can safely resume coitus
 - ⇒ Perineum is comfortable
 - ⇒ Bleeding diminishes

Contraception

- Choice of contraceptive
 - ⇒ Woman's preference
 - ⇒ Medical history
 - ⇒ Breast feeding or not



Contraception..

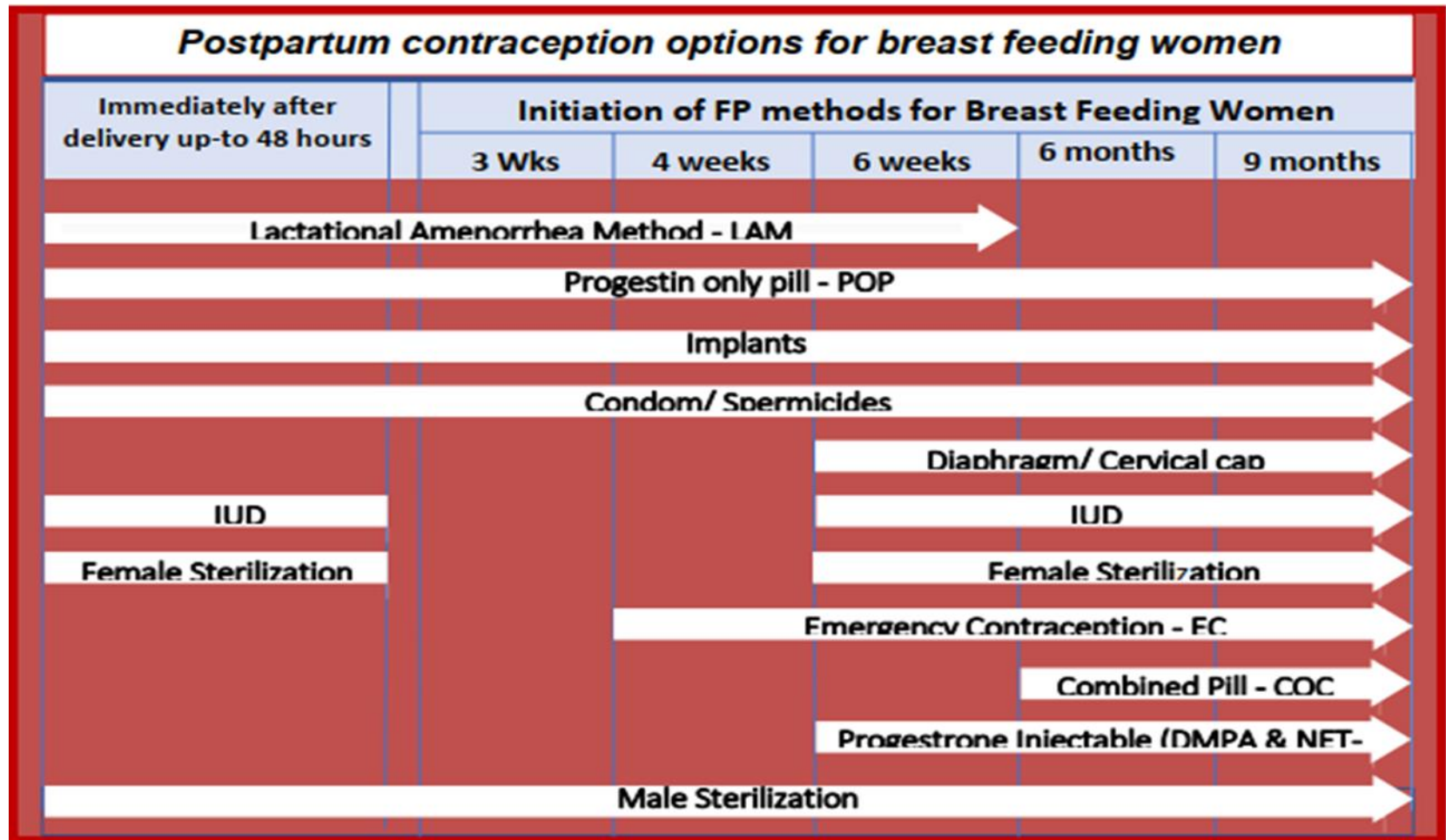


Figure 30. *Post partum contraception options for breast feeding women.*

Contraception..

Postpartum contraception options for non-breastfeeding women					
Immediately after delivery up-to 48 hours	Initiation of FP methods for Non-Breastfeeding Women				
	3 Wks	4 weeks	6 weeks	6 months	9 months
	Progestin only pill - POP				
	Implants				
	Condom/ Spermicides				
			Diaphragm/ Cervical cap		
IUD			IUD		
Female Sterilization			Female Sterilization		
		Emergency Contraception - EC			
	Combined Pill - COC				
	Progestrone Injectable (DMPA & NET-ET)				
	Male Sterilization				

Figure 31. *Post partum contraception options for non-breast feeding women.*

Sleep and rest

- The mother should have sufficient sleep and rest.
- Keep a quiet comfortable atmosphere with out disturbance.
- Hypnotics may be needed and it is given with out hesitation.
- Undue anxiety, sleepless ness and loss of appetite should be rewarded as serious.
- Rest is usually encouraged during the day preferably in prone position as this aids drainage from the uterus and vagina.

Ambulation

- Mothers benefit a feeling of well being
- Reduces the incidence of thrombi embolic disorders.
- Maintains good circulation, decrease venous thrombosis
- Restores muscle tone of the abdominal wall & pelvic floor
- Promotes for normal drainage of lochia
- Helps in emptying the bladder, bowels and uterus
- Restores her body figure

Nutrition

- Good balanced diet should be taken
- Protein foods are important particularly for breast feeding.
- Excess fruit should be avoided, it may cause diarrhea.
- The daily fluid intake should be from 2.5-3 liters of which at least 600ml should be milk.
- Mother needs 600kcalory/day food for next 6months

Advise on danger signs

Mother	Infant
• Fever • Abdominal pain • Foul-smelling lochia • Pain in the perineum or draining pus • Urine dribbling or pain on micturition • Breasts swollen & red	• Vomiting every thing • Hot or cold to touch • Abnormal body movement • Decrease or unable to breast feeding • Pusy discharge from eye or umbilical stump • Grunting

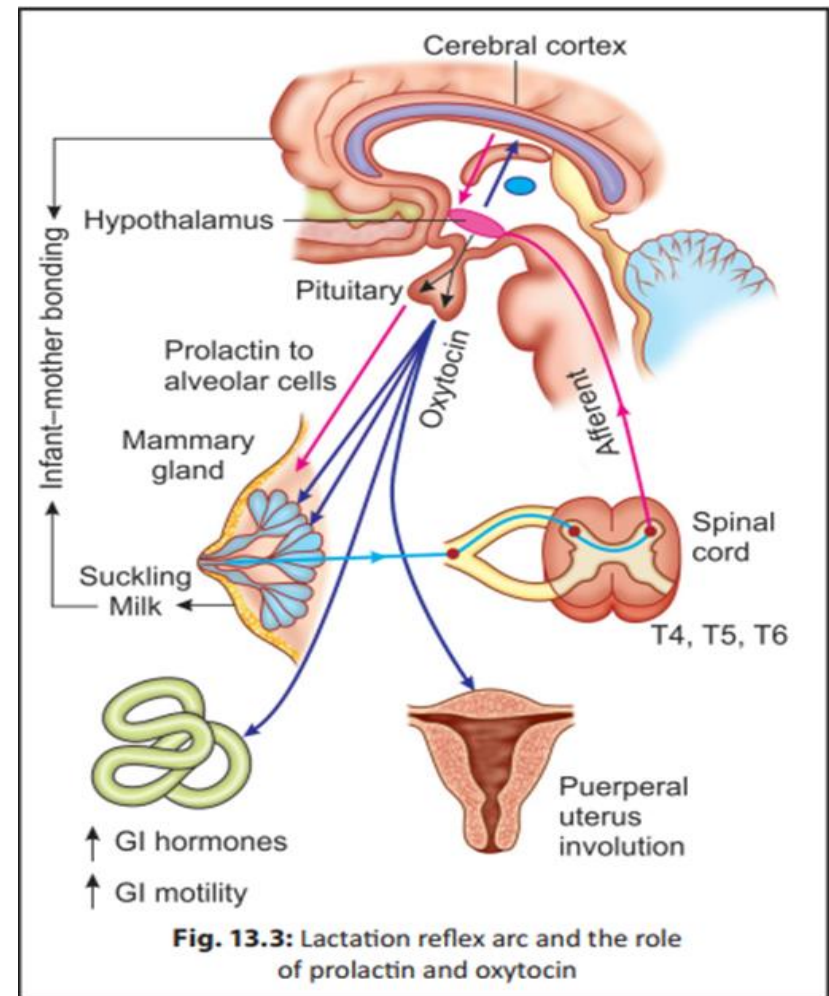
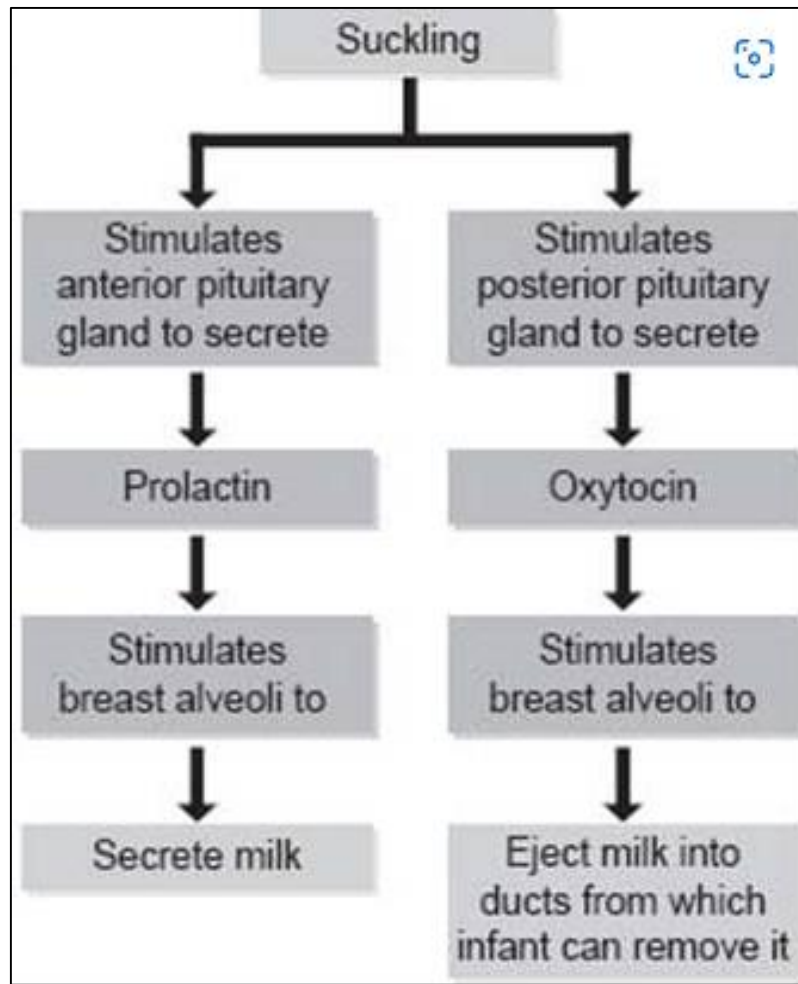
Time of discharge

- Could be made 6hours postnatal
- Mother should be well educated about
 - ⇒ Cares that should take for her self
 - ⇒ Infant feeding &care
 - ⇒ Identification of danger signs in either the infant or herself

Physiology of lactation

- lactation starts following delivery
- The preparation for effective lactation starts during pregnancy.
- The physiological basis of lactation is divided into four phases:
 1. Preparation of breasts (mammogenesis).
 2. Synthesis and secretion from the breast alveoli (lactogenesis).
 3. Ejection of milk (galactokinesis).
 4. Maintenance of lactation (galactopoiesis).

Physiology of lactation...



Physiology of lactation...

- Composition of breast milk
 - ⇒ Breast milk is 90% water
 - ⇒ 10% solids consisting of CHO, proteins, fats, minerals, vitamins
 - ⇒ Secretory IgA, immunoglobulin found in colostrum and breast milk, has antiviral, antibacterial, antigenic-inhibiting properties
 - ⇒ Helps meet changing needs of baby
 - ⇒ Foremilk – high water content, vitamins, protein
 - ⇒ Hindmilk - higher fat

Breastfeeding

- Early skin to skin contact of mother and baby and immediate initiation of breast feeding
 - ⇒ Initiate breastfeeding within 1hours
 - ⇒ Incase breast feeding can't be started due to either maternal or newborn illness, feeding the baby has to be initiated if possible by milk sucked from the mother herself.

Breastfeeding...

- Advantages to breast feeding:
 - ⇒ Promotes bonding between mother & baby.
 - ⇒ High nutritional value for infant.
 - ⇒ Promotes uterine involution through release of oxytocin
 - ⇒ Reduces cost of feeding & preparation time.

Breastfeeding...

- Women should be
 - ⇒ Encouraged to exclusive breast feeding for six month
 - ⇒ Complimentary feeding start at 6 month
 - ⇒ Breast feeding should be continued up to 2 years.
 - ⇒ Educated about common breastfeeding problems

Breastfeeding and breast care

- Postpartum education and counseling includes:
 - ⇒ Good position and attachment of the baby at the breast
 - ⇒ Exclusive breast feeding.
 - ⇒ Breast feeding on demand

Position and attachment

Good position	• The body and head of the baby should be straight • The baby should face the mother • The baby should be close to the mother • The baby should be fully supported
Good attachment	• The baby's chin should touch the breast • The baby should have a wide open mouth • The baby's lower lip should be turned outward • There should be more breast areolar tissue visible above the breast than below it

Management of breast feeding

- Additional advice:
 - ⇒ Choose position that is comfortable and effective
 - ⇒ Use both breasts at each feed; do not limit time at either
 - ⇒ Ensure adequate sleep/rest – take nap when baby sleeps
 - ⇒ Ensure adequate food/fluid intake – glass of fluids per feed; extra meal per day
 - ⇒ Feed 8-12 times per day

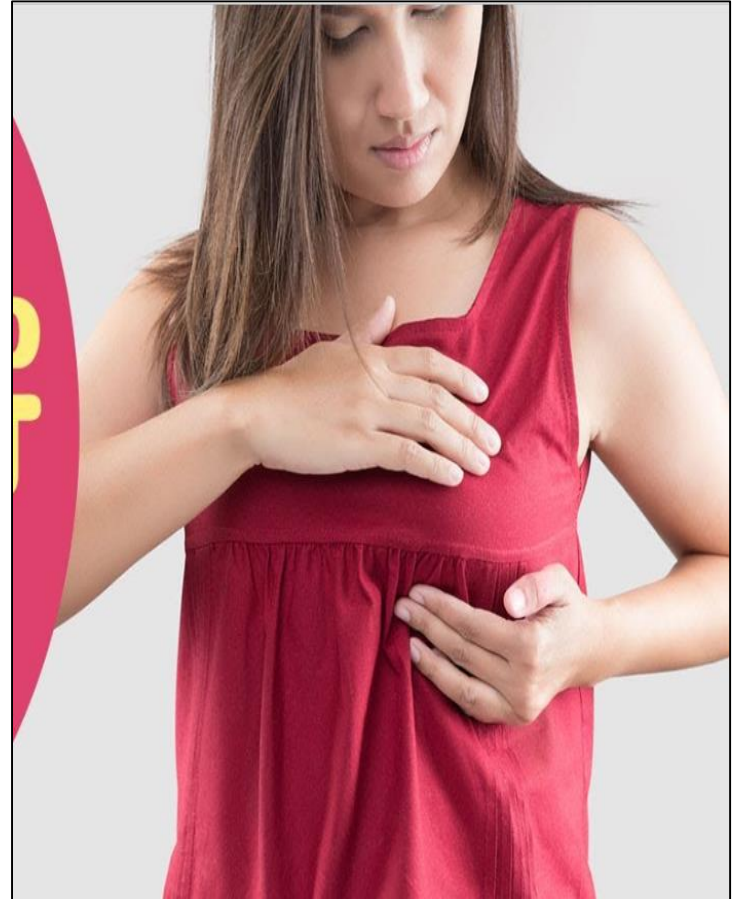
Management of breastfeeding...

- Breast care:

- ⇒ To prevent engorgement, breastfeed every 2-3 hours
- ⇒ Wear supportive (but not tight) bra or binder
- ⇒ Keep nipples clean and dry
- ⇒ Wash nipples with water only once per day – no soap
- ⇒ After breastfeeding, leave milk on nipples and allow to air dry

Management of breastfeeding...

- Breast engorgement
 - ⇒ Is breast swelling that to feel warm, heavy and tender
 - ⇒ Due to excessive build up of milk
 - ⇒ Occurs in the first 3 days after childbirth.



Management of breastfeeding...

	Management
Breastfeeding	• Encourage to breastfeed more frequently • Apply warm compresses just before breastfeeding • Support breasts with a binder or brassiere • Apply cold compress to the breasts between feedings • After feeding encourage the woman to express milk • Antipain :Ibuprofen 400mg PO TID base
Not breastfeeding	• Support breasts with a binder or brassiere • Cold compresses to the breasts to reduce swelling and pain • Avoid massaging or applying heat to the breasts • Avoid stimulating the nipples • Antipain :Ibuprofen 400mg PO TID base

Management of breastfeeding...

- Fissured Nipples:

- ⇒ Nipples may become fissured(cracks)

- ⇒ Feeding become painful and difficult.

- ⇒ Are a portal of entry for bacteria.

- ⇒ Further breast feeding should be stopped

- Milk can be expressed manually until the nipples heal

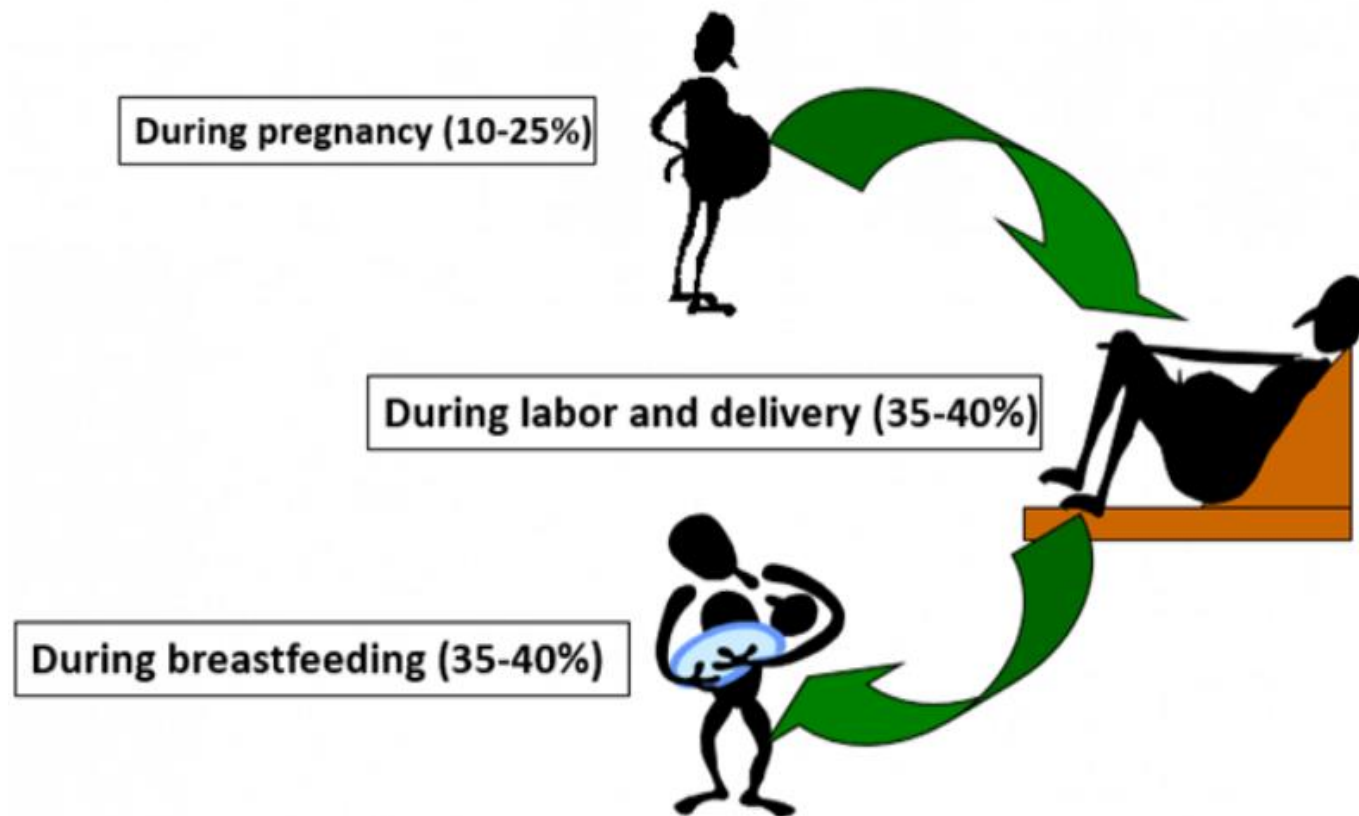
- ⇒ Contraindication with HIV positive mother

Mother to child HIV transmission(MTCT)

- Prevention of Mother to Child Transmission (PMTCT)
- Is the prevention of transmission of HIV virus from the mother to the fetus and child during pregnancy, childbirth and breastfeeding



Estimated risk of MTCT



Factors that affect the rate of MTCT

Maternal Factors	Infant factors	Obstetric and delivery practices
• High maternal viral load. • New or recently acquired maternal HIV infection. • Low CD4 count. • Advanced maternal disease. • Viral or parasitic placental infections during pregnancy, labor and childbirth. • Maternal malnutrition. • Nipple fissures, cracks, mastitis and breast abscess. • Poor ART adherence. • Genital tract infections like herpes simplex.	• First infant in multiple birth • Pre-maturity and low birth weight • Longer duration of breastfeeding • Mixed feeding during the first six months of life • Oral diseases in child	• Ante-partum procedures (amniocentesis cephalic version). • Rupture of membrane for more than four hours. • Vaginal delivery compared to CS. • Injuries to birth canal during child birth (vaginal and cervical tears). • Invasive childbirth procedures (e.g. episiotomy). • he first twin in vaginal delivery of multiple pregnancies. • Delayed infant drying with clean towels and eye care • Routine vigorous infant airway suctioning. • Instrumental deliveries (vacuum & forceps). • Fetal birth trauma.

Adverse of effect of HIV

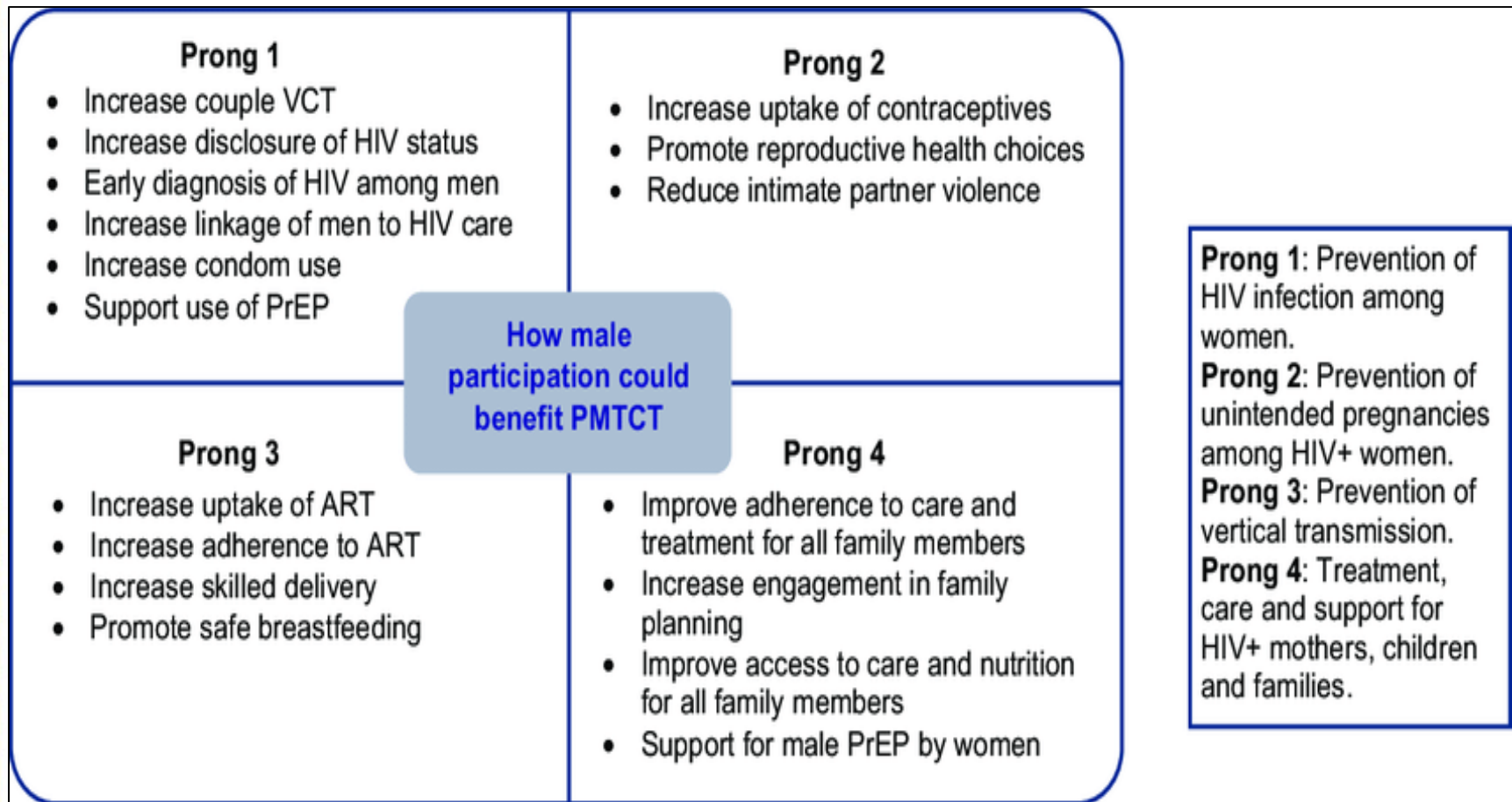
Effect of HIV on pregnancy

- Low birth weight
- Intrauterine growth restriction
- preterm delivery

Effect of pregnancy on HIV

- There is no known effect of pregnancy on HIV.

Prevention of MTCT of HIV



Diagnosis

- All pregnant should have screening for HIV.
- For positive determine CD4 count and Viral load.
- Symptoms and signs of opportunistic infections
- Clinical stage of the disease assigned.
- Repeat HIV counseling and testing at 28 to 36 weeks or during labor as appropriate.

Management

- Preconception care:
 - ⇒ Baseline CD4 and viral load.
 - ⇒ Avoiding unintended pregnancy(contraception)
 - ⇒ Prevention of malaria(ITN)
 - ⇒ Screening & treatment for opportunistic infections & STIs.
 - ⇒ Initiate ART/ link to PMTCT unit

Management...

- Discuss on future plan for pregnancy
- Prophylaxis for opportunistic infections:
 - ⇒ Cotrimoxazole for stages 2,3,4 and $CD4 \leq 350$
- Discuss on partner involvement & screening.
- Avoid pregnancy for 6 months after recovery from any chronic infections
 - ⇒ TB

Management

- Advise on general health
 - ⇒ Good nutrition
 - ⇒ Adequate caloric intake
 - ⇒ Consumption of iron rich foods (beans, lentils, meat, liver)
 - ⇒ Iron and folate for three month
 - ⇒ Intake of iodized salt.

Management

- All HIV positive mother will be initiated on HAART for life
⇒ **TDF, 3TC and DTG.**
- HIV positive woman already on ART at time of pregnancy should continue and stay on the same regimen.
- Pregnant women with WHO clinical stage 1 and 2 can safely be initiated on ART in ANC
- WHO stage 3 and 4 and opportunistic infections treat at ART clinic

Management....

- **Intrapartum care:**

- ⇒ Avoiding invasive procedures whenever possible:
- ⇒ Avoid artificial rupture of membranes
- ⇒ Avoid routine episiotomy.
- ⇒ Limit instrumental delivery
- ⇒ Avoid repeated vaginal examinations during labor.
- ⇒ Treat chorioamnionitis with appropriate antibiotics.
- ⇒ Provide essential newborn care (ENC)

Management....

- Mode of delivery:-

⇒ For women on HAART,

⇒ If the viral load is > 1000 copies/ml ⇒ C/S at 38 weeks

⇒ If the viral load is ≤ 1000 copies/ml ⇒ vaginal delivery.

Post-partum care

- Continue initial ART
- Formula feeding should be considered (AFASS) criteria,
- Exclusive breastfeeding for six months
- Complementary feeding should start at 6th month.
- Breastfeeding should not more than two years.
- Give NVP + AZT syrup, Cotrimoxazole
- DBS for DNA/PCR should be done at sixth week of life

Post-partum care

Table 19. Enhanced Post-natal Prophylaxis (e-PNP) for HIV Exposed Infants.

Infant age / weight		Formulation	Dosing
0-6 weeks	< 2500g	NVP 10mg/ml +	10 mg (1ml) once daily +
	> 2500g	AZT 10mg/ml NVP 10mg/ml +	10 mg (1ml) twice daily 15mg (1.5ml) once daily +
		AZT 10mg/ml	15mg(1.5ml) twice daily
6-12 week		NVP 10mg/ml	20mg (2ml) Once daily

Table 20. Dosage of Co-trimoxazole preventive therapy in infants and children.

Age	Preparation of the Co-trimoxazole suspension and tablets		
	Suspension per 5 ml 200/40 mg	Pediatric tablet 100/20mg	Single strength adult tablets(400/80 mg)
< 6 months	2.5 ml	1 tablet	½ tablet
6 months - 5 years	5ml	2 tablets	½ tablet

Assignment(25%)

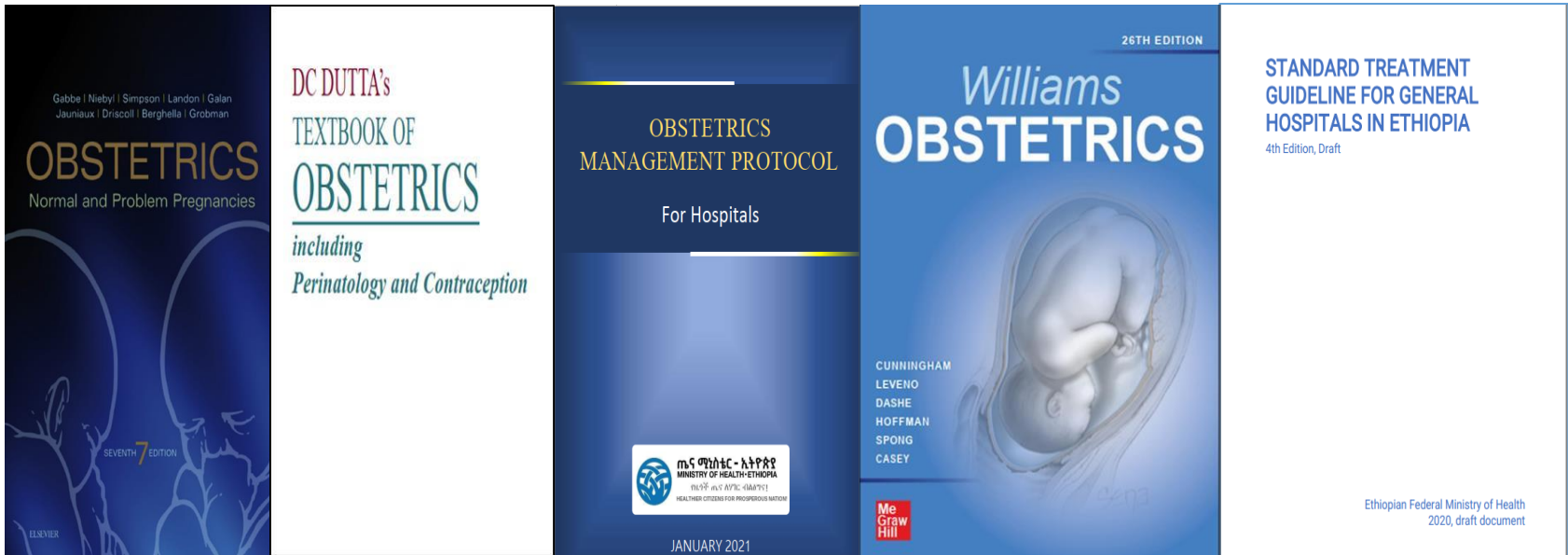
- Discuss about medical disorder of pregnancy

1. Gestational Diabetes
2. Heart disease
3. Bronchial ashma

- Content of assignment

- ⇒ Objective
- ⇒ Definition
- ⇒ Incidence
- ⇒ Risk factor
- ⇒ Etiology
- ⇒ Clinical presentation
- ⇒ Investigation
- ⇒ Diagnosis
- ⇒ Management
- ⇒ Complication on pregnancy
- ⇒ Reference

Reference



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